# Joint ML-KEM Key Generation against Algorithm Substitution Attacks

Full specification, simulator and reductions accompanying draft v43

Yan Shao, Yuejun Liu, Yongbin Zhou, Wei Cheng,
Jingdian Ming and Liehuang Zhu

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**Abstract**

We analyze joint ML-KEM generation for a managed receiver with a replaceable worker and a protected online helper. The deployment keeps complete-key generation outside the protected application and restricts the helper to shared custody. The protocol publishes $\rho$ before activation. Both direct and staged generation permit pre-activation aborts. The staged placement moves public-matrix work outside its second circuit and adds an authenticated sufficient-state handoff. Two separate recovery experiments preserve this observation schedule. An external extractor receives replaced Alice's full view, while passive Bob receives his complete view in an honest-Alice execution. Both recovery experiments test a fixed candidate private key on fresh encapsulation. A concrete PRG-based string commitment replaces deterministic share digests, with statistical binding over a public Toeplitz setup and computational hiding for honest Alice. Component simulators and public-key emulation reduce both recovery probabilities to honest-key KEM security with reservation and expected-time truncation costs. Bob-view privacy retains all component coins and messages. Its hiding hybrids charge latent anchors before output-dependent simulation, even when a call stops before delivery or activation. The corrected Lindell–Pinkas construction realizes the ordinary stage circuits. Commitment openings add explicit storage and input costs. The C38 circuits and the new generic-matrix U control have deterministic cost inventories, clear functional controls and native EMP measurements. A causal public-history wrapper exposes protocol status while retaining private command processing inside the trusted application. Its strict-time overhead is included in the recovery reductions. The LP realization and the native component evidence retain separate security and timing interpretations.

## 1 Statement, parameters and scope

This document proves two separate security goals for the commitment-anchor construction C38. Replaced Alice is handled by the framework and recovery reduction below. Prescribed Alice and passive Bob are handled by Sec. 11, which includes Bob's entire execution view. The measured hash-anchor construction H is retained only as reference cost evidence. Its native binary supplies the archived digest-anchor circuit measurements.

The security parameter is $\lambda$. State lengths, circuits and reservation budgets $\kappa, \tau$ are polynomially bounded. Public setup fixes indexed collision-resistant $H_c, H_p$, the PRG $G$ and its fixed domain, and the canonical public Toeplitz setup $T$. The latter is sampled once independently of initial advice and is immutable throughout the service budget. All encodings are injective and domain separated. Section 9 specifies the ordinary-circuit commitment and its private opening. Context fixes the key

handle, generation SID, profile, owner and setup indices. Each request fixes its operation SID, stage and public constants. Bob's owner never replaces its original commitment on a caller's instruction.

For Alice-direction security, Alice is a static classical malicious worker with arbitrary inputs, coins, state, messages and abort choices. Bob's code, randomness, shares and owner are protected. Channels are confidential and authenticated. Calls are sequential, with atomic nonrollback state and truthful termination. Only declared public scheduling and aborts are visible. At each invocation boundary the next application stage uses its prescribed records and outputs, rather than earlier component work tapes. This is an application dependency restriction. Bob may retain every earlier tape for the passive-view experiment; neither theorem assumes their physical erasure. Fresh current-call component and simulator coins are independent after entry.

The second experiment uses prescribed Alice with uniform contributions, openings and pads. Passive Bob may inspect all his inputs, tapes, messages and intermediate configurations. His own encapsulated session keys are allowed. The experiment excludes Alice's voluntary disclosure, a full Decaps oracle and hidden-result-dependent downstream feedback. Malicious Bob, concurrency, rollback, side channels and a post-quantum claim for classical LP are separate questions. Trusted delivery and application KDF/AEAD are outside the output-share boundary.

Fixed ML-KEM-768 has a concrete recovery bound with shared-key length 256. The asymptotic statement separately requires a growing secure KEM family with the same public-prefix and bounded-refusal contracts. PRG, hash and LP parameters then grow with $\lambda$, and $\sigma_{\mathrm{bind}}$ grows at least linearly. Fixed ML-KEM, SHAKE-based PRG and framework hashes receive explicit concrete assumptions. C38 has statistical binding over its public setup and computational hiding under its separate PRG assumption. Its Bob-view proof charges each latent anchor before output-dependent simulation, including a successfully computed candidate whose invocation is interrupted before delivery. The stopped-prefix argument preserves complete-view hiding and both expected-time truncation costs for these interrupted candidate outputs.

**Uniform public specialization.** A stage may be implemented by one uniform polynomial-time circuit generator $\mathsf{Gen}(1^\lambda, \mathsf{stage}, \mathsf{pub})$. This generator is deterministic (or canonically seeded by the same public transcript) and obeys an exact totalized-function invariant. Its inputs contain only the authenticated public parameters and hash indices; its output is an ordinary Boolean circuit for the same totalized stage function, with polynomial gate and interface bounds. Public matrix entries can therefore be compiled as constants rather than private multiplication inputs. This also fixes the public Toeplitz coefficients as an XOR-only linear map on the private digest. Both parties recompute the canonical circuit and descriptor before submitting private inputs. Full-string agreement, or the descriptor fingerprint already charged to $\epsilon_p$, fixes the common circuit. There is no caller-supplied secret-dependent compiler state. The U control retains T's two stages, domain tags and public descriptor while disabling public-matrix constant specialization. Its runtime public $\rho$ wires are fixed by the original descriptor; any private-holder copy is compared to that original public string inside the function before acceptance. These copies are public data, not new secret inputs or authority to change $\rho$. An honest copy gives exactly $F_T$; an unequal copy has only the specified fixed-zero failure output. The same commitment checks, output gating and owner rules therefore apply. This equality check is counted by the U inventory. Compilation belongs to the strict-time outer application; call security and running-time bounds are uniform over the generated circuits of the public size bound. Conditioning on the public descriptor in the forward lemma thus leaves the same standalone component guarantee. This permitted specialization does not assert correctness of any native compiler. The independent circuit construction below gives a quantitative realization.

# 2 Mathematical protocol

## 2.1 Deterministic computation and bounded sampling

For a 64-byte joint seed $w = d \parallel z$, let $\mathsf{SeedExpand}(w) = (\rho, \sigma, z)$ use the ML-KEM-768 internal expansion [1]. Let $M(\rho)$ expand each matrix polynomial with four SHAKE128 rate blocks, retain the first 256 valid coefficients of the resulting 448 candidates in stream order, and return $\perp$ if insufficient candidates occur. This is a fully specified bounded function. On acceptance it equals the corresponding FIPS stream prefix. An honest sampling refusal is a specified abort, not an integrity violation. We do not assert that keys conditioned on adversarial acceptance are uniform.

Define the sufficient states

$$S_T = \sigma \parallel z, \qquad S_L = \hat{s}[3][256] \parallel \hat{e}[3][256] \parallel z.$$

The latter uses canonical 12-bit coefficients. Deterministic $F_T(\rho, S_T)$ performs noise sampling, NTT and public-matrix completion; $F_L(\rho, S_L)$ performs the remaining matrix products and serialization. On honest states and successful $M(\rho)$ both return exactly the same $(ek, X)$ as the unsplit bounded KeyGen. Here $X = dk_{\mathrm{PKE}} \parallel z$ is the 1184-byte private core. These identities follow by substitution of the same intermediate values in the specified algorithms; they do not assert equivalence of an optimized native circuit.

Every stage is defined on its complete fixed-length input domain. For an arbitrary L-state, the fixed Boolean circuit defines a deterministic total extension even on noncanonical 12-bit coefficients. The proof imposes no particular dummy-pair output on that extension. The specified public sampling-refusal and authentication flags are unchanged; there is no extra private-format validity flag. A changed noncanonical state cannot be accepted outside the global bad-setup or hash-collision events, because honest generation is canonical.

For $X = dk_{\mathrm{PKE}} \parallel z$, form the full secret-key encoding

$$dk = dk_{\mathrm{PKE}} \parallel ek \parallel H(ek) \parallel z.$$

$\mathsf{TotalDecaps}(X, ek, c)$ executes complete FIPS decapsulation with this $dk$, including re-encryption and implicit rejection. It is total on all fixed-length inputs: invalid private-core encodings return the fixed string $0^{256}$ internally, without a public format-validity bit. Precisely, this guard checks that each of the 768 packed 12-bit coefficients of $dk_{\mathrm{PKE}}$ is less than 3329; every 32-byte $z$ is format valid. It does not test an additional private/public-key relation. This totalization matches the concrete C38 private-core guard. Honest generated cores use ordinary FIPS decapsulation. All statements about outputs refer to designated interface shares, not arbitrary values that a malicious Alice may later publish.

## 2.2 Anchors and public agreement

With fresh Bob masks $B_S, B$, define $A_S = S \oplus B_S$ and $A = X \oplus B$. Alice supplies opening inputs $s_S, s_F$. The transfer and final anchors are

$$D_S = \mathsf{Com}(\mathsf{transfer}_R, \mathsf{ctx}, \rho, A_S; s_S), \tag{1}$$

$$C = \mathsf{Com}(\mathsf{final}, \mathsf{ctx}, H_p(ek), A; s_F). \tag{2}$$

Commitments include the original descriptor in their digest. The semantic split profile $R \in \{T, L\}$ is domain separated. U is a generic-circuit representation of T and uses the same T domain

and descriptor, not a third commitment domain. In the strict theoretical variant, public descriptors must equal Bob's canonical descriptors. To cover digest-agreed interfaces, we also allow checking equality of their $H_p$ fingerprints; a distinct accepted descriptor is explicitly charged to an $H_p$ collision. Descriptor agreement is completed before invoking 2PC. Bob's canonical mathematical function and public constants determine the invoked circuit. Public parameters are inputs to a finite set of padded, uniform stage families; equivalently, the generator above may specialize them under its exact-function invariant. Profile, context, hash indices, public key and ciphertext never select unrelated nonuniform families. The agreed operation selects a stage family; any remaining public-input copies are checked against the canonical descriptor. In the proof a distinct hash-agreed descriptor is detected and charged before using a component theorem that presumes a common function. This check never gives Alice authority to update Bob's stored context, constants, public key or anchors. The concrete C38 prefixes use fixed domain tags and fixed-length public fields. Their unambiguous encoding, Toeplitz setup bits, PRG identifier and exact function are pinned by this descriptor, rather than selected by private caller state. The full-string variant is a special case with no descriptor-collision event.

### 2.3 Stage functionalities and outputs

Each function below is a deterministic circuit/function evaluated by 2PC. Fresh coins are explicit private inputs, sampled locally by the prescribed caller. Each opening input is a fixed-length PRG seed; every bit string of that length is valid. The public setup encoding and original descriptor are checked before private invocation. There is no scalar-format or opening-sampling refusal branch. Public validity checks precede acceptance; failed checks release a fixed failure flag and zero/empty payloads, including a zero masked relay, and Alice does not unpad or deliver that failed payload. An earlier released prefix remains visible. Ciphertext validity and implicit rejection stay private, with no corresponding public flag. Private-output designations specify the mathematical interface. The realized experiment observes Alice's view and Bob's prescribed application outputs. Bob full-view privacy is treated separately in Sec. 11.

**Direct generation D.** Alice provides private $w_A$ and final opening $s_F$, while Bob independently samples uniform $w_B, B$. Inside the function compute $w = w_A \oplus w_B$, the bounded KeyGen, $A = X \oplus B$ and $C$ on those same generated $A$ wires. Alice's private output is $A$; intended public outputs on success include $ek, C, \rho$ and the success flag. On bounded-sampling refusal D releases only the fixed failure flag and zero payloads, not the candidate $\rho$ or $ek$. Bob keeps $B$ and the original public record. A candidate for which bounded sampling fails never becomes an active handle.

**First split call.** For T, U or L, compute $(\rho, S_R)$ from $w_A \oplus w_B$, and produce $A_S, D_S$ as in Eq. (1), with Alice's opening input $s_S$. Alice privately receives $A_S$; Bob retains $B_S, \rho, D_S$ and the pending context. Public $\rho$ is allowed before a later abort. The intermediate mask is independent of final masks and output pads. Both parties may compute $M(\rho)$ publicly. Its refusal terminates the attempt before call two. The fixed second-stage function uses precisely this public matrix and context. U computes the same matrix in its generic circuit from the public $\rho$ wires and checks Bob's holder copy against the original descriptor before accepting. Its bounded sampling-validity wire is also checked. The same public matrix precheck refuses an insufficient bounded prefix before call two. Thus accepted T/U inputs have identical outputs, while a changed public copy has the specified zero payload. U retains T's 64-byte state and original commitment tags.

**Second split call.** On Alice's *actual* reinput $(x_S, s'_S)$, verify its commitment against the original $D_S$. On failure, release no final key-dependent payload. Otherwise compute $(ek, X) = F_R(\rho, x_S \oplus B_S)$, then $A = X \oplus B$ and $C$ on those same final wires using Alice's fresh final opening input $s_F$. Alice privately receives $A$ and the prescribed public candidate outputs; Bob retains the original record and $B$. Only protected final completion activates the handle. An interrupted candidate may have reached Alice but is never an active key.

**CheckedUse.** For an active handle, Bob retrieves its original $(\mathsf{ctx}, ek, C, B)$, samples fresh uniform $R$, and fixes the public request $(h, c, \mathrm{SID})$. In one call the function authenticates actual $(x, s'_F)$ under Eq. (2) and *consumes that same variable* in

$$U = \mathsf{TotalDecaps}(x \oplus B, ek, c) \oplus R. \tag{U}$$

Alice privately receives $U$; Bob retains $R$. Public Auth reports only descriptor/share authentication. Private canonical-core errors and implicit rejection remain inside TotalDecaps. A use cannot enroll a replacement $C$ for the previously accepted handle.

### 2.4 Owner transition rules

Every generation/use attempt spends a fresh SID and its quota before launch. Generation has at most one D call or two ordered T/U/L calls; a stage is never reused. Owner states distinguish reserved/pending, active, completed and burned attempts. Only protected completion of the final generation call activates a key. Outputs may be delivered before durable completion; pending state never permits a second execution. Failure burns the reservation. Recovery requires positive termination of the old worker, retains the original key record and spent history, and never resumes an orphaned transfer. Restoring a local share does not replace the owner record. These rules are part of the mathematical protocol, not properties inferred from an SQLite test.

## 3 Ideal service and security experiments

**Definition 1** (Exact-origin service $\mathcal{F}^R_{\mathrm{orig}}$)**.** *The ideal owner enforces the transition rules above. Opening inputs are fixed-length seeds, all of which are valid. It saves the original share and the image of its effective opening seed. For every reserved generation attempt it accepts the effective Alice contribution and opening inputs, and samples a uniform joint seed $w$ internally. It computes the specified mathematical state/core, keeping $S_R, X$ secret. For T/U/L it independently samples uniform $A_S$ and sets internal $B_S = S_R \oplus A_S$. It releases only the prescribed $(\rho, A_S, D_S)$, computing the commitment from $A_S$ and the effective $s_S$. It accepts continuation only for actual $x_S = A_S$, original descriptor and $G(s'_S) = G(s_S)$. The original seed encoding itself need not be unique. Completion independently samples uniform $A$, sets $B = X \oplus A$ internally, and creates the original $C$ using effective $s_F$. The exact public output schedule is the declared profile schedule: in particular $\rho$ may precede abort, and candidate output may precede activation. The adversary retains its abort/delivery choices. An uncompleted candidate stays inactive/burned.*

*For use, the service accepts only actual $x = A$, the exact original descriptor and an opening seed satisfying $G(s'_F) = G(s_F)$. On acceptance it samples a fresh uniform Alice output $U$, sets $R = \mathsf{TotalDecaps}(X, ek, c) \oplus U$ internally, and delivers the prescribed party outputs. It never gives Alice $X, S_R, B_S, B, R$. Restore and recovery leave these original records unchanged. The leakage consists of profile, public identifiers and parameters, lengths, released $\rho, ek$, public authentication/refusal and completion/abort schedule. Private ciphertext validity is excluded from the declared public leakage.*

The ideal service is an analysis functionality, not a third party deployed in the architecture. The profile's public leakage is intentional; no timing equivalence of D and T is asserted. The ideal service can delay public $\rho$ for D until that profile releases it. All output scheduling capabilities may be left unused by a simulator, which mirrors the actual protocol's order.

Let $\mathsf{Bad}_{\mathrm{orig}}$ mean a delivered designated share pair violates $U \oplus R = \mathsf{TotalDecaps}(X_h, ek_h, c)$ for the originally generated active handle, or an accepted handoff changes its original sufficient state. An audit may retain original values as ghost state to evaluate this event; they are never extra Alice outputs. For a real 2PC realization, the reference values use the effective-input schedule guaranteed by its extraction/simulation theorem. Values published by Alice after the protocol are outside the designated application interface-share relation.

Fix polynomial reservation bounds and write $Q_R = d_R\kappa + \tau$ for the deterministic slot budget. A shorter trace is padded by uninvoked inactive slots: an inactive slot has no messages, state change, component error or running time. A partially executed or externally interrupted invocation remains an invoked component slot, including any output-dependent prefix, even when the owner subsequently burns it. An abort need not force later independent reservations to be inactive. Slot indices always refer to this fixed budget, not to a random stopping time.

For every strict polynomial-time Alice $\mathcal{A}$ there is a single expected-polynomial-time simulator $\mathcal{S}$ independent of the final strict polynomial-time observer $\mathcal{D}$. Security holds for every polynomial-length auxiliary input and honest initial state, including correlated inputs, with the usual uniform-in-input bounds. Fixing the machines and polynomial size bounds fixes the polynomials and negligible functions below; these do not depend on the particular prior history. The initial auxiliary information and honest initial state are fixed, or sampled efficiently, before the fresh hash indices and Toeplitz setup are generated. Any nonuniform advice is independent of these fresh setup values. Later application states may depend on the indices through the prescribed execution; this is not permission for collision-dependent advice at setup. All computational assumptions, including collision resistance, the PRG, component DDH, PRF/encryption and knowledge/zero-knowledge components, use the same nonuniform strict-PPT convention. Initial advice is independent of freshly sampled hash/Toeplitz setup and challenge randomness; correlations arising through the prescribed execution are allowed. This convention also applies to strict-time observers and truncated collision finders. Component security below remains uniform over every bounded entry snapshot. Its pointwise use after an expected-time prefix does not strengthen the ordinary hash assumption. For slot $i$, write $\epsilon_{\Pi,i} = \epsilon_{\Pi,A,i}$ for the Alice-direction joint call simulation error against the residual-machine and observer wrappers defined below, including abort/delivery, the corrupted continuation state and honest application output. It is uniform over all their auxiliary snapshots.

Let $\epsilon_c = \epsilon_H + 2^{-\sigma_{\mathrm{bind}}}$. Here $\epsilon_H$ is the whole-execution first-changed-anchor $H_c$ collision finder probability, and $2^{-\sigma_{\mathrm{bind}}}$ bounds the single global bad-Toeplitz-setup event. The union over all message/seed pairs already covers every handle, so the setup term has no second session multiplier. Let $\epsilon_p$ sum real pre-call descriptor collisions and remaining all-simulated $H_p$ collisions, zero where full-string agreement is used. Stop before invoking a component under distinct accepted descriptors. The remaining simulated experiment always has a common canonical function. These are whole-execution probabilities, without a second handle multiplier. Expected-time search reductions are truncated using Lemma 7.

# 4 Lemmas and simulator

**Lemma 1** (Fresh joint seeds)**.** *Conditioned on any prior history and effective Alice contribution $w_A$, the joint seed $w_A \oplus w_B$ is uniform when $w_B$ is fresh uniform and private.*

*Proof.* The map $w_B \mapsto w_A \oplus w_B$ is a bijection. The assertion is made at the input boundary of an ideal 2PC call, before outputs and selective abort. It holds for adversarially chosen $w_A$ and for correlated prior sessions. It does not assert uniformity conditioned on later acceptance. □

**Lemma 2** (Joint masked-view simulation)**.** *At a fresh mask boundary, condition on prior state and history, including public outputs determined before sampling. The newly sampled Alice-facing share or use output is uniform. The intermediate and final shares at their first sampling boundaries, together with successive fresh use outputs, admit the joint simulation below, even for correlated generated states. Their anchors are generated jointly with the shares; disclosed anchors are never used to claim that those shares remain conditionally uniform.*

*Proof.* Condition only on pre-sampling state/history. Apply the bijections

$$B_S = S_R \oplus A_S, \qquad B = X \oplus A, \qquad R = \mathsf{TotalDecaps}(X, ek, c) \oplus U.$$

Each mask has its own independent uniform tape. Hence the induced joint distribution of $A_S, A, U$ is the product-uniform distribution even though $S_R, X, ek$ may be correlated. Anchors are deterministic functions of the same simulated shares, effective private opening inputs and public metadata. After disclosure the stored share and anchor are retained, never resampled or claimed uniform conditional on their disclosure. Reuse of $B$ across uses is harmless because the original $A, B$ relation is fixed and every use has a new $R$. Induct on messages/operations: each next Alice action is a function of an identically distributed view, so adaptive ciphertexts, reinputs, restores and aborts are preserved. The induction therefore preserves aborts that depend on the full preceding view and the public reservation history throughout every fresh invocation. □

**Lemma 3** (First wrong transfer breaks binding)**.** *An accepted $x_S \neq A_S$ under original $D_S$ gives an $H_c$ collision outside the global bad-setup event. Otherwise completion preserves the original key.*

*Proof.* The ghost execution records $(A_S, s_S)$ at commitment creation and extracts actual $(x_S, s'_S)$ from the new call. Lemma 10 applies under the fixed descriptor. If $x_S = A_S$, equality of anchors gives $G(s'_S) = G(s_S)$, precisely the ideal opening-equivalence relation. XOR cancellation gives $x_S \oplus B_S = S_R$, and the fixed completion returns the original key. D has no transfer event. □

**Lemma 4** (First wrong use breaks binding)**.** *Absent descriptor collisions and commitment-binding failures, checked use consumes original $A$ and returns authorized complete decapsulation shares.*

*Proof.* A distinct accepted descriptor gives an $H_p$ collision. With descriptor and $ek$ fixed, accepting $x \neq A$ supplies both openings of original $C$ and invokes Lemma 10, outside its global bad-setup event. For $x = A$, acceptance instead requires only $G(s'_F) = G(s_F)$; distinct seed encodings do not change the consumed share. Bob's original $B$ reconstructs $X_h$. Equation (U) gives authorized Decaps, including implicit rejection. Public Auth conceals that private branch, and the same checked input is consumed by complete decapsulation on that restored state. □

**Lemma 5** (Owner invariants and budget)**.** *With at most $\kappa$ generation and $\tau$ use reservations, the number of 2PC invocations satisfies*

$$q \leq Q_R = d_R\kappa + \tau, \qquad d_D = 1, \quad d_T = d_U = d_L = 2.$$

*The original anchor of an active handle is invariant under every recovery trace, and no such trace authorizes a second execution under an already spent SID.*

*Proof.* Induct on transitions. Reserving spends a fresh SID/quota monotonically. Each generation reservation permits only the fixed ordered stage list, with each stage launch recorded before execution. A use permits one launch. Completion, abort and worker termination do not decrement spent counters. Recovery tests termination, preserves those counters and tombstones, and does not resume pending stages. Only final completion changes an inactive candidate to active, without replacing its stored original fields. Thus every launch has one paid slot and the claimed injection into at most $d_R\kappa + \tau$ slots holds, including failed calls and early delivered outputs. A candidate output followed by failed completion cannot create an active handle. A delivered use followed by failed completion still cannot repeat its already spent application request. □

**The simulator in the ideal-2PC hybrid.** The simulator runs Alice, mirrors owner public transitions, and passes each effective Alice input to the ideal service. It uses the ideal public outputs and uniform Alice-facing shares and effective opening inputs to compute transfer/final anchors. If a private-output adapter is used, it retains the extracted Alice pad $P$ and forms the visible masked output $A_S \oplus P$, $A \oplus P$, or $U \oplus P$ on success. On a declared failure it instead forms the fixed zero relay; no failed payload is unpadded or delivered. It never queries original $S_R, X, B_S, B, R$. It mirrors the public descriptor checks, stage order, refusals and delivery/abort timing. It checks exact equality of the saved Alice-facing share and equality of the saved opening image $G(s)$, so it can reject altered shares without learning any underlying secret. Distinct seed encodings with equal image remain valid. Failed calls emit the same fixed failure payloads. The collision reduction, unlike this simulator, may run the honest mathematical algorithms and record full ghost state: this is a different proof algorithm, not information supplied to the adversary.

# 5 Forward composition with expected-time call simulation

Expected-time simulation cannot be inserted into an arbitrary composition argument: a later simulator can rewind an earlier expected-time simulation and invalidate its running-time bound [2]. The following lemma proves only the forward, call-local structure needed here. It does not invoke the general expected-time composition theorem or its additional superpolynomial hardness premise.

**Lemma 6** (Forward call-local composition)**.** *Consider a polynomial-slot sequential application with a strict polynomial-time adversary, polynomially bounded retained application state, and a straight-line mathematical stage oracle. Suppose one fixed universal residual-machine wrapper has, for each stage family, a call simulator with the following standalone guarantees, uniformly over all polynomial-length auxiliary states and both parties' inputs:*

1. *Its standalone real/ideal experiment jointly covers the corrupted continuation state and honest application output, with call error at most $\epsilon_{\Pi,i}$. It extracts the current effective corrupted input or abort, and performs at most one effective ideal-stage invocation.*

2. *It uses fresh simulator coins at entry, may rewind only the current residual invocation, and returns one committed continuation. It neither rewinds the application prefix nor exposes speculative extraction transcripts or ideal outputs to later stages.*

3. *Counting the strict-time residual adversary's steps and all local rewinding, its conditional expected running time is at most a common polynomial $p(\lambda, m, c)$ for every entry snapshot and honest input of the public size bound $m$.*

*Then the forward simulator is expected polynomial time. For every strict polynomial-time observer its error over $Q_R$ fixed slots is at most*

$$\sum_{i=1}^{Q_R} \epsilon_{\Pi,i}.$$

*If every uniform call remainder is at most $\nu(\lambda)$, summing over the fixed slot budget bounds the total remainder by $Q_R\nu(\lambda)$, which is negligible for polynomial $Q_R$.*

*Proof. Snapshots and resumption.* Sample the strict-time adversary's polynomial-length random tape once. At a call boundary let $s_A$ consist of its configuration, transcript, remaining clock budget and residual tape. A single clocked machine $\mathcal{A}^\star$, on auxiliary input $(i, \mathrm{stage}, s_A)$, resumes this configuration only for the current call and outputs its complete continuation configuration at that call's boundary. It is strict polynomial time on every message sequence, including messages arising in rewinding. A rewound branch uses the same residual adversary tape; only the local simulator's prescribed coins change. This faithfully preserves correlations with all earlier choices, instead of replacing Alice by a freshly randomized adversary at each call.

Write $s_B$ for honest retained application state and current private input, and $s_{\mathrm{obs}}$ for the observer's state and permitted initial auxiliary information. Neither $s_B$ nor private auxiliary values are given to $\mathcal{A}^\star$ or to the ideal simulator. They may be auxiliary information of the external call distinguisher, as allowed by joint-output standalone security. The honest continuation is a polynomial-time function of $s_B$, the joint call output and fresh future honest coins. It never requires discarded component-private coins. Thus the *joint* continuation distribution is covered, not only Alice's marginal transcript. An extracted effective input is a private ideal-side request, not an added real-side observer output. The later collision finder records these requests only in the all-simulated experiment; no hybrid comparison gives real parties an input extractor. The snapshot does not fix any current-call honest component coins: those coins, and the current simulator tape in the comparison experiment, are sampled after entry. Fixing Bob's application input is permitted by input-uniform security; conditioning on his fresh component tape would not be justified.

*Forward simulator.* At slot $i$, run the same call simulator against $\mathcal{A}^\star(i, \mathrm{stage}, s_A)$ with fresh private simulator coins. Pass its single effective stage request to the commitment-check mathematical stage oracle for that component; give back exactly its prescribed reply. This oracle checks the supplied commitment relation, not exact-origin equality. Commit the returned corrupted configuration, apply the honest/owner output transition and continue forward. Retain only that committed state, the prescribed application records and public transcript. The ideal service is never rolled back and no previously delivered output is rerun. Inactive slots take zero time. This construction depends on $\mathcal{A}$, not on the final observer.

*Hybrid error without an expected-time distinguisher.* Let $G_j$ simulate slots $1, \ldots, j$ as above and run every remaining slot by the real component. $G_0$ is real and $G_{Q_R}$ is the forward simulation with the same commitment-check stage oracle. The replacement of that service by $\mathcal{F}_{\mathrm{orig}}$ occurs only in the later collision reduction. To compare $G_{i-1}$ and $G_i$, fix their common complete entry snapshot $(s_A, s_B, s_{\mathrm{obs}})$. Their earlier simulated prefix may have taken expected time to generate, but it is not rerun by this comparison. The standalone adversary is the same strict-time $\mathcal{A}^\star$ with its snapshot as auxiliary input. The standalone distinguisher receives the joint output, performs the relay and owner transition, and runs only *real* components in all later slots, then runs the final observer. All these operations are strict polynomial time. There are no future expected-time call simulators in this distinguisher. Uniform auxiliary-input security bounds the conditional difference by $\epsilon_{\Pi,i}$ for *every* fixed snapshot. Averaging this pointwise bound over their

common, possibly expected-time-generated prefix gives the same bound; it does not assume that the prefix distribution has a strict-time sampler. The triangle inequality over the deterministic $Q_R$ slots proves the stated error. Slot index and adaptive public inputs are auxiliary inputs to the same universal wrappers, so a common negligible remainder applies to all slots.

*Running time.* Let $\mathcal{G}_{i-1}$ contain all committed state before slot $i$, including previously used simulator coins, and let $T_i$ count all steps of the new local simulation. Its new tape is independent of $\mathcal{G}_{i-1}$. The uniform entry-state bound gives

$$\mathbb{E}[T_i \mid \mathcal{G}_{i-1}] \le \mathbf{1}_{\{\text{slot } i \text{ active}\}} p(\lambda, m, c) \le p(\lambda, m, c).$$

The tower property and finite linearity of expectation yield

$$\mathbb{E}[T_{\mathcal{S}}] \le p_{\text{outer}}(\lambda, m) + \sum_{i=1}^{Q_R} \mathbb{E}[T_i] \le p_{\text{outer}}(\lambda, m) + Q_R p(\lambda, m, c).$$

The conditional bound applies even to rare entry histories; it is not a bound conditioned on later success. The outer simulator never uses a component simulator as a rewound oracle, so the known generic expected-time oracle-composition counterexample does not arise. □

**Lemma 7** (Expected-time collision extraction)**.** *Let $\mathcal{C}$ be a hash-collision finder whose expected running time is uniformly bounded by a polynomial $P(\lambda)$ and whose success probability is $\eta(\lambda)$. For every polynomial cutoff $t(\lambda)$, the strict-time truncated finder $\mathcal{C}^{\le t}$ satisfies*

$$\eta(\lambda) \le \mathrm{Adv}_{\text{search}}(\mathcal{C}^{\le t}, \lambda) + \frac{P(\lambda)}{t(\lambda)}.$$

*Consequently the corresponding ordinary strict-time search assumption makes $\eta$ negligible.*

*Proof.* Truncate after $t$ steps; a lost successful run must have exceeded $t$. Markov's inequality bounds this probability by $P/t$. For the asymptotic claim, if $\eta \ge 1/r$ infinitely often for some polynomial $r$, take $t = 2Pr$. Then the strict-polynomial-time finder succeeds with probability at least $1/(2r)$ on those parameters, contradicting the search assumption. No fixed polynomial cutoff is asserted to make $P/t$ negligible; the negligibility conclusion is this contradiction argument. The finder returns only its polynomial-length, efficiently checked collision witness. □

# 6 Framework theorem and concrete theoretical realization

**Theorem 1** (Framework security)**.** *For every strict polynomial-time Alice there is an expected-polynomial-time simulator for $\mathcal{F}^R_{\text{orig}}$, independent of the final strict-time observer. In the ideal-stage hybrid its joint simulation error is at most $\epsilon_c + \epsilon_p$. If the component $\Pi$ has the call-local properties of Lemma 6, the real/ideal distinguishing advantage is at most*

$$B_R = \sum_{i=1}^{Q_R} \epsilon_{\Pi,A,i} + \epsilon_H + \epsilon_p + 2^{-\sigma_{\text{bind}}}. \tag{3}$$

*Honest execution returns the specified generated key and complete decapsulation, or the specified sampling/component abort. In the ideal-stage experiment, every designated output satisfies exact-origin correctness except on the hash-collision or global bad-setup events. The realized protocol has simulation-based exact-origin security for the joint Alice view and honest application outputs.*

*Proof.* First consider commitment-check ideal stage calls. Lemma 2 couples all shares and anchors with the exact-origin ideal service. The only differences between their acceptance rules are commitment checks versus original-share equality with the saved opening-image equivalence. For the same share, both accept exactly the seeds with that image. Stop at the first accepting changed transfer, full share or descriptor. Lemmas 3–4 give a hash collision outside the one global bad-setup event under the fixed descriptor. A reduction simulates the whole execution and stops at that first event, so there is no need to guess its handle or multiply a global advantage by a second session factor. Outside those events the coupled transcripts and owner public states are identical. The same algebraic lemmas and owner induction prove exact-origin correctness in this ideal-stage experiment. Honest inputs cancel masks, pass the original anchors and execute the stipulated functions.

For the real protocol, first stop before invoking a component if distinct descriptors pass pre-call fingerprint agreement. The public reduction runs the real service and outputs those encodings as a collision, at the pre-call part of $\epsilon_p$. In the remaining game the function is common. Lemma 5 supplies the fixed $Q_R$ slots. Apply Lemma 6 to obtain the all-simulated *commitment-check* stage experiment, at error at most $\sum_{i=1}^{Q_R} \epsilon_{\Pi,i}$. Only then apply the straight-line exact-origin simulator to the committed effective stage requests. The joint masked-view coupling and first-collision argument cost the remaining $\epsilon_H + \epsilon_p + 2^{-\sigma_{\mathrm{bind}}}$. This proves Eq. (3). The ideal-stage transformation never rewinds a call simulator or consumes speculative extraction output. No independence assumption about failures, selective aborts or adaptive reservations is used.

The ghost hash-collision finder is a separate proof algorithm. It samples honest stage inputs and original state itself, runs the forward call simulations, and records their effective requests. It can therefore detect the first unequal accepted share without supplying any private state to the actual ideal simulator. Equal resulting digests give a hash collision; unequal digests and equal anchors witness the already bounded bad-setup event. Its expected time is polynomial by Lemma 6 plus polynomial owner, mathematical-stage and hash computations. Lemma 7, rather than an unsupported strict-time assertion, proves that its success probabilities are negligible under ordinary collision-resistance assumptions. The separate setup-event probability is bounded information-theoretically.

The correctness statement does not manufacture a strict-time observer that extracts hidden real inputs. The direct wrong-transfer/use probability statement applies to the ideal-stage, extracted-request experiment. The real protocol securely realizes that exact-origin service in the ordinary joint-output simulation sense. Effective corrupted inputs are supplied by the published component simulator; the ideal service keeps the original state resulting from them. No separate quantitative claim about an unobservable real-world ghost event is inferred solely from transcript indistinguishability. Arbitrary post-protocol shares Alice publishes are not designated interface outputs. □

### 6.1 A realized component, not an unresolved backend residual

Use the corrected full Lindell–Pinkas construction [3], dated October 29, 2014: Protocol 4.1/Theorem 4.2, Proposition 3.9 and Appendices A/B. Set Appendix B's aggregation parameter $L = \lambda$ and use a prime-order cyclic group of order greater than $2^L$; no fixed 40-bit auxiliary parameter is inherited. All knowledge and encryption parameters scale with $\lambda$. The required assumptions are DDH, the specified zero-knowledge/knowledge primitives and the stated garbling encryption; the paper gives a PRF-based encryption instantiation. Do not substitute EMP/KOS or optimized garbling without a separate argument.

**Qualification for the forward lemma.** For corrupted garbler $P_1$, the published simulator completes local extraction before its sole ideal-function request (Theorem 4.2, proof Steps 1–8). Its main simulation is straight-line; Section 5.1 confines rewinding to the zero-knowledge sub-

protocols. Use the construction's witness-extended emulation, which includes aborting/rejecting executions, rather than repeated extraction conditioned on success. Apply it to the fixed clocked residual wrapper. LP Protocol 4.1 explicitly takes the circuit description $C$ as common auxiliary input (p. 16); Theorem 4.2 and its corrupted-$P_1$ simulator (pp. 17–18) are one circuit-generic algorithm. The wrapper therefore takes this public circuit as part of its input. Its polynomial bounds depend on the declared circuit-size/input bounds, not on the specific public constant values. Both the real and simulated wrappers first recompute the same deterministic public specialization. This gives a uniform expected-time polynomial for every generated circuit and entry auxiliary state in the standalone theorem's input convention. Specifically, the emulator's analysis fixes the prover's auxiliary input and random tape: an accepting first execution's probability cancels its inverse-probability extraction cost. Lemma 3.1, Claim 3.2 and Section 3.3.1 of the cited emulator construction [4] give a uniform polynomial when the resumed prover is clocked strict time. Definition 2.1 has uniform arbitrary-input security. A residual wrapper that outputs its full continuation configuration therefore obtains the required joint state/output ensemble by Theorem 4.2. The next application stage uses only application input/output, not past component-private tapes. The Bob-view experiment retains those tapes as observations, as proved separately in Sec. 11. Fresh local simulator tapes are sampled after entry. These facts instantiate all three hypotheses of Lemma 6: no earlier call, owner transition or ideal reply can be rewound. They do not assert a stronger general-purpose composition property of LP.

The published protocol has evaluator output. Assign Alice to its garbler role and protected Bob to evaluator. For a target function $f(a, b) = (o_A, o_B)$, with fixed public output lengths, let $\beta$ be its declared public stage-success flag in the metadata of $o_B$; for a stage without such a failure branch set $\beta = 1$. Evaluate

$$g((a, P), b) = \big(o_B,\ \beta \cdot (o_A \oplus P)\big), \tag{4}$$

where multiplication by $\beta$ gates every relay bit. Alice unpads and delivers the private result only when $\beta = 1$. When $\beta = 0$, both the relay and the other failed payloads are their fixed zeros. This matches the concrete circuit's final output gating. Use a fresh Alice-private pad $P$. Inputs are padded to the same public length as required by the component. Bob relays *only* the fixed-length masked Alice component and intended public metadata; he never relays the whole output containing Bob's designated private component $o_B$.

**Lemma 8** (Evaluator-output adapter)**.** *Against malicious Alice and protected Bob, Eq.* (4) *and its authenticated relay realize the joint Alice-view and honest-application-output ensemble with the transformed component's simulation error and prescribed abort semantics.*

*Proof.* The call simulator extracts $(a, P)$, including adversarially selected $P$. The outer simulator obtains the ideal private result and public flag and computes exactly the relay $\beta \cdot (o_A \oplus P)$. In a failed stage this is zero and requires no private result. It need not know Bob's private output. An honest Bob sends the correct relay only after accepting the computation; aborts before that point are preserved. Subsequent Alice postprocessing is part of her view. Uniform fresh $P$ is the honest algorithm, but malicious Alice may choose any pad without invalidating this one-direction proof. Section 11 separately uses the evaluator simulator and honest uniform pads to establish Bob-view key privacy. Fairness remains outside both privacy-with-abort results. □

The component's conservative cut-and-choose term is $2^{1-c/4}$, for the number $c$ of garbled circuits. Write all its other proved negligible terms as $\nu_{\mathrm{LP},i}(\lambda, c)$, including aggregation soundness ($2^{-L}$ plus its reduction term), OT/DDH, knowledge/zero-knowledge, encryption and ex-

traction terms. For polynomial circuit sizes and $c$ they are negligible under the stated assumptions. The simulation/extraction convention is that of the cited standalone theorem, including its expected-polynomial-time witness-extended emulation; no stronger strict-time simulator is claimed. These terms are not empirical estimates or a native implementation residual.

**Corollary 1** (Negligible theoretical framework error)**.** *Let $\bar{Q} = \max(1, Q_R)$, take $\sigma_{\mathrm{bind}} \geq \lambda$ in the growing family, and choose*

$$c = 4\big(\lambda + \lceil \log_2 \bar{Q} \rceil + 1\big).$$

*The theoretical construction above satisfies*

$$B_R \leq 2^{-\lambda} + \sum_{i=1}^{Q_R} \nu_{\mathrm{LP},i}(\lambda, c) + \epsilon_H + \epsilon_p + 2^{-\sigma_{\mathrm{bind}}} = \mathrm{negl}(\lambda). \tag{5}$$

*Proof.* Substitute the component bound and Lemma 8 into Theorem 1. Since $Q_R \leq \bar{Q}$,

$$Q_R 2^{1-c/4} \leq \bar{Q}\, 2^{-\lambda - \lceil \log_2 \bar{Q} \rceil} \leq 2^{-\lambda}.$$

Inactive slots contribute zero. The clocked universal wrapper and finite uniform stage families give one common negligible bound $\nu_{\mathrm{LP},i} \leq \nu_{\mathrm{LP}}$, so the sum is at most $Q_R \nu_{\mathrm{LP}}$, not an unqualified sum of unrelated negligible functions. Lemma 7 bounds the expected-time collision finders under ordinary collision-resistance assumptions. The separate setup-event probability is bounded information-theoretically. Thus every term is negligible. This establishes the Alice-direction theoretical realization under standard assumptions; the separate Bob-direction result additionally uses the latent-anchor and stopped-prefix proof in Sec. 11. Neither result relies on an empirical estimate of any native EMP error term. □

For illustration, $\lambda = 128$, $\kappa = 1$, $\tau = 12$ and T/U/L give $Q_R = 14$ and $c = 532$; the *summed cut-and-choose statistical term* is at most $2^{-128}$. This is not an unconditional 128-bit total-security claim: C38 additionally has a global binding-setup term at most $2^{-128}$ at $\sigma_{\mathrm{bind}} = 128$, and all other cryptographic errors and hash terms remain in Eq. (5). The EMP timings are not timings of this 532-circuit theoretical realization.

## 7 Optional numerical random-oracle bound

**Corollary 2** (Explicit hash-model probability)**.** *If the two framework hashes are modeled as independent random oracles with output lengths $\ell_c, \ell_p$, choose deterministic hard query caps for each whole-execution collision-finder experiment, and let $Q_c, Q_p$ be the sums of these caps for the respective oracle. Let $\delta_{\mathrm{cap}}$ be the sum of probabilities that any such experiment exceeds its cap in the uncapped construction. Then*

$$B_R \leq \sum_i \epsilon_{\Pi,i} + 2^{-\sigma_{\mathrm{bind}}} + \frac{\binom{Q_c}{2}}{2^{\ell_c}} + \frac{\binom{Q_p}{2}}{2^{\ell_p}} + \delta_{\mathrm{cap}}.$$

*The hashes in this optional idealization are independent of component primitives. This is an Alice-direction bound, so it needs no commitment PRG hiding term. The separate Bob-view analysis includes the computational hiding loss $\Gamma_H(t_H)$ derived in Sec. 11.*

*Proof.* Under lazy sampling, the $j$th new input collides with an earlier output with probability at most $(j-1)/2^\ell$, even when inputs are adaptive. Summing up to a deterministic cap gives $\binom{Q}{2}/2^\ell$ in a capped experiment. Every wrong accepted input/descriptor supplies a hash collision outside the global bad-setup event by the previous reductions. Keep that setup term separate and sum hash bounds for all finder experiments; $\sum_j \binom{Q_j}{2} \le \binom{\sum_j Q_j}{2}$ yields the displayed conservative expression. Coupling each capped experiment to its uncapped version adds $\delta_{\mathrm{cap}}$. Both queries and abort choices may depend adaptively on all observations in the preceding protocol transcript. □

Setting $\ell_c = \ell_p = 512$ produces a numerical *model* bound for the chosen hard query budgets. Expected polynomial running time alone does not make these deterministic caps or set $\delta_{\mathrm{cap}} = 0$. A numerical zero-tail bound therefore refers only to an explicitly capped idealization; an uncapped realization retains the tail term. No assertion that actual SHA3/BLAKE2b are proven random oracles, or that EMP achieves this probability, follows. The standard model result in Corollary 1 does not require this idealization.

# 8 Machine checks and evidence boundary

The retained v24 `Theory.ec` checks 19 algebraic/interface and finite probability-accounting lemmas with arbitrary positive intermediate word length and arbitrary deterministic completion. They cover actual reinput binding, completion, same-input use, an explicit wrong-output collision witness, honest correctness, joint masks, correlated-event union bounds and call-budget arithmetic. Cold Z3/CVC5 quorum-two checks pass; five semantic mutations fail and one equivalent rewrite passes. The existing whole-lifecycle, transfer/owner and late-cut entries also pass cold checks.

These archived results concern the earlier digest-anchor interfaces. The C38 commitment binding/hiding and full Bob-view reductions are written proofs, not added to the 19 checked-lemma count. The archived finite-group checks validate the earlier Pedersen identities only and do not verify the current Toeplitz or PRG claims. These results mechanize useful steps of the earlier proof. They do not mechanize the full cryptographic sequential-composition reduction, prove DDH/PRF/hash assumptions, or certify any native implementation. The complete cryptographic argument is Theorem 1, its simulator and Corollary 1, using the cited published component theorem. Detailed evidence and complete reproduction records for the archived digest-anchor interface checks are retained in `data/theory-v24-proof.json`.

The v39 finite stopped-prefix diagnostic enumerates 47,989 toy event schedules with up to six potential new-anchor slots. It checks the fixed slot cap, latent-value reuse and the difference between a functional-zero output and interruption before decoding. Its mutations reject charging only delivered or activated anchors and resampling a latent value on rewind. The deliberately biased one-bit toy generator exhibits why undelivered output-dependent messages still require a hiding charge. The reproducible diagnostic `audit_anchor_prefix_v39.py` checks this finite accounting model. Cryptographic security follows from the written component, stopping and PRG reductions with explicit component and hardness assumptions.

The archived v37 finite-group controls concern the Pedersen-anchor mathematical interface, including its perfect-hiding property. They are not controls for the new Toeplitz/SHAKE construction or its computational PRG assumption. C38's Toeplitz rank and binding arguments and complete Bob-view PRG hybrids are the written proofs in Secs. 9 and 11.

Current native C38 clear-function checks, complete protocol probes and cost measurements are reported in the distinct experiment module and its raw evidence. They concern the exact public setup, PRG mapping, input orientation and guard definitions used there. These tests sup-

plement functional and cost evidence; they do not establish an LP refinement or supply a native complete-view simulator. Historical H measurements retain their own evidence boundary.

The archived EMP H D/T/L measurements use deterministic digest anchors. They provide reference placement costs and exclude the new commitment circuitry and opening interfaces. They establish neither the theoretical realization's elapsed times nor EMP's simulation error. Its source-level conditional ledger remains archival implementation analysis in `supplement/backend-notes-v20.tex`. A future native security claim requires its own qualification and refinement. These implementation obligations are separate from the mathematical protocol and do not appear among the premises of the theorem proved here.

# 9 A circuit-oriented Naor-style string anchor

The concrete C anchor uses a PRG-based string commitment, following Naor's commitment method and its linear-map extension [5, 6]. This is an instantiation of an established commitment technique, not a new commitment primitive. Its binding is statistical over protected Bob's public setup; its hiding is computational. These properties replace, respectively, the discrete-log binding term and perfect-hiding argument of the archived v37 group anchor. The same stage constructs the anchor on the produced share, and a later stage checks and consumes the same actual restored share.

**Parameters, setup and bit encoding.** Let $k$ be the opening-seed length, $n$ the framework digest length, and $\sigma_{\mathrm{bind}}$ the statistical binding parameter. Put

$$m = n + 2k + \sigma_{\mathrm{bind}}, \qquad N_T = m + n - 1.$$

Fix a deterministic polynomial-time generator $G : \{0,1\}^k \longrightarrow \{0,1\}^m$ independently of the setup below. Its pseudorandomness is an explicit assumption: for a fresh uniform seed, $G(s)$ is computationally indistinguishable from a fresh uniform $m$-bit string against the declared nonuniform strict-PPT adversaries at the stated computational resource bound.

Before generation, protected/passive Bob samples $N_T$ independent uniform bits $t_0, \ldots, t_{N_T-1}$. They define the public $m$-by-$n$ Toeplitz matrix

$$T_{j,i} = t_{j+n-1-i}, \qquad 0 \le j < m, \quad 0 \le i < n. \tag{T38}$$

The setup index, complete setup string, parameters and PRG identifier are immutable authenticated public constants of the owner and every relevant descriptor. The setup is sampled once for the service budget; restoration and reruns load that same string rather than replace it. Bob's complete setup tape is included in the passive-view experiment. There is no hidden setup trapdoor. The initial advice is independent of this fresh setup, as for the freshly sampled framework hash indices.

For the concrete profile, $k = 128$, $n = 512$ and $\sigma_{\mathrm{bind}} = 128$, so $m = 896$ and $N_T = 1407$. Pack consecutive bits least-significant-bit first within each byte. The setup occupies 176 bytes; bit 7 of its last byte is zero and is not a sampled bit. A noncanonical encoding is refused before use. The digest uses the same bit convention when forming the input vector to $T$. All $2^{128}$ opening encodings are valid 16-byte seeds; no scalar sampling or rejection loop is needed.

The concrete deterministic generator is

$$G(s) = \mathsf{SHAKE256}(\mathsf{tag}_G \parallel s, 112), \tag{G38}$$

where $\mathsf{tag}_G$ is exactly the 16 ASCII bytes `ASA2PC-C-v38-PRG` and $s$ is exactly 16 bytes. The input is thus 32 bytes. SHAKE's prescribed padding and 112-byte output fit within one 136-byte rate block and require one Keccak-f[1600] permutation. This inventory is not a proof that

this concrete SHAKE mapping is a PRG. We state its PRG assumption separately from collision resistance of $H_c, H_p$, without identifying SHAKE with a random oracle. An asymptotic claim uses a growing secure PRG family and growing $k, \sigma_{\mathrm{bind}}$, not fixed SHAKE parameters alone.

**Commitment and check.** Use the original injective, domain-separated framework encoding and set

$$d(A) = H_c\big(\mathsf{enc}(\mathsf{domain}, \mathsf{ctx}, p, A)\big) \in \{0,1\}^n, \qquad \mathsf{Com}_T(\mathsf{domain}, \mathsf{ctx}, p, A; s) = G(s) \oplus Td(A). \tag{C38}$$

Here the transfer descriptor includes its original profile and public $\rho$; the final descriptor fixes the original public key as in the framework. Honest Alice chooses a fresh independent uniform seed for each new anchor. She retains $(A, s)$; Bob retains $(B, C)$ and the immutable record. Replaced Alice may choose arbitrary seed inputs. A check computes Eq. (C38) on its actual $(x, s')$, compares all $m$ anchor bits, and consumes that same $x$ only on success. Descriptor agreement, activation, burning and recovery retain the original owner rules.

**Lemma 9** (Uniform image of a nonzero difference)**.** *For every fixed $\delta \in \{0,1\}^n \setminus \{0^n\}$ and every $v \in \{0,1\}^m$, a uniform setup satisfies $\Pr_t[T\delta = v] = 2^{-m}$.*

*Proof.* Let $i_*$ be the largest index with $\delta_{i_*} = 1$, and put $a = n - 1 - i_*$. Condition on all setup bits outside $t_a, \ldots, t_{a+m-1}$. Row $j$ of $T\delta$ is $t_{a+j}$ plus terms $t_{a+j+i_*-i}$ with $i < i_*$. Each such term has a strictly larger setup index. Thus the $m$ equations, solved in descending row order, form a triangular system with diagonal one in the $m$ remaining uniform bits. Every $v$ has exactly one solution, proving the claim. This argument proves the exact rank condition using the independent setup bits; independence of different matrix entries is not required. □

**Lemma 10** (One global statistical binding event)**.** *With probability at least $1 - 2^{-\sigma_{\mathrm{bind}}}$ over the single public setup, no two distinct digest vectors and any two opening seeds yield the same anchor. This statement holds simultaneously for every message, seed, handle and subsequent adaptive request, even for unbounded choices after setup. Consequently, two accepted distinct shares under a fixed original descriptor yield an $H_c$ collision outside that event.*

*Proof.* Call a setup bad if some $s, s'$ and nonzero $\delta$ satisfy $T\delta = G(s) \oplus G(s')$. For a fixed triple its probability is $2^{-m}$ by Lemma 9. There are at most $2^{2k}$ seed pairs and $2^n - 1$ nonzero differences. Therefore

$$\Pr[\mathsf{Bad}_T] \le 2^{2k}(2^n - 1)2^{-m} < 2^{-\sigma_{\mathrm{bind}}}.$$

The union already covers every possible adaptive choice, so reuse of this setup adds no handle or request multiplier. If two distinct shares have distinct digests, their two accepted openings contradict the complement of this event. If their digests agree, their injective share encodings give an $H_c$ collision. The proof places no distributional or pseudorandomness condition on malicious opening seeds. It requires only that $G$ is fixed independently of the fresh Toeplitz setup. □

**Lemma 11** (Computational hiding at a fresh anchor)**.** *For every fixed setup and message, replacing $G(s)$ for a fresh independent uniform seed by an independent uniform $m$-bit string makes the anchor uniform and independent of the message. A strict-time test's change in acceptance is bounded by the corresponding PRG distinguishing advantage. The same holds with auxiliary information independent of that fresh seed, including Bob's complete setup tape and complementary share.*

*Proof.* The reduction XORs its PRG-or-uniform challenge with $Td(A)$. The uniform branch is exactly uniform for every fixed $T, d(A)$. The reduction may fix arbitrary auxiliary information before the fresh seed. Adaptive fresh anchors are handled by chronological latent-slot hybrids. Their charge precedes every output-dependent simulated message, including a prefix of a call interrupted before decoding or owner activation. Repeated checking of one original anchor creates no fresh PRG sample. Complete component views and expected-time postprocessing require the additional accounting in Sec. 11; its stopped-prefix lemma lifts this marginal argument to the complete view of every completed or interrupted invocation under the fixed slot budget. □

**Opening equivalence and the exact-origin service.** A PRG need not be injective. For the *same* share, accepted seeds need not be identical: acceptance gives precisely $G(s') = G(s)$. The C exact-origin service therefore saves the original share $A$ and seed image $G(s)$, and accepts a later opening precisely when the actual share is $A$, the original descriptor agrees, and $G(s') = G(s)$. It does not require equality of the incidental seed encodings. This change preserves the original-share and original-key relation. The Alice-facing simulator knows its sampled share and extracted original seed, and can test this condition without the underlying secret key. The first accepted changed share supplies the hash collision above, outside the global setup event. No claim of opening uniqueness is used.

Consequently the Alice-direction framework bound becomes

$$B_R^{\mathrm{C38}} = \sum_{i=1}^{Q_R} \epsilon_{\Pi,A,i} + \epsilon_H + \epsilon_p + 2^{-\sigma_{\mathrm{bind}}}. \tag{A38}$$

The hash terms remain whole-execution search-reduction probabilities; the existing expected-time truncation argument applies to those finders. No PRG hiding assumption is needed against replaced Alice, because her simulator extracts her seed and computes each anchor exactly. Hiding is used in the separate honest-Alice/passive-Bob experiment.

**Circuit, storage and evidence boundary.** The setup is public and canonical. Its Toeplitz coefficients are constants of the ordinary circuit, so $Td$ is a linear XOR network, not a private matrix multiplication with $mn$ AND gates. The descriptor and uniform specialization invariant pin those constants before private inputs. Construction or checking additionally evaluates $G$ and XORs its $m$ output bits with $Td$. Checking compares all $m$ bits; relative to the old $n$-bit digest check this adds $m - n$ AND gates in an ordinary binary equality tree, besides the PRG circuitry. The actual inventory includes the implementation's XOR gates, constants, padding and output interfaces.

At the concrete parameters a private opening takes 16 bytes, a public anchor 112 bytes, and the shared setup 176 bytes. Alice's expanded and retained-seed records have 1200 and 80 bytes before common metadata, a 1120-byte difference and 93.33% reduction. Bob's complementary share payloads remain 1184 and 64 bytes. Each constructed or checked anchor adds 128 Alice opening-input bits. A T/L generation constructs one transfer anchor, then checks it and constructs the final anchor; a checked T or SR use checks one original anchor. Reused anchors keep their original seeds; distinct new anchors use fresh seeds.

Concrete C functionality and cost measurements must identify this exact mapping, setup and circuit inventory. They do not retrospectively convert the retained H measurements into C measurements, and measurements alone do not establish the native backend's component simulation guarantee, which requires a separate protocol-level security reduction.

# 10 Joint generation and private-key recovery under substitution

## 10.1 Purpose, roles and the protected boundary

In unilateral ML-KEM generation, replaced KeyGen code controls its randomness and public outputs. A replacement equipped with the attacker's trapdoor may encode information from which an external extractor recovers the generation seed or a useful decapsulation key. Classical ASA work formalizes this covert replacement threat [7, 8]. Honest ML-KEM supplies the underlying KEM security assumption. Our defense brings Bob into generation. The parties compute $w = w_A \oplus w_B$ inside malicious-secure 2PC, with fresh secret uniform $w_B$, and return only $A = X \oplus B$ to Alice while protected Bob retains uniform $B$. The mathematical generation function returns only the prescribed shares. Bob's complete protocol view is a separate object, treated in Sec. 11. Splitting a key after compromised unilateral generation leaves the attacker with the complete key already exposed.

Alice is the replaceable worker. Bob is protected in the Alice-direction experiment. Honest Alice and passive Bob define a separate experiment in Sec. 11. In both, the encapsulator may know his own session key while the decapsulation key is the private recovery target. The service returns separate Decaps shares $U, R$. Trusted delivery, KDF/AEAD and hidden-result-dependent feedback require separate analysis. Alice's voluntary disclosure is excluded from the honest-Alice Bob experiment, as in ordinary KEM secrecy. It remains allowed in the malicious-Alice experiment without giving the external extractor Bob's protected outputs.

## 10.2 Recovery experiments and honest-key assumptions

Fix polynomial reservation budgets $\kappa \geq 1, \tau$, a public profile and the framework assumptions, including the immutable public Toeplitz setup. The adversary is the pair of a strict-PPT substituted Alice worker $\mathcal{A}$ and a strict-PPT decoder $E$. Their initial advice may include an attack trapdoor. It is independent of fresh honest key-generation coins, hash indices, the Toeplitz setup and challenge coins as required in Sec. 3. The decoder $E$ receives Alice's full view, covering an external extractor that sees only the public key and attacker-selected exfiltration strings. The generation and use counts include aborted and failed reservations. A transport interruption of a mathematically successful call preserves its already observed prefix and burns its reservation; it is not a functional failure with replacement zero payloads. The experiment measures success over all attempts, including selection and activation of favorable keys.

**Definition 2** (Offline private-key recovery experiment)**.** *Run the real service with substituted Alice, retaining its full view $V_A$ and the declared public record $L$. At the end of all service interaction, run $E(V_A, L)$ to obtain a handle $h$ and candidate private core $\widehat{X}$. An inactive handle, invalid candidate encoding, public key failing Encaps input checks or absent candidate loses. The public bounded matrix-sampling predicate must also pass. On that domain fixed-bound Encaps equals standard Encaps. Other keys lose, keeping the checker strict polynomial time. Otherwise assemble a standard candidate decapsulation key $\widehat{dk}$ from $\widehat{X}$ and the authenticated public $ek_h$, including $H(ek_h)$. After fixing the candidate, sample independently*

$$(c^*, K^*) \leftarrow \mathsf{Encaps}(ek_h), \qquad b = [\mathsf{Decaps}(\widehat{dk}, c^*) = K^*].$$

*The experiment returns $b$ and $p_{\mathrm{rec}}^{\mathrm{real}}$ is its probability of returning one. The checker keeps the fresh ciphertext/key pair private from the worker and decoder. The ideal experiment substitutes the exact-origin service and its fixed simulator, using the same final checker and budgets. Its success probability is denoted $p_{\mathrm{rec}}^{\mathrm{ideal}}$ throughout this section.*

This test measures possession of a usable offline private key, including an equivalent key that decapsulates honest traffic. Alice's own share is already available to the attacker and is insufficient for this test. Verification uses the public handle, public key, candidate and fresh checker coins. It is therefore a strict-PPT observer of the existing real/ideal output ensemble. Equality with the hidden original core is a separate audit relation and would require an additional argument.

A decoder recovering $d \parallel z$ can run deterministic KeyGen and assemble the candidate. A decoder recovering $\sigma \parallel z$ can apply $F_T$ with the public $\rho$. A decoder recovering $dk_{\mathrm{PKE}}$ alone can append any canonical $z$ and still attack honest encapsulations. The experiment also covers this useful recovery target. Correctly recovered original keys pass except for honest KEM correctness failures. Consequently the recovery target includes the private-key leakage that would enable passive decryption of future traffic, even if the original rejection seed itself remains unavailable.

For a growing KEM family let $\ell(\lambda)$ be its shared-key bit length. Use the IND-CPA distinguishing convention

$$\mathrm{Adv}_{\mathrm{KEM}}(D) = \left|\Pr[D(ek, c, K) = 1] - \Pr[D(ek, c, Z) = 1]\right|,$$

where $(ek, dk)$ is honestly generated, $(c, K) \leftarrow \mathsf{Encaps}(ek)$, and $Z$ is independent uniform in $\{0,1\}^{\ell}$. The adversary can process $ek$ before requesting this one challenge. The proof assumes this honest-key security for the exact underlying KEM and its public parameters. ML-KEM's existing security analyses supply the underlying hardness and hash-model assumptions [9]. Our offline recovery reduction uses this existing KEM result through IND-CPA security.

For the ML-KEM profiles the early public $\rho$ is part of $ek$, and bounded matrix refusal is a deterministic polynomial-time predicate of $\rho$. Successful bounded generation agrees with ordinary KeyGen under the same seed. These public-prefix properties are required when applying the argument to any growing KEM family with a split generation interface. Selection and activation are included in the experiment, so its analysis uses the unconditioned candidate distribution. Fixed ML-KEM-768 supplies a concrete KEM assumption and $\ell = 256$. The asymptotic result separately assumes a growing secure KEM family with the same public-prefix/refusal properties and polynomial-size circuits satisfying the uniform stage contracts. Increasing LP or hash parameters alone leaves the fixed KEM assumption unchanged.

### 10.3 Public-key-only ideal emulation

**Lemma 12** (Alice-facing ideal view from public keys)**.** *Given an honest public key for each generation reservation, their public prefixes and the service's public setup, one can emulate every prescribed Alice-facing ideal response without its private key, seed or Bob's share. The emulation is exact for the marginal $(V_A, L)$ under the delivery boundary above, including restoration, SR, public sampling refusals and the worker's selective abort choices.*

*Proof.* For reservation $j$ take an independent honest candidate $ek_j$. In the original ideal service the joint seed is fresh uniform regardless of the effective Alice contribution. The honest public-key distribution is therefore the correct unconditioned candidate distribution. Release $\rho_j$ and later $ek_j$ only at the original profile's prescribed times. Evaluate the public bounded sampling predicate from $\rho_j$ and preserve refusal. D and split profiles retain their original failure and public-output schedules. The emulator releases the complete public key only at the prescribed stage. In D, a bounded-sampling refusal exposes only its failure flag and zero payloads; the candidate prefix is not released. In T/U/L an already released $\rho$ remains visible when the later public matrix check refuses.

Every effective opening is a fixed-length PRG seed and is valid. For T/U/L inputs sample independent uniform $A_{S,j}$ and compute $D_{S,j}$ from that share, effective opening seed $s_{S,j}$ and public context. For final generation sample independent uniform $A_j$ and compute $C_j$ on

the same share, effective opening input and public descriptor. Original mathematical masks $B_{S,j} = S_j \oplus A_{S,j}$ and $B_j = X_j \oplus A_j$ establish this distribution, but the emulator never needs to evaluate them. In SR retain the original seed-share record and public descriptor instead of the expanded record. Sampling shares and then computing commitments with their extracted openings reproduces their joint distribution, including the dependence introduced by the anchor.

At restoration, transfer and use accept precisely the saved original share and descriptor and an opening seed with the original image $G(s)$. These values are available to the Alice emulator. A PRG need not be injective: distinct seed encodings with equal image are permitted and preserve the original consumed share. No opening-uniqueness assertion is used. On a successful use sample independent uniform $U$ and preserve public Auth. The actual hidden $R = \mathsf{TotalDecaps}(X, ek, c) \oplus U$ stays outside Alice's view and the permitted owner's future decisions. Ciphertext validity remains internal. The emulator thus handles invalid ciphertexts through the same uniform-output rule. The emulator runs the same public reservation, activation, burn and recovery rules. Alice's next action and abort choice have the same distribution given the resulting view. Induction over the fixed reservation budget proves the marginal equivalence, including key-dependent selective abort, refusal at the public matrix predicate, and the same reservation and activation history. □

The lemma establishes the marginal view needed for recovery. Bob's hidden application outputs retain the joint simulation supplied by the framework theorem. This marginal emulation depends on the stated exclusion of private-result-dependent application feedback.

### 10.4 Reduction with retries and expected-time simulation

**Theorem 2** (Offline key recovery bound)**.** *Under Theorem 1, for every fixed strict-PPT $(\mathcal{A}, E)$ there is an expected-time KEM reduction with polynomial uniform expected prefix-time bound $P(\lambda)$. For each polynomial cutoff $t$, its truncated strict-PPT version $D_t$ satisfies*

$$p_{\mathrm{rec}}^{\mathrm{real}} \le B_R + \kappa \left( \mathrm{Adv}_{\mathrm{KEM}}(D_t) + 2^{-\ell} + \frac{P(\lambda)}{t(\lambda)} \right). \tag{6}$$

*The time of $D_t$ is $t$ plus polynomial candidate/challenge processing. The single Alice-direction simulation gap $B_R = \sum_i \epsilon_{\Pi,A,i} + \epsilon_H + \epsilon_p + 2^{-\sigma_{\mathrm{bind}}}$ covers the complete execution. The setup term is global. This direction needs no PRG hiding term: its simulator extracts the replaced Alice's opening seeds and computes every anchor exactly. If $\kappa = 0$ the experiment has no active handle and wins with probability zero because activation never occurs.*

*Proof. Transfer the efficient recovery test.* For fixed $\mathcal{A}$, take the one framework simulator independent of the observer. The decoder and fresh-encapsulation verification form a strict-PPT observer, so $p_{\mathrm{rec}}^{\mathrm{real}} \le p_{\mathrm{rec}}^{\mathrm{ideal}} + B_R$. The observer uses the declared record, decoder's candidate and fresh encapsulation. The same statement applies to SR using its retained-seed ideal service and $2\kappa + \tau$ call budget. All choices of a target handle occur before the checker samples its fresh encapsulation.

*Embed before any selection.* The KEM reduction receives honest challenge public key $ek^*$. Independently choose $J$ uniformly from the $\kappa$ reserved generation slots, including slots that may ultimately stay unused. Place $ek^*$ in slot $J$ and generate independent ordinary honest keys for all other slots as they are needed. Use Lemma 12 to answer the framework simulator's ideal requests. The simulator may rewind its current corrupted call but commits one ideal-stage request. The forward lemma gives a polynomial bound on its global expected prefix cost. All responses that it actually receives are the same Alice-facing values as from the original ideal service. The public-key emulator supplies all required responses.

Once the decoder returns $(h, \widehat{X})$, output zero unless $h$ is active and belongs to slot $J$ with a valid candidate encoding. Otherwise request the KEM challenge $(c^*, T)$, assemble $\widehat{dk}$ and output $[\mathsf{Decaps}(\widehat{dk}, c^*) = T]$. The challenge is requested after the candidate is fixed. Since every slot has the same independent honest public-key distribution, the hidden index $J$ is independent of the entire unmarked ideal execution and final selection. In the real-key challenge branch the untruncated reduction therefore accepts with probability exactly $p_{\mathrm{rec}}^{\mathrm{ideal}}/\kappa$. In the random-key branch $T$ is independent uniform after fixing the candidate. Even conditioned on $c^*$, a deterministic candidate decapsulation equals $T$ with probability $2^{-\ell}$. Inactive targets and unused slots can only reduce this probability.

*Apply a strict-time assumption.* The KEM assumption quantifies over strict-PPT adversaries, whereas the framework simulator may run in expected polynomial time. Bound the expected pre-challenge cost of this complete reduction by a uniform polynomial $P$. This includes the public emulator, ordinary nonchallenge key generations, owner and decoder. Stop that prefix after $t$ steps and output zero on cutoff. Markov's inequality bounds loss in its real-key acceptance by $P/t$. The random-key branch still accepts with probability at most $2^{-\ell}$, since stopping and candidate generation precede the challenge. Therefore

$$\frac{p_{\mathrm{rec}}^{\mathrm{ideal}}}{\kappa} \leq \mathrm{Adv}_{\mathrm{KEM}}(D_t) + 2^{-\ell} + P/t.$$

Combine with the single real/ideal gap to obtain Eq. (6). The unconditioned experiment already includes public sampling refusal and attacker selection of favorable active keys. Its bound therefore applies directly over all reservations. The factor $\kappa$ accounts for all attempts, including candidates discarded after their public prefix was observed by the substituted worker. □

**Corollary 3** (Asymptotic and fixed-parameter interpretations)**.** *For a growing IND-CPA-secure KEM with the public-prefix properties above, polynomial budgets, negligible $B_R$ and negligible $\kappa 2^{-\ell}$, every strict-PPT substituted Alice and decoder has negligible offline recovery probability. For ML-KEM-768 Eq.* (6) *is a concrete bound with $\ell = 256$, depending on the chosen honest-KEM and protocol assumptions at the stated resource budget.*

*Proof.* If real recovery were nonnegligible, then outside negligible $B_R$ and $\kappa 2^{-\ell}$ its ideal success divided by polynomial $\kappa$ would be nonnegligible on an infinite parameter set. Choose a sufficiently large polynomial cutoff so $P/t$ is smaller than half that nonnegligible advantage. The resulting strict-PPT $D_t$ contradicts IND-CPA security. The cutoff is chosen in this contradiction argument, so its error is smaller than the hypothesized nonnegligible advantage. This establishes recovery security with an expected-time simulator under the ordinary strict-polynomial-time KEM assumption. The term $P/t$ remains explicit for a fixed polynomial cutoff. □

### 10.5 Recovery guarantee and scope

The new result establishes the recovery baseline that pure real/ideal simulation leaves unspecified. Joint generation supplies a fresh honest-key candidate distribution before selective abort. Secret sharing and malicious 2PC prevent the replaced worker from obtaining an additional secret-dependent view. Original anchors then ensure that transfer, restoration and use remain tied to that generated key. These are three distinct parts of the defense. The 64-byte cut and retained-seed choice determine implementation costs rather than creating a new source of key-generation entropy.

Private-core equality is a different audit relation from the efficient test. For honestly generated candidates, an exactly recovered original core (or its original seed) gives the same Decaps algorithm as the genuine key. If $\delta_{\mathrm{KEM}}$ is the average honest correctness failure, a union

bound over at most $\kappa$ independent candidate keys bounds failures for an adaptively selected candidate by $\kappa\delta_{\mathrm{KEM}}$. Thus exact recovery in that honest-candidate experiment is bounded by the functional recovery test plus this correctness term. This pointwise correctness implication connects the separate equality audit to the efficient recovery test. The real/ideal distinguisher still uses only its prescribed observable ensemble. The main security claim uses the directly observable functional recovery experiment and already covers useful equivalent keys.

The attacker may obtain Alice's contribution, share, transcript, public key and full bounded state. Bob's random contribution and complementary shares remain protected in this experiment. The allowed interface also keeps full Decaps results and additional secret-dependent external outputs outside the attacker's view. An encapsulator can generate its own known $(c, K)$ from public $ek$ using ordinary public-key computation. Malicious Bob, concurrent composition, rollback, processor side channels, a compromised trusted application and quantum attacks on classical LP are outside the model. EMP measurements establish the reported finite observations. The recovery bound follows from the written reduction and its cryptographic assumptions.

# 11 Private-key privacy from Bob's complete C view

## 11.1 A separate complete-view experiment

The two recovery goals concern separate executions. The Alice-direction theorem allows replaced Alice and protects Bob. Here Alice follows the prescribed algorithm with fresh independent seed contributions, commitment seeds and output pads. Bob follows his prescribed algorithm and retains his entire view. This includes every application and component random tape, all inputs, sent and received messages, intermediate configurations, public records, complementary shares, masks, and local Encaps coins and shared keys. His setup tape includes every sampled Toeplitz bit. There is no erasure assumption and no secret setup logarithm in C.

Calls are classical, static, sequential and fresh at the component boundary. The original non-rollback owner, reservation quotas, authenticated confidential channels, truthful termination and public availability rules remain premises. Later stages use prescribed application records rather than hidden component tapes; Bob may nevertheless retain and inspect those tapes. Permitted scheduling can depend on his view and public history. Recovery restores Alice's original share and its seed or an equivalent opening image. Arbitrary corruption of Alice's restored record, processor side channels, a full Decaps oracle and feedback from a reconstructed private application result are outside this experiment. Alice's voluntary disclosure is excluded by the honest-Alice role. Trusted delivery, combination of result shares, KDF and AEAD remain separate interfaces.

**Definition 3** (C complete-Bob-view offline recovery)**.** *Run at most $\kappa$ generation and $\tau$ use reservations with prescribed Alice and passive Bob. Give $(V_B^{\mathrm{all}}, L)$ to a strict-PPT decoder $E_B$ with initial advice independent of fresh honest seeds, keys and challenge coins. It fixes a handle and candidate core $\widehat{X}$. Apply the encoding, activation and public bounded-matrix checks of Definition 2. An invalid or inactive target loses. Only after the candidate is fixed, sample an independent fresh $(c^*, K^*) \leftarrow \mathsf{Encaps}(ek_h)$, assemble $\widehat{dk}$ using the original public $ek_h$, and test $[\mathsf{Decaps}(\widehat{dk}, c^*) = K^*]$. Denote its probability by $p^{\mathrm{real}}_{\mathrm{rec,B}}$.*

The checker challenge is independent of Bob's earlier local encapsulations. Thus his knowledge of his own session keys is permitted without supplying a candidate private key for future traffic. The experiment gives neither Alice's output share nor a combined Alice/Bob view to the decoder.

### 11.2 Qualified component views and observable output delivery

We require the component theorem for the actual two-output stage interface, including its prescribed abort and delivery semantics. The Bob-facing simulator receives Bob's input/entry state and his designated application output and public metadata, and returns his complete continuation view, including component coins and intermediate configurations. It is uniform over the stage circuits, bounded entry states and arbitrary fixed inputs. The complete-view slot error is $\epsilon_{\Pi,B,i}$; unused slots have zero error. New component coins and simulator tapes are fresh after entry. The simulator never rewinds a completed invocation or an owner transition.

Alice is the garbler and Bob the evaluator, preserving the output adapter of Eq. (4). Honest Alice supplies a fresh independent uniform pad $P$ for every call's fixed-length private output. Bob obtains $Y = \beta \cdot (o_A \oplus P)$, his intended output and public metadata, where $\beta$ is the public stage-success flag from the adapter; he relays only the designated masked Alice component. On success the relay $Y$ is uniform conditional on all mathematical stage variables and anchors. On failure it is the fixed zero string and Alice does not unpad or deliver it. Fresh use masks and output pads are distinct random variables. The complete component theorem, rather than the relay marginal alone, covers all pad-dependent messages and Bob's coins.

Theorem 4.2 of corrected LP treats the corrupted evaluator separately [3]. Taking that party to follow prescribed Bob and to output its entire configuration gives the complete evaluator view. Its input/output simulator, uniform circuit interface and call-local subprotocol simulation supply the forward qualification described for the framework. No native EMP simulation theorem is inferred from the mathematical LP theorem or from a functional test.

For the computational-hiding hybrids below, the output-simulation algorithm must additionally be total with a uniform expected-polynomial-time bound on every syntactically valid designated output string of the public size bound, including outputs outside the stage function's image. A synthetic anchor need not have an opening. We use these calls only as transcript generators, not as a claim that an impossible mathematical execution is secure. A component instantiation must justify this total extension from its simulator algorithm, or cap it and retain the corresponding tail. For the corrected LP evaluator simulator (Theorem 4.2, pp. 22–23), its Step 5 invokes Lemma A.1 to construct a fake circuit with a specified output $z$. This construction accepts every string of the declared output length, without a preimage of $z$ under the stage function. The later simulated input-consistency proof uses the simulator's chosen zero input, not a witness for that stage output. The local OT and knowledge/ZK simulation procedures likewise do not search for an input producing $z$. The fixed-length fake circuit and the same call-local subprotocol runtime analysis therefore give the required total extension. This is a property of that simulator algorithm, stronger than merely citing standalone security on valid ideal outputs. A different native component must qualify its own simulator or retain an explicit capped tail.

### 11.3 Interrupted prefixes and latent anchor slots

A mathematical stage failure and an interrupted invocation have different meanings. The former is the function's $\beta = 0$ output, whose gated payloads are fixed zero. A transport or scheduling interruption can instead stop a $\beta = 1$ invocation before its output is decoded, relayed, delivered or recorded as active. Its earlier messages may already depend on the candidate successful output. Burning its owner reservation does not remove those messages from Bob's retained view.

For the passive-Bob experiment, represent an invocation by its canonical continuation with external interruption disabled. It uses the same entry state, private inputs and component tapes as the interrupted invocation. An event log records Bob-visible sent/received messages, his coin

draws and his local intermediate configurations in their protocol order. A permitted strict-time stopping policy scans this log with the preceding public history and its own fixed coins, and selects its first stop event. The projection retains exactly Bob's view at that event and marks the owner invocation interrupted; the unused continuation and its owner outputs are discarded. The policy sees exposed logical protocol events and Bob configurations, not the simulator's wall-clock runtime, internal instructions or rewind attempts. After the first stop, the fixed passive continuation suppresses that external stop only for analysis; it cannot update the actual owner. If a random tape is allocated at entry, the projection retains that allocated tape; otherwise it retains the coins actually generated by the stopping event. For clocked prescribed Bob the log is reconstructed by replay from his inputs, retained tape and messages. This uses the qualified complete view, rather than an additional leakage interface. This is the same tape convention as in the real experiment. A completed invocation is the projection that retains the whole log. Subsequent reservations use the projected owner state, never the discarded continuation state.

**Lemma 13** (Stopped-prefix closure)**.** *Let a complete-call real/simulated view ensemble include the event log above and satisfy its complete-view distinguishing bound for every strict-time observer. Applying the same strict-time stopping projection to both ensembles preserves that bound, including the joint entry state and owner transition. This holds for adaptive stopping determined by the visible prefix, without conditioning on completion. For a total output simulator, the projection is also defined for synthetic output strings outside the stage function's image.*

*Proof.* Until the first stop, disabling the external interruption changes neither prescribed party's next instruction nor the messages or coins already observed. The continuation and the interrupted real execution therefore have identical projected views, path by path. A distinguisher for the projected ensembles, composed with this projection and the clocked policy, is a strict-time distinguisher for the complete ensembles with the same advantage. The projection also computes the prescribed burn/no-activation transition from the stop event. It reveals no discarded output and makes no test of whether a hidden candidate would later be completed. This proves the joint claim. The last statement is algorithmic: a total simulator returns a syntactically valid event log for each designated output string, and the same bounded projection acts on that log. It does not assert that such a synthetic string has a real preimage. A full simulation may take expected time, but its returned event log has the wrapper's polynomial size bound, so this postprocessing is strict polynomial time. □

The complete-call qualification includes the clocked wrapper and its full continuation configurations. It is needed even when Bob normally follows his prescribed algorithm. In corrected LP's evaluator simulation, the ideal output is obtained before the output-dependent fake evaluation circuits are sent. The stopping projection may retain those circuits while discarding final decoding or delivery. Lemma 13 therefore justifies prefix simulation; it does not justify skipping a hiding hybrid merely because an owner record was never activated.

Put $Q_R = d_R\kappa + \tau$, with $d_D = 1$ and $d_T = d_U = d_L = 2$. Reserve $N_R = d_R\kappa$ potential new-anchor slots in generation/stage order; SR uses the conservative $N_R = 2\kappa$ bound. At an invoked slot, fix the mathematical designated output before invoking any output-dependent view simulation. If it contains a fresh successful anchor, allocate that *latent* anchor then and charge its slot, even if a later interruption prevents every final delivery. Freeze its value, the candidate output and its independent pads throughout that call's local simulation, including its rewinds. An owner-completed anchor is saved for later checks; an interrupted one is retained only where it already affected the view and cannot activate a record. A functional $\beta = 0$ output contains no latent anchor: its fixed-zero payload is supplied to the simulator even if the

underlying circuit computes unused candidate wires internally. An uninvoked slot or such a zero-output slot has identical adjacent distributions and zero hiding loss. Repeated uses/restores preserve the original anchor and add no fresh PRG challenge. Consequently the charge count is bounded by $N_R$, rather than by the number of released anchors or activated records.

### 11.4 Computational anchor replacement with expected-time tails

**Lemma 14** (Complete-view anchor hybrid)**.** *After replacing all component calls by their qualified complete-Bob-view simulators, including the stopped-prefix projection of Lemma 13, enumerate the $N_R$ potential latent-anchor slots above. Replace their allocated fresh anchors in this order by independent uniform $m$-bit strings, before any output-dependent simulator step. Keep each value fixed within the call and on all later repeated appearances. Use the same stopping policy and the prescribed honest restoration/authentication flags in every synthetic transcript generator. For anchor slot $j$, let its PRG reduction's expected runtimes in its PRG and uniform branches be bounded by a common polynomial $P_{H,j}$. For every polynomial cutoff $t_H$, the total distinguishing gap is at most*

$$\Gamma_H(t_H) = \sum_{j=1}^{N_R} \left( \mathrm{Adv}_{\mathrm{PRG}}(D^H_{j,t_H}) + \frac{2P_{H,j}}{t_H} \right). \tag{H38}$$

*Conditional on any preceding history in which a slot is uninvoked or functional-zero, its adjacent hybrids coincide. The displayed bound still sums over the full fixed budget, without conditioning on which slots complete. A successful latent anchor is charged even when its invocation is interrupted. The reductions and polynomial bounds are fixed for the specified strict-time observer and wrappers. Under ordinary strict-PPT PRG security the underlying untruncated distinguishing gap is negligible in a growing family, by a sufficiently large polynomial cutoff contradiction. The explicit bound $\Gamma_H(t_H)$ remains cutoff dependent.*

*Proof.* For one adjacent hybrid the reduction samples the legitimate honest mathematical keys and stage variables itself. It can compute the current digest $d$ and all earlier/later application data. It receives one PRG-or-uniform string $Z$ and supplies $Z \oplus Td$ as the current latent anchor, before any output-dependent message or configuration is simulated. In the PRG branch the unknown challenge seed need never be materialized: the component view simulators use Bob's input and designated outputs, and honest future restoration follows the saved original-share policy. Its flags are known without revealing or testing the original private seed. All other latent anchors use independent explicit seeds or the uniform strings specified by the hybrid. Their values and all fresh output pads are sampled once for the call, not resampled on a local rewind or after a stop decision. Fresh pads reproduce each successful masked relay; functional failures reproduce their fixed zero relay. Apply the same strict-time stopping projection to the completed synthetic log. A transport interruption of a successful call retains the prefix for its successful candidate output; it is never replaced by a functional-zero execution. Later owner transitions use only the projected state. The current seed is independent of the digest and preceding mathematical history. Whether the slot is invoked is decided before its new output simulation. A functional-zero or uninvoked slot ignores $Z$ in both branches. In a latent-success slot, replacing $Z$ by uniform makes $Z \oplus Td$ uniform for the fixed preceding state. Thus the two challenge branches induce the two adjacent synthetic complete-view distributions, including any prefix whose stopping event depends on that anchor.

The public setup can be fixed or sampled by this reduction. Knowledge of all its coins changes neither PRG pseudorandomness nor the XOR argument. The reduction may retain earlier component views in its auxiliary state. After replacement, component output simulators are called

with possibly synthetic outputs; this is why their total-extension runtime condition is explicit. Uniform conditional expected-time bounds on every syntactically valid output, the fixed slot budget and the forward tower-property argument bound both whole branches by $P_{H,j}$. Each new local simulation has its bound conditional on its preceding synthetic entry history; the tower property gives the unconditional bound for the whole reduction. This does not bound elapsed prefix time conditional on a rare resulting transcript. The reduction charges the full local simulation even when the projection later selects a short prefix; there is no completion-conditioned runtime claim. Strict stopping/owner processing and the final strict-time observer are included.

Truncate the whole reduction after $t_H$ steps and output zero on cutoff. Markov's inequality bounds the change in each branch's acceptance by $P_{H,j}/t_H$. Its distinguishing gap is therefore at most the strict-PPT PRG advantage plus $2P_{H,j}/t_H$. Sum adjacent hybrids. If their total gap were nonnegligible, one of polynomially many adjacent gaps would be nonnegligible, and a sufficiently large polynomial cutoff would preserve half that gap, contradicting ordinary PRG security. No fixed polynomial cutoff is claimed to make its explicit tail negligible by itself. □

The initial component replacements cost

$$\Delta_{\Pi,B} = \sum_{i=1}^{Q_R} \epsilon_{\Pi,B,i}.$$

They use the forward hybrids with strict-time real suffixes, as in Lemma 6. The PRG hybrids are a separate expected-time reduction with their own cutoffs; one must not silently apply a strict-time PRG assumption to an expected-time transcript generator. The statistical binding event is not needed for prescribed Alice's Bob-view privacy.

**Lemma 15** (Public-key-only synthetic Bob view)**.** *There is an expected-polynomial-time emulator $\mathcal{M}_B$ that samples and retains Bob's actual setup/application tapes and uses an independent honest candidate public key for each generation reservation, but no candidate private key, generation seed or commitment opening. For every fixed strict-time observer and cutoff $t_H$, its complete-view gap is at most*

$$\Delta_B(t_H) = \Delta_{\Pi,B} + \Gamma_H(t_H). \tag{B38}$$

*Proof.* First perform the component replacement and anchor hybrids above. In the last synthetic generator every allocated latent anchor is independent uniform and its repeated appearances are identical, including those allocated before an interrupted successful invocation. Such a string need not be a real commitment to the unseen share. The generator supplies only prescribed Bob-visible outputs to the total component output simulators; it does not claim knowledge of an opening or invoke a secret-output oracle.

Now sample Bob's full Toeplitz setup tape, canonical public setup, contributions $w_B$, complementary masks and fresh use masks from his prescribed algorithm. Honest Alice's fresh uniform $w_A$ makes $w_A \oplus w_B$ uniform independently of these Bob values. Each candidate public key therefore has exactly the ordinary honest-key distribution, jointly with Bob's inputs. Obtain $\rho$ from that candidate $ek$ and release the prefix and complete public key only at their profile's prescribed stages. The bounded matrix predicate is public in $\rho$; preserve its refusal, all reservation slots, interruptions and activation rules without conditioning on success.

For each invoked generation stage, first determine its mathematical success flag from the public bounded-matrix predicate and prescribed original-record checks. On success allocate its latent anchor uniformly before calling the total output simulator. Save the same value

for all output-dependent simulated messages, local rewinds and the stopping projection. On functional failure supply the fixed-zero output, including its zero anchor payload. Generate each successful Alice-output relay uniformly with fresh independent pads; functional-failure relays are fixed zero. Then run the complete-call simulator and apply the same strict-time stopping policy to its log. If interrupted, retain the resulting prefix and burn the reservation; discard the unused candidate record. If the prescribed completion occurs, retain its original anchor and activate only as the owner rules permit. The emulator never substitutes a zero-output call merely because delivery or activation did not occur.

Prescribed Alice supplies her original share/opening on continuation, so descriptor/owner policy determines the public authentication flags. Equivalent opening seeds have the same image and do not alter those flags or the consumed original share. For an accepted use, the hidden result is $\mathsf{TotalDecaps}(X, ek, c) \oplus R$; its padded Alice component is uniform even for rejecting ciphertexts. The total simulators and their stopping projections supply the corresponding intermediate configurations and interrupted prefixes, without knowing the hidden result or an opening of a synthetic anchor. No erased tape is needed.

SR uses the original seed-share record and regenerates internally; its flags and synthetic outputs require no seed from the emulator. Run Bob's local Encaps using the public algorithm and retain its fresh coins, ciphertext and known shared key. Permitted public scheduling may depend on these values and his simulated history. Hidden-result-dependent application feedback remains outside the interface. Induction over the fixed slot budget gives exactly the last synthetic generator's Bob marginal, including his full setup tape. Only the previously charged component and PRG changes separate it from the real execution. Uniform output-simulator runtime bounds give the asserted uniform expected running time, including calls that reuse a synthetic anchor. □

### 11.5 Offline recovery reduction

**Theorem 3** (C private-key recovery from Bob's complete view)**.** *For every fixed strict-PPT $E_B$ in Definition 3, the qualified C realization has expected-time public-key emulation and a KEM reduction with a uniform pre-challenge expected-time bound $P_B$. For every polynomial hiding cutoff $t_H$ and KEM cutoff $t_K$,*

$$p^{\mathrm{real}}_{\mathrm{rec,B}} \le \Delta_{\Pi,B} + \Gamma_H(t_H) + \kappa \left( \mathrm{Adv}_{\mathrm{KEM}}(D^B_{t_K}) + 2^{-\ell} + \frac{P_B}{t_K} \right). \tag{R38}$$

*The KEM assumption, public-prefix/refusal contract and concrete ML-KEM-768 interpretation are those of Theorem 2. For $\kappa = 0$ recovery is zero. In a growing secure family, negligible component errors and the stated PRG assumption imply negligible recovery.*

*Proof.* The decoder followed by the fresh-encapsulation checker is a strict-time observer. Lemma 15 transfers its success to the synthetic public-key emulator with one additive cost $\Delta_B(t_H)$. Choose $J$ uniformly among all $\kappa$ reserved generation positions before selection, embed the KEM challenge public key in position $J$, and generate ordinary independent public-key candidates elsewhere. All slots have the same unmarked honest-candidate distribution. Thus $J$ is independent of the complete simulated view and final target, including public-prefix refusal and unused or interrupted reservations.

Run the emulator and decoder. Unless the selected target is an active slot $J$ with a valid candidate, output zero. After that candidate is fixed, request the single KEM challenge $(c^*, T^*)$ and test its candidate Decaps result against $T^*$. In the real-key branch its acceptance is the emulated recovery probability divided by $\kappa$. In the random-key branch the fresh uniform $T^*$ matches a fixed candidate's deterministic decapsulation with probability at most $2^{-\ell}$, even conditioned on $c^*$ and all prior local encapsulations.

The final public-key emulator, owner, ordinary key generation and decoder have pre-challenge expected runtime at most $P_B$. Stop that prefix at $t_K$ and output zero on cutoff. The real-key acceptance loss is at most $P_B/t_K$. Since cutoff and candidate selection precede the challenge, the random-key bound remains $2^{-\ell}$. Rearranging gives Eq. (R38). The PRG cutoffs belong to separate view-hybrid reductions and are not multiplied by $\kappa$ or charged again as a KEM simulation loss. The asymptotic statement follows by the same nonnegligible-advantage cutoff contradiction for both reductions. □

Original-core equality remains a separate audit relation; the same $\kappa\delta_{\mathrm{KEM}}$ correctness qualification connects exact original-core recovery to this functional checker. A Bob-only dealer still fails the complete-view experiment because its generation tape contains the complete key or a reconstructing seed. Erasing it later does not remove it from $V_B^{\mathrm{all}}$. The two theorems do not combine party views, protect deliberate honest-owner disclosure, establish malicious-Bob robustness, or turn a native functionality/performance test into a component security proof.

## 12 An interrupted candidate with no delivered anchor

The latent-slot rule in Lemma 14 is needed even when both parties follow their prescribed algorithms. Consider one invoked generation slot $a$ whose mathematical stage output has $\beta = 1$. Before simulating its view, sample its fresh opening seed and private output pad, and fix

$$C_a = G(s_a) \oplus Td_a, \qquad z_a = (1, C_a, Y_a, w_a).$$

Here $d_a$ is the stage digest, $Y_a$ is the padded Alice output, and $w_a$ denotes the remaining designated evaluator output and public metadata. This is schematic output notation for the existing adapter. The successful candidate and all its pads are fixed even though the owner has not activated a record.

Corrected LP gives a concrete order of events. Its evaluator simulator obtains the designated output $z$ in Step 3, constructs fake evaluation circuits for $z$ using Lemma A.1 in Step 5, and sends the circuits in Step 6. The later garbler-input-label delivery occurs in Step 9 [3], Theorem 4.2, pp. 22–23. Thus a permitted external stop can occur after Bob receives these circuits and before final decoding or application delivery.

Table 1: A successful candidate interrupted before delivery.

| Event | Bob's retained complete view | Owner state |
|---|---|---|
| Candidate fixed | No new observation; the generator privately fixes the candidate and pads. | Reservation launched, no active record. |
| Circuits received | The output-dependent fake evaluation circuits, coins and current configurations are retained. | Completion pending. |
| External stop | This prefix survives before decoding, relay or activation. | Reservation burned, no delivered anchor or active record. |

Receiving the circuits does not mean that Bob has decoded $C_a$. It means that the complete view already includes an object generated using the successful candidate output. Burning the reservation changes the owner state while preserving that observation.

To state the projection explicitly, let $\mathcal{L}_a(z_a)$ be the canonical complete continuation log with external interruption disabled. Fix a stopping policy $\pi$ and its coins. On each log sep-

arately, let $\theta_\pi(h, \mathcal{L})$ be its first stop event, computed from exposed logical events, Bob configurations and the preceding history $h$. At the stop in Table 1, write

$$\mathrm{Stop}_\pi(h, \mathcal{L}_a(z_a)) = \big(h, \mathcal{L}_a(z_a)_{\le \theta_\pi(h, \mathcal{L}_a(z_a))}, \mathsf{burn}, \mathsf{inactive}\big).$$

The same policy and coins run separately on each hybrid log; the resulting stop indices may differ. The unobserved continuation supplies neither an actual output nor a later owner transition. The policy cannot inspect the hidden candidate, the simulator's running time or its internal rewinds. This is exactly the stopping projection of Lemma 13.

On this trace the delivered-anchor and active-record indicators are both zero, whereas the fresh latent-anchor indicator is one. The missing comparison in a delivery-only proof is the adjacent pair

$$\begin{aligned} &K_a(G(s_a)) \quad \text{and} \quad K_a(U_m), \\ &\qquad K_a(Z) = \mathrm{Stop}_\pi(h, \mathsf{Sim}_B(b_a, z_a[Z \oplus Td_a])) \,, \end{aligned}$$

where $b_a$ includes Bob's input and entry state, $U_m$ is fresh uniform, and the bracket replaces only the anchor component. The simulator returns the complete continuation log. Charging only delivered or activated anchors would assign this comparison zero cost solely because delivery was interrupted. Output dependence supplies no equality between the two prefix distributions. It identifies the PRG hybrid that still has to be paid before stopping is applied.

The existing proof pays that hybrid when the fresh successful anchor is allocated, before the first output-dependent simulation step. The same strict-time projection then preserves its distinguishing bound. The total output-simulation qualification makes $K_a$ defined also when its uniform synthetic anchor has no opening. Keep the candidate, anchor and independent pads fixed through local rewinds and the stop decision. For completed deliveries, later repeated uses or restores retain their original value and add no fresh PRG challenge. The interrupted candidate in this example remains burned and cannot be restored. Summing over the fixed potential-slot budget covers this trace as well as completed calls. Uninvoked slots and functional $\beta = 0$ outputs have zero hiding loss because their designated payloads ignore the challenge. An interruption of a successful candidate retains its successful-output prefix.

This example diagnoses a missing step in an incomplete recovery reduction. A concrete attack would additionally require an efficient distinguisher or decoder with nonnegligible unconditional advantage. The conditional event order alone supplies no such advantage. The example uses the existing complete-view experiment, stopping rule and component qualifications, and introduces no additional security goal or assumption.

# 13 Recovery through the early-public observation interface

This section makes explicit the common proof obligation created by early public $\rho$. It collects the consequences of Theorems 2 and 3 for the specified C service. All component qualifications, uniform conditional expected-time bounds and delivery restrictions of those theorems remain premises. The deductions concern this service rather than a generic compiler for arbitrary leakage or concurrent protocols.

## 13.1 The reservation and stopping envelope

Sample the public setup once according to the original service experiment. All probabilities below average over that setup as well as fresh keys, protocol coins and decoder coins. Fix a public profile, at most $\kappa$ generation reservations and $\tau$ use reservations, with their original owner

rules. An invoked T/U/L generation exposes $\rho$ at its prescribed first stage. It may then publicly refuse bounded matrix sampling, be interrupted, or continue to key activation. D releases its public information according to its single-stage schedule. Fresh candidates are honest-key distributed before selection under the appropriate honest party's fresh contribution. Every reserved slot remains indexed, including a slot never invoked. The early prefix belongs to the eventual public key, so the added obligation concerns the release schedule and candidate selection.

The Alice experiment permits substituted Alice to choose subsequent strict-time actions from its complete history. The Bob experiment keeps both party algorithms prescribed and permits the declared stopping policy to inspect Bob's logical protocol prefix. The latter policy cannot inspect simulator runtime, internal rewinding or hidden output values. Each transport interruption burns its reservation and preserves the view already observed. Functional refusal instead has the specified zero payload. The experiments therefore distinguish a successful candidate whose delivery is interrupted from a functional-zero call.

Let $Q$ be a public subset of the bounded-sampling-qualified prefixes, with positive probability under honest generation. Suppose an attacker continues a candidate precisely when its early prefix lies in $Q$. Honest generation then yields a continued-prefix law conditioned on $\rho \in Q$, already including the mandatory public sampling predicate. The proof uses the unconditional recovery event over all reservations and charges this selection directly. It requires no assertion of uniformity after adversarial acceptance.

**Proposition 1** (Recovery with the staged prefix and selective abort)**.** *For the observation envelope above, let $p_A$ and $p_B$ be the useful-key recovery probabilities in the two separate experiments of Definitions 2 and 3. For each fixed strict-PPT worker/decoder and the cutoffs of the underlying theorems, there are truncated strict-PPT KEM reductions $D_{A,t_K}, D_{B,t_K}$ such that, for $j \in \{A,B\}$ and $\kappa \geq 1$,*

$$p_j \leq \Delta_j + \kappa \left( \mathrm{Adv}_{\mathrm{KEM}}(D_{j,t_K}) + 2^{-\ell} + P_j/t_K \right), \tag{7}$$

*where $\Delta_A = B_R$ from Theorem 2 and $\Delta_B = \Delta_{\Pi,B} + \Gamma_H(t_H)$ from Eq.* (B38)*. The gap $\Gamma_H$ charges all potential latent anchor slots, including interrupted successful calls, and contains a runtime tail for each of its two PRG worlds. The view-transfer gap is paid once outside the reservation factor. The global bad-setup term in $\Delta_A$ is charged once under the original setup sampling, using the pointwise binding argument on the good-setup event. For $\kappa = 0$ both recovery probabilities are zero.*

*Proof.* For A, first apply the framework simulation and the Alice-facing public-key emulator of Lemma 12. It reconstructs exact anchors using the extracted effective openings. This transfers the final strict-time recovery checker with cost $B_R$. For B, complete-view component replacement, the latent-anchor hybrids and Lemma 15 transfer the checker with cost $\Delta_{\Pi,B} + \Gamma_H(t_H)$. The B emulator retains Bob's actual application coins and synthesizes a complete evaluator view, including stopped prefixes. Its synthetic anchor can lack an opening, which is why the qualified total output simulator is required.

In either emulated experiment choose an independent hidden $J \leftarrow \{1, \ldots, \kappa\}$ and place the challenge public key in slot $J$. Release its prefix, public refusal and later public key only at the original times. Other slots receive independent honest public keys. The visible execution has the same unmarked distribution for every $J$. Its stopping decisions, interruptions and final selected handle therefore reveal no challenge-slot mark. In the real-key challenge branch the reduction accepts with probability $p_j^{\mathrm{em}}/\kappa$, even when the selected handle depends on earlier discarded prefixes. Slots that remain unused or inactive simply cannot be the successful selected handle.

Only after the decoder fixes a candidate does the reduction request the KEM challenge and compare deterministic candidate decapsulation with the supplied shared key. In the random-key branch this test succeeds with probability at most $2^{-\ell}$. The pre-challenge expected-time bound $P_j$ and its strict cutoff give an additional $P_j/t_K$ loss in the real-key branch. Combining the single view transfer with this reduction gives Eq. (7).

The B view transfer remains valid after an interruption because the latent anchor is fixed before output-dependent simulation. The same strict-time stopping projection applies to both complete-call views by Lemma 13. It retains the observed prefix and the burn transition while discarding the unobserved continuation. Rechecking an original anchor adds no fresh hiding hybrid. Charging only activated records would miss observed prefixes. The potential-slot budget in $\Gamma_H$ also covers successful candidates stopped before decoding, delivery or activation. □

### 13.2 What the bound adds and what it assumes

The proposition makes the selection cost explicit in the unconditional experiment. For a fixed direction $j$, define $\mathcal{E}_j$ in that same real experiment as the event that the decoder returns an active target handle and a syntactically valid candidate, with the target public key passing the Encaps and public bounded-matrix checks. This event is fixed before the fresh checker challenge. Put $q_j = \Pr_{\mathrm{real,j}}[\mathcal{E}_j]$. Recovery implies $\mathcal{E}_j$, so for $q_j > 0$ its conditional recovery probability is at most the right-hand side of Eq. (7) for that direction divided by $q_j$. The event, probability and bound all refer to the same setup sampling and real experiment. Arbitrarily rare eligibility can therefore yield a weak conditional bound while total recovery remains small. The paper's claim uses the unconditional experiment.

This analysis connects public specialization to recovery from two different full views. A uses protected Bob and a substituted Alice. B uses honest Alice and passive Bob. The proofs cover neither the union of the two private views nor a malicious Bob's arbitrary deviations. The trusted owner, confidential authenticated channels, logical stopping model and absence of private-result feedback are inherited interface assumptions. Corrected LP supplies the qualified theoretical component; native functionality and timing remain separate evidence.

## 14 Single-worker substitution and the exact-origin theorem

The ASA setting replaces Alice's entire worker implementation while Bob's code, randomness, complementary share and persistent owner remain protected. The attacker may choose Alice's generation inputs, retain every message and share, reload altered state, deviate during any call and choose public aborts. The service boundary is generation, restore and complete decapsulation. The following experiment makes the connection between this threat and Theorem 1 explicit.

### 14.1 Observations, outputs and the experiment

Let $V_A$ be Alice's full corrupted view and bounded continuation state, including her auxiliary input, local coins, sent and received messages, candidate shares and delivered outputs. Let $O_B$ contain Bob's prescribed private application outputs. The declared public record $L$ contains the profile, hash indices, identifiers, request descriptors and lengths, released $\rho, ek$, anchors and public authentication/refusal and completion/abort schedule. Public anchors and masked relay messages are the prescribed interface values already covered by the original experiment. Private ciphertext validity and the reconstructed Decaps result stay inside the function. The experiment's checker

may inspect joint outputs $(V_A, O_B, L)$, while the replaced worker receives only its own view and the public record. In particular, giving $O_B$ to the checker grants no additional output to Alice.

Write $Z$ for permitted initial checker auxiliary information under the quantifiers of Sec. 3. It obeys the same independence from fresh hash indices and challenge randomness. Every comparison uses the same initial input ensemble, owner contracts, reservation budget and public profile. The real experiment $\mathsf{Real}^R_{\mathcal{A}}$ runs the mathematical protocol with its specified 2PC realization and returns $(V_A, O_B, L, Z)$ to a strict-PPT checker $W$. The ideal experiment $\mathsf{Ideal}^R_{\mathcal{S}}$ runs the exact-origin service of Definition 1 against its simulator and returns the corresponding joint tuple. For SR use its retained-seed service from Lemma 17. The ideal service preserves the original record throughout restore and recovery and retains the profile's declared abort and output schedule.

**Definition 4** (Single-worker substitution advantage)**.** *For one strict-PPT replaced worker $\mathcal{A}$, its ideal simulator $\mathcal{S}$ and a strict-PPT attack checker $W$, define*

$$p_{\mathrm{real}}(W) = \Pr[W(\mathsf{Real}^R_{\mathcal{A}}) = 1], \tag{8}$$

$$p_{\mathrm{ideal}}(W) = \Pr[W(\mathsf{Ideal}^R_{\mathcal{S}}) = 1], \tag{9}$$

$$\mathrm{Adv}^R_{\mathrm{sub}}(\mathcal{A}, \mathcal{S}, W) = |p_{\mathrm{real}}(W) - p_{\mathrm{ideal}}(W)|. \tag{10}$$

*The service resists single-worker substitution at its declared interface if for every $\mathcal{A}$ one simulator $\mathcal{S}$, independent of $W$, makes this advantage negligible for every such checker and admissible input ensemble. An attack's success in excess of its ideal-service success is the measured security loss. An ideal behavior already authorized by that service is part of the baseline, including permitted selective abort and public-key-dependent information.*

This is a service-specific simulation formulation of one-sided substitution. The changed-$z$ witness identifies the stateful authorization failure that the service excludes. Classical ASA formulations additionally specify an undetectability experiment and a particular leakage target [7, 8]. Here the replaced worker may behave arbitrarily, so stealth is unnecessary for the above security guarantee. Any efficiently recognized stealth restriction can be included in $W$. A particular secret-recovery or exfiltration test is meaningful only with its declared target and permitted checker information. The theorem compares that test with the same ideal interface. It retains public-key and abort observations in the explicitly defined ideal baseline.

### 14.2 Reduction and the meaning of a successful attack

**Corollary 4** (Substitution advantage from framework simulation)**.** *Under Theorem 1, for every strict-PPT $\mathcal{A}$ there is one expected-PPT $\mathcal{S}$ such that, for every strict-PPT checker $W$,*

$$\mathrm{Adv}^R_{\mathrm{sub}}(\mathcal{A}, \mathcal{S}, W) \le B_R.$$

*Consequently, if a specified attack test satisfies $p_{\mathrm{ideal}}(W) \le \eta_W$, then*

$$p_{\mathrm{real}}(W) \le \eta_W + B_R. \tag{11}$$

*The LP instantiation in Corollary 1 makes the additional advantage negligible under its stated assumptions. The same statement applies to SR using its seed-retention lemma and call budget $2\kappa+\tau$.*

*Proof.* Fix $\mathcal{A}$ first and take the simulator of Theorem 1. The checker $W$ is a permitted strict-time observer of that theorem's joint output ensemble. Public records are components of this same ensemble or polynomial-time projections of it. Applying the theorem with observer $W$ yields

Eq. (10) bounded by $B_R$. The simulator was fixed before the observer, so this is the required quantifier order. The one-sided inequality $p_{\text{real}} \leq p_{\text{ideal}} + B_R$ gives Eq. (11). Substituting the proved LP bound or SR lemma establishes the corresponding instances. This mapping preserves the existing error accounting and call-local rewinding discipline over the fixed service budget. □

For a concrete view-based recovery test, let a strict-PPT decoder $E(V_A, L)$ output a candidate string and let a strict-PPT predicate $P(E(V_A, L), O_B, L, Z)$ recognize the declared target. Take $W$ to run $E$ then $P$. Both are strict time, and Corollary 4 bounds its extra recovery success by $B_R$. For example, the checker can test a predicate of Bob's prescribed outputs without revealing those outputs to $E$. Its ideal success $\eta_W$ must be proved or stated for that target and input distribution. This construction covers efficient postprocessing of the worker view and avoids an unconditional secret-guessing claim based solely on masking.

**Original-state integrity and ghost audits.** In the hash-check ideal-stage experiment, define $\mathsf{Bad}_{\text{orig}}$ exactly as in Sec. 3. It records an accepted changed handoff or an incorrectly delivered designated share pair relative to the original handle. Original state is legitimate ghost state of this proof experiment. The transfer/use lemmas and owner induction give

$$\Pr_{\text{stage}}[\mathsf{Bad}_{\text{orig}}] \leq \epsilon_c + \epsilon_p.$$

The exact-equality ideal service has zero such state violations. In the real 2PC realization, the theorem supplies effective inputs through simulation and extraction. A hidden effective input or newly generated ghost key is not automatically an efficiently observable variable of $W$. Therefore the real-protocol claim is the joint-output simulation bound above. A specific functional audit receives a real success bound only when its checker is a permitted efficient test of that joint experiment. Arbitrary shares published later by malicious Alice remain outside the designated-output relation.

**Which channels are covered.** The protected owner fixes the original descriptor and record. The compromised worker sees its own messages, inputs and state, public lengths and parameters, public authentication and the profile's completion/abort schedule. Uniform shares and output masks are delivered through the specified private interfaces. Ciphertext validity remains internal. Processor side channels, additional physical observations, rollback of Bob and downstream share reconstruction require a richer experiment. A classical LP component supplies the real protocol in this corollary. The native EMP measurements retain their separate backend qualification boundary.

# 15 Choosing an authenticated state boundary

A history-sensitive service must specify both the generation that a later restore/use continues and the actual state consumed by that request. We express these requirements as an authenticated sufficient-state condition, then use it to compare placement and retention costs. This application invariant complements the established malicious-secure 2PC component. The condition below is a direct correctness corollary of Theorem 1.

## 15.1 What state must determine

Let $v$ be the original generated state, $p$ its immutable public descriptor, and $u$ an admissible future request. Write $f(p, v, u)$ for the complete authorized operation. A retained state $S =$

$\mathcal{R}(p, v)$ is *continuation sufficient* if a fixed, total, polynomial-time function $F$ satisfies

$$F(p, \mathcal{R}(p, v), u) = f(p, v, u) \quad \text{for every admitted } (p, v, u). \tag{12}$$

The future-request domain is common to compared states and determined by the public descriptor and owner state. The protocol fixes any sampling refusal. The definition permits noninjective $\mathcal{R}$ without a minimum-length requirement. Equivalently, for any two admissible $v, v'$ with the same $p$ and $\mathcal{R}(p, v) = \mathcal{R}(p, v')$, all future authorized outputs must agree for every common admitted request. Necessity follows because a deterministic $F$ receives identical inputs on these two states. Sufficiency follows by defining $F$ on each representation class, provided the selected polynomial-time completion exists. Each profile must explicitly supply that efficient completion.

For decapsulation, "every request" includes all fixed-length ciphertexts. The state must determine both successful decapsulation and implicit rejection. A changed $z$ may change $J(z \parallel c)$ on a rejecting ciphertext while preserving the successful branch. Such a representation fails Eq. (12). Ciphertext rejection remains internal to the specified function under the declared leakage interface.

**Corollary 5** (Authenticated sufficient states)**.** *Within the scope of Theorem 1, suppose an original boundary state $S$ satisfying Eq.* (12) *is shared as $a \oplus b$ with a fresh independent uniform Bob mask. Protected Bob retains immutable $(p, b, C)$ with anchor*

$$C = \mathsf{Com}(d, a; r), \qquad d = d(p),$$

*Alice retains its private opening $r$. For Bob-view privacy she samples it independently and uniformly as in Sec. 9. Here $d(p)$ is a fixed public anchor context (it may encode only a projection of $p$). Bob and the agreed invocation descriptor independently pin the entire original $p$, including every field omitted from $d$. A later mathematical stage authenticates its actual reinput $(x, r')$ against this original $C$ and consumes that same $x$ in $F(p, x \oplus b, u)$. For the same share a seed is acceptable when its PRG image equals the original seed image; seed encoding equality is unnecessary. Its descriptor fixes $F, p, u$ and the declared output/abort interface before invocation. Except on the global descriptor collision and commitment-binding failure events, an accepted stage computes $f(p, v, u)$ for the original generated state. If it creates another persistent boundary, that boundary's share and anchor are formed on the same computed wires before activation.*

*With the original fresh output masking, declared leakage, owner rules and call-local component premises, the existing simulation proof applies to such a profile with its fixed call budget. The bound remains the sum of its call errors and the whole-execution collision-finder advantages and the global statistical setup error over the entire bounded execution.*

*Proof.* First work with commitment-check ideal mathematical stages. If a distinct descriptor passes fingerprint agreement, the two unequal descriptor encodings give an $H_p$ collision. Otherwise $F, p, u$ are the stipulated ones. The first accepted $x \neq a$ gives two openings of the original commitment on different encoded share values. Lemma 10 gives an $H_c$ collision outside the one global bad-Toeplitz event. Absent both an $H_c$ collision and the global bad-setup event, $x = a$ and $x \oplus b = S$. Equation (12) gives the original authorized operation. An outgoing anchor on those same computed wires establishes the same invariant at the next boundary. Immutable records and the owner induction preserve it through restore, failures and further admitted requests.

This algebra establishes integrity in the ideal-stage experiment. Alice-view privacy follows from fresh uniform share/output masks and the joint-view coupling of Lemma 2, including the declared public schedule. Lemma 6 reaches the commitment-check stage experiment. The first-binding-failure transformation then reaches the exact-origin service, using Lemma 7 for expected-time reductions. Bob full-view privacy additionally uses the PRG assumption and honest uniform

opening seeds and the evaluator-view proof in Sec. 11. Together these steps lift stage integrity to the service under the stated simulation guarantees and the protected owner's persistent storage, declared nonrollback recovery and spent-reservation contracts throughout the service lifetime. □

### 15.2 How the four profiles instantiate the condition

| Profile | Original authenticated state | Fixed public data | Completion from checked state |
|---|---|---|---|
| D | No intermediate handoff; $X$ at use | $ek$, context | Complete $\mathsf{TotalDecaps}(X, ek, c)$ |
| T | $\sigma \parallel z$ at handoff; $X$ at use | $\rho$, context; then $ek$ | Noise/NTT and matrix completion; then complete $\mathsf{TotalDecaps}$ |
| L | $\hat{s} \parallel \hat{e} \parallel z$ at handoff; $X$ at use | $\rho$, context; then $ek$ | Matrix completion; then complete $\mathsf{TotalDecaps}$ |
| SR | Original $\sigma \parallel z$ retained at use | Original $ek$, $\rho$, context | Secret-noise/NTT regeneration $\mathsf{Regen}$ followed by complete $\mathsf{TotalDecaps}$ |

Each row additionally retains Bob's corresponding original share and anchor and Alice's corresponding private opening. T/U/L need an opening at both handoff and final boundaries; SR retains its seed opening for use. The request descriptor binds the current ciphertext and SID. T/U/L implement Eq. (12) by the specified $F_T, F_L$ identities. SR implements it by $\mathsf{Regen}(S) = X$ and the original retained $ek$. Its anchor context fixes $\rho$ and the generation context. Protected final activation and subsequent descriptor agreement fix the final $ek$, produced after this anchor. The functional identity in Lemma 17 proves this relation. The current commitment/opening integrity follows from this corollary; the older lemma's hash-anchor security argument concerns H. SR regenerates the secret polynomials needed for use, leaving $\hat{e}$ and matrix products outside that computation. D forms its final anchor inside the original generation call. Every profile allows malicious Alice to retain all earlier outputs.

### 15.3 Service integration obligations

The conditions identify the implementation obligations of this construction. First, replacing the original anchor on restore permits a changed state to become freshly authorized even with ideal components. The wrong-$z$ witness in the main paper demonstrates this history failure. Second, authenticating $a$ but consuming a separate caller input $x'$ permits changed computation without a commitment-binding failure. Third, keeping a checked share while changing $p$ can change its meaning or selected completion. Fourth, authenticating a representation that omits a future-relevant value does not satisfy Eq. (12), regardless of anchor strength. Finally, rollback, reuse of a spent SID or resuming an orphaned stage violates the owner premise. The protected owner must enforce that premise in addition to cryptographic input checking. Alternative protocols may enforce these service invariants differently.

### 15.4 A reproducible selection procedure

The corollary can be used as an admissibility filter before optimizing a boundary. Its input is the historical service relation, custody policy, candidate representations and a declared workload. Its output is an admissible resource frontier, or a refusal when no candidate has the required completion and interface argument. The following four steps produce a checkable record for this service from explicitly supplied completions and proof obligations.

1. Fix the request domain and original-handle relation, including invalid ciphertexts, public aborts and the output-share interface. The relation includes the original rejection map as well as the public key and the complete successful branch.

2. For each candidate, supply $\mathcal{R}, F, p$ and establish Eq. (12). Keep an explicit counterexample when it fails. For ML-KEM, retaining only $\sigma$ fails because two keys with equal $\sigma, ek$ and different $z$ can disagree on rejection. Both $X$ and $\sigma \parallel z$ have explicit efficient completions.

3. Check the lifecycle. Create the original anchor on the produced wires, fix their meaning with the immutable descriptor and consume the same actual reinput that authentication checks. Activation, spent IDs and failure recovery must satisfy the owner contract. The proof lifts these identities under the component and platform assumptions. Finite interface tests supplement this mathematical argument.

4. On the remaining candidates, record private storage, generation work, per-use work/input width, complete-request traffic and sampled memory under the same backend configuration. Apply the application's storage and use budget only to these candidates. Report ties or unresolved latency ordering when uncertainty or measurement scope prevents a stronger choice.

| Candidate | Admissibility result | Diagnostic or completion |
|---|---|---|
| Keep $ek$ or successful branch | Insufficient for the declared domain | A changed $z$ changes rejection |
| Keep $\sigma$ alone | Insufficient retained state | Same $\sigma, ek$, distinct rejection maps |
| Fresh enrollment on restore | Breaks original-handle history | Freshly enrolled changed-$z$ witness |
| Check $a$, consume $x'$ | Breaks actual-input correspondence | Choose $x' \neq a$ after a valid check |
| Keep $X$, opening and original $C$ | Admissible under the stated assumptions | Complete CheckedUse (T) |
| Keep $\sigma \parallel z$, opening and original $D_S$ | Admissible under the same assumptions | Regen then complete CheckedUse (SR) |

The first four rows identify failures of the service specification or integration obligations. The last two represent the same original continuation under the same rejection and custody goals. The archived full-domain witness and seed-circuit controls exercise the relevant concrete cases. The written identities and reductions establish admissibility over the full domain.

### 15.5 Selecting the boundary and retained lifetime

The numeric gate and timing examples in this subsection are retained H-version references. They exclude the randomized commitment circuitry. For the new construction, Alice stores the additional $u$-byte opening per live boundary; Bob stores the 112-byte commitment. The proof of admissibility above concerns the commitment version, while any resource selection must include its own measured costs for this randomized construction.

Once the invariant is met, the design variables are the state boundary and its lifetime. A generation handoff authenticates its transferred share when produced and when consumed. Moving a boundary after an expanding transform can therefore increase authentication work even if it reduces the largest arithmetic stage. For the earlier H mathematical circuits, Eq. (28) gives

$$G_L - G_T = 232959 + 2(L_h - T_h),$$

which makes the boundary cost explicit instead of assigning a universal percentage advantage to a 64-byte state. T also permits the public matrix to be specialized outside private computation. The quantified public-input alternative separates that benefit from the authentication cost of the state boundary in the selected mathematical construction.

For a fixed workload with $N$ stored keys, one completed generation per key and $n_h$ completed uses of key $h$, two simple coordinates of the resource objective are

$$\mathsf{PrivateStorage}_j = N\ell_j, \qquad \mathsf{GateWork}_j = NG_{\mathrm{gen,j}} + \sum_{h=1}^{N} n_h G_{\mathrm{use,j}}.$$

Here $\ell_j$ counts required private-record bytes for the party being budgeted, including its openings when the commitment version is used, excluding common public records and spent-request history. Failed attempts add their actual executed stage work. The choice also changes private-input bits, call counts, round structure and backend memory. Complete-request measurements capture these costs alongside gate work. The common conservative setup for T and SR isolates the lifetime choice. SR retains 1120 fewer private-share bytes per party, adds 944316 reference ANDs per use, and reduces padded private input by 8960 bits per party. The smaller input can lower OT/input-checking costs. The complete native-use experiment in Sec. 23.3 measures these effects together under the complete-request timing boundary.

For the H reference, let $\mathcal{B}$ be a per-party budget for prescribed private shares, excluding the common record header, and require $N$ simultaneously live keys. If $64N \le \mathcal{B} < 1184N$, SR is the only storage-feasible member of that H pair. In the commitment version Alice instead requires $(64+u)N$ or $(1184+u)N$ bytes. The per-key difference remains 1120 bytes, but its percentage and feasibility thresholds change. PRG and linear-network costs and the new use latency also require replacement in the objective below. If both fit, let $w_s, w_t, w_b$ be the application's declared nonnegative costs per stored byte over the horizon, per second of use and per peer byte. Conditioned on completed conservative T setup, define a storage-and-use objective $J$ over all $N$ keys, excluding setup latency. For equal use mix and $n$ completed uses per key, its per-key incremental value is

$$(J_{\mathrm{SR}} - J_T)/N = -1120w_s + n\big(w_t\Delta t + w_b\Delta b\big).$$

Here $\Delta t, \Delta b$ are SR-minus-T differences under the matched measurement boundary, with uncertainty retained for $\Delta t$. A hard temporary-memory cap must also be checked independently against garbling/preprocessing memory. For positive use penalty, the break-even count is $1120w_s/(w_t\Delta t + w_b\Delta b)$. For a nonpositive use penalty, this two-coordinate objective does not impose a use-count crossover. Without declared weights or constraints, the resource frontier records the available tradeoffs. The reproducible tool `select_state_v30.py` accepts the key count, per-party private-share budget, use count, valid fraction and optional explicit prices. It returns storage-feasible T/SR profiles, six-block projected cost intervals and a conditional choice or unresolved ordering. It takes the written admissibility argument as an input assumption. Workload totals and crossovers are projections from this equation and the measured complete-use costs. Failed reservations add their actual execution costs.

A generic compiler protects its supplied operation. This procedure checks the historical policy and state sufficiency before selecting a concrete boundary and lifetime from the proof obligations and measured costs. It operates within the adversarial model of the generic components. The 64-byte choice is justified by its complete-operation identity and measured lifetime tradeoff.

This secondary design contribution connects the original-handle invariant to concrete computation/storage choices. Authenticate sufficient state before its representation grows, or retain

it and pay reconstruction at each use. The application selects between these valid placements according to its custody policy and workload. A protected single-holder service remains appropriate when that policy permits reconstruction. The corollary uses the same protected-Bob assumption as the joint-generation framework and its private-key recovery theorem.

## 16 Joint-generation service and application delivery

**A managed receiver with restricted helper custody.** Consider a remotely maintained receiver for queued management commands from several senders. Its application obtains ML-KEM operations from an updateable cryptographic worker Alice and a separately protected gateway Bob. The gateway may originate some ciphertexts, but its custody policy forbids a reusable private key for all senders' traffic. The senders only use the public ML-KEM interface; the two workers perform the joint computation. This illustrates the deployment policy studied here. The application can retain a protected output combiner, and Bob can retain atomic nonrollback owner records, while complete-key generation remains outside that protected application. Replacement affects Alice's cryptographic worker. It gives the attacker all of that worker's state, coins, inputs and messages, while the application combiner and Bob remain protected. It does not describe compromise of the receiver's entire application environment.

The service exports standard ML-KEM public keys and accepts standard ML-KEM ciphertexts. Bob contributes to generation and stays online for every checked Decaps. The service requires resistance to usable-private-key recovery from replaced Alice with protected Bob, and separately from passive Bob's complete view with honest Alice. The second requirement concerns recordings made while Bob follows the protocol, including setup and component tapes. Erasure cannot remove information from this complete view. Bob may know his own Encaps session keys; a reusable private key would additionally let him decapsulate independently generated traffic. This distinction motivates restricting helper custody rather than assigning complete KeyGen to Bob.

The two goals concern separate executions. Alice can voluntarily send her share to Bob in an execution where she is malicious. That execution is outside the passive-Bob experiment. Private-key custody also retains the protected application boundary: the workers provide separate masked results, and the application combines them. Transport, downstream key derivation and AEAD have their own security obligations. Continued Bob availability, protected owner state and this delivery interface are costs of the chosen service policy. Section 17 gives an explicit public-status wrapper for the existing no-private-feedback boundary.

**Architectural alternatives under that policy.** Alice-only KeyGen followed by sharing preserves any private-key knowledge already encoded by an ASA. Having protected Bob generate and later split or delete the key prevents Alice-side generation from choosing it, but Bob's complete generation view already contains the key. A trusted local module is another viable option if it retains the complete key for subsequent use or exports only prescribed shares. This moves complete-key generation and custody into a protected component. We neither implement that option nor compare its cost with the two-worker service.

Direct malicious-secure 2PC with the specified shared-output and committed-state interface is already a sufficient realization. Existing input-consistency tools can implement that interface. The staged T profile is an implementation choice within it, justified by specializing the public matrix and by measured circuit costs. The new proof obligation comes from changing the observation schedule before activation. Section 19.4 separates that specialization benefit from staging overhead; the measured host-dependent latency is a separate result. An appropriate guarded

Table 2: Architecture choices for the managed-receiver policy.

| Choice | Relation to the two custody goals | Boundary or additional obligation |
|---|---|---|
| Alice KeyGen, then sharing | Prior ASA knowledge persists. | Complete KeyGen remains replaceable. |
| Bob generates, then splits | Bob's full view contains the key. | Deletion does not change that view. |
| Trusted local KeyGen | Can meet both with protected key retention or shared outputs. | Complete generation moves inside a protected component. |
| Direct committed-state 2PC (D) | Realizes both goals under the written qualifications. | Online Bob and the original-state service remain required. |
| Staged 2PC (T) | Preserves the same goals. | Early public $\rho$ and interrupted calls require the stated recovery analysis. |

implementation of the whole historical service is also an alternative under its own setup and verification assumptions. No general superiority over such alternatives is claimed.

**Lifecycle and interfaces.** The application first requests joint generation. The owner reserves the generation SID and quota. The selected D/T/U/L profile creates its original shares and anchor on the same computed wires. Only protected final completion activates handle $h$. Alice may back up and reload her share and private commitment opening. Bob retains the original public descriptor, complementary share and anchor. Restore preserves these records and all spent reservations.

For a later request $(h, c)$, the owner allocates a fresh operation SID before launch. The component binds that request's descriptor, authenticates the actual restored share and computes complete decapsulation on the same share after commitment verification. Both valid and invalid fixed-length ciphertexts are permitted. Public Auth reports share/descriptor authentication. Ciphertext validity remains internal. The two workers retain the prescribed separate output shares. Client transport, share combination, key derivation and AEAD use are downstream interfaces with separate security obligations. Failure burns the reservation, and recovery requires truthful worker termination before an independent fresh request can start.

In this workflow, the original-share relation protects future behavior that public-key and pairwise tests leave unchecked. An altered $z$ can change implicit rejection while preserving the public key and successful branch. AWS KMS provides an example of an immutable-material restore policy [10]. The comparison concerns immutable material alone. Our two-worker custody policy and threat model are specified independently for this service.

**Joint generation and subsequent design choices.** Joint generation and the recovery reduction in Sec. 10 establish the principal ASA defense. State authentication keeps this jointly generated key authoritative on later uses. A security-constrained method for selecting the boundary and retained lifetime then addresses the cost of that defense. The admissibility record in Sec. 15 supplies a full-domain completion, original-input correspondence and owner invariant before any cost ranking. It rejects incomplete representations and history-changing restoration even when the individual computations are ideally secure. For admitted candidates, the same record exposes the storage/use tradeoff. T materializes $X$ once, whereas SR stores $\sigma \parallel z$ and reconstructs the needed secret core at each use. Matched complete requests, rather than gate counts alone, supply the empirical resource comparison. This secondary design contribution is the service-specific method and its ML-KEM instantiations, using established seed representations and authentication primitives.

Self-Guarding [11] uses trusted setup and input-private verifiable computation to protect

a supplied functionality under its standalone substitution model. Its reusable construction (Construction 4, Definition 17 and Theorem 3) supports repeated runs of the specified function. A Restore API must still identify the earlier generation that it authorizes. Protecting a function that freshly enrolls altered state preserves that policy. The Self-Guarding compiler could instead protect a suitable standalone encoding of a correctly specified historical service. Our construction uses classical malicious-secure 2PC under static Alice corruption and protected Bob. Its security follows from the stated component assumptions and reductions, rather than an instantiation of that compiler or a stronger ASA guarantee.

**Reading the resource results.** T authenticates a 64-byte sufficient state before NTT growth. L demonstrates the larger authentication cost of a later state boundary. SR retains the small state for its whole lifetime and reconstructs only the secret polynomials at each use. These alternatives let an application distinguish persistent-share bytes, generation work, repeated-use gate work and private input width. In C38, SR saves 1120 private share bytes per party; Alice retains a 16-byte opening in either profile. The complete native experiment in Sec. 21 includes the new commitment circuitry, with fresh preprocessing and matching public keys and ciphertexts. Its memory measurements retain shared-mapping and sampling qualifications. The ordinary emitted circuit inventory also gives a reproducible LP table-count projection; native EMP timing is not LP timing.

# 17 A public-trace application boundary

The managed receiver can use the key service through an asynchronous private command queue. Its gateway acknowledges receipt of the public key-service request, rather than successful command execution. A protected application receives the separate result shares, combines them and performs any downstream key derivation, AEAD processing and queue admission. The key-service controller has no input from that private processing branch. Senders still use the ordinary public ML-KEM interface; they do not become additional parties in the two-worker computation.

## 17.1 Public and private state transitions

The public controller uses only the original public record, its public inputs and fresh controller coins independent of the service secrets. An operation SID and its quota are reserved under the original owner policy before worker launch. The fixed-format receipt $\mathsf{Admit}(h, \mathsf{SID}, c)$ confirms that request's admission to the service, before any private result is processed. Public refusal of an inactive handle, exhausted quota or invalid public request format follows the original admission rules. The two tracks are

$$\mathsf{Ingress} \xrightarrow{\text{Reserve/Admit}} \mathsf{Reserved} \xrightarrow{\text{Launch}} \mathsf{Running},$$

$$\mathsf{Running} \xrightarrow{\text{public protocol event}} \{\mathsf{PublicDone}, \mathsf{AuthFail}, \mathsf{Burn}\},$$

$$(U, R) \xrightarrow{\text{protected join}} K \xrightarrow{\text{private processing}} \mathsf{PrivateQueue}.$$

The last track runs only after the prescribed protected completion and delivery of both shares. Its branches are internal to the trusted application. They do not release an event to either worker or the public controller. A use failure burns the operation reservation while preserving the active key's original record, as in the existing service.

In particular, a completed, authenticated TotalDecaps call can emit PublicDone for either a valid or a rejecting fixed-length ML-KEM ciphertext. Public Auth authenticates the original service

Table 3: State transitions and visibility in the public-trace wrapper.

| Event | Meaning and source | Information withheld |
|---|---|---|
| Admit | Public handle, allocated SID and ciphertext are admitted under the owner policy. | No result has been combined or processed. |
| PublicDone | The prescribed protocol and owner operation completed with public Auth equal to one. | Ciphertext validity, AEAD acceptance and command success. |
| AuthFail | The original share or descriptor authentication failed in the key service. | This is not a KEM or AEAD validity flag. |
| Burn | An interrupted operation keeps its observed prefix and spends its reservation. | Undelivered shares and any latent private result. |
| Private queue branch | Protected join, downstream checking, enqueue or execution status. | All branch values remain outside the workers' views. |

input; it does not disclose re-encryption equality. A receipt such as *command accepted*, an execution error or an AEAD-dependent retry would have a different meaning and is outside this wrapper.

### 17.2 A sufficient condition for admissible feedback

Let $L_{\le n}$ be the original public service history at logical event $n$. It includes admitted public requests, owner transitions and the public protocol flags already covered by the service experiment. Let $\omega$ be a separate controller tape sampled independently of the honest service seeds, setup and challenge coins. A public-trace wrapper $W$ is admissible when a strict-PPT causal map $\Phi_W$ generates every additional public event from $(L_{\le n}, \omega)$:

$$Z_{W,\le n} = \Phi_W(L_{\le n}; \omega). \tag{W43}$$

The controller's future request policy and logical event schedule obey the same condition, with polynomial bounds and the original reservation quotas. Public requests may depend on the permitted public history. The substituted worker's original abort choices and the Bob-view stopping policy retain their existing observation rules.

For any two counterfactual private application outcomes with the same public history and controller tape, the added event values, their presence, encodings, logical order and the controller's next request must therefore coincide. Private queue admission or execution cannot suppress a receipt, change its length, choose a later SID or trigger a key-service retry. This contract is about the declared logical view. It supplies no physical timing or constant-time implementation guarantee.

**Lemma 16** (Public-trace wrapper closure)**.** *Fix an admissible $W$ and the component, owner, stopping and delivery qualifications of the two recovery theorems. Compose its public controller with either service experiment, retain the private application branch inside the protected boundary, and include the added public trace in the decoder's view. The two usable-private-key recovery bounds retain their form. For $j \in \{A, B\}$,*

$$p^W_{\mathrm{rec},j} \le \Delta^W_j + \kappa \left( \mathrm{Adv}_{\mathrm{KEM}}(D^W_{j,t_K}) + 2^{-\ell} + P^W_j / t_K \right). \tag{R43}$$

*Here the original component and binding error expressions are evaluated for observers including $W$. The reductions' uniform expected-time bounds include its polynomial work. In particular,*

$$\Delta^W_B = \sum_i \epsilon^W_{B,i} + \sum_{a=1}^{N_R} \left( \mathrm{Adv}_G(D^W_{a,t_H}) + 2P^W_{H,a}/t_H \right), \qquad N_R \le d_R \kappa.$$

*Here $N_R$ is a fixed potential-slot list, padded to $d_R\kappa$ if desired, rather than the random number of delivered anchors. Uninvoked and functional-zero slots have identical adjacent distributions. No new anchor slot or reservation factor is charged solely for a wrapper event.*

*Proof.* Fix the independent controller tape and proceed through logical events. At each step, $W$ reads an already covered public prefix. Equation (W43) determines its next added event and controller action without the reconstructed private result. Causality resolves adaptive requests one event at a time, so the combined controller is an allowed strict-time observer and scheduler for the original sequential service. The argument also holds after a permitted interruption because the controller reads the projected public owner state.

Sample $\omega$ in each real, simulated and public-key-emulated experiment. Apply the same causal map to the public prefix in each world. A distinguisher for the enlarged view can be composed with this strict-PPT map, preserving the original view-transfer bounds. The private join and downstream branch can be omitted from those transcript generators: they supply no observation or controller input. In particular, the public-key emulators need neither a private key nor an AEAD accept bit to produce the added trace.

The controller does not change which generation or use operations must reserve a slot. A new generation anchor is still charged before its first output-dependent simulation, and an interrupted successful candidate is still charged before stopping. Hence the same $\kappa$ and $N_R$ budgets apply. Include $W$'s bounded work in the uniform reduction runtime polynomials $P_j^W$ and $P_{H,a}^W$ before applying the original strict-time cutoffs. The two PRG tails, single view transfer and random-reservation KEM reduction give Eq. (R43), after averaging over independent controller coins. □

This lemma instantiates the existing no-private-result-feedback interface. It does not extend the service theorem to arbitrary application feedback. An application that returns AEAD acceptance, exposes command effects to the decoder or chooses subsequent key-service requests from those effects needs its own composed security analysis. Bob's legitimately known local Encaps keys remain permitted; neither worker receives the joined result of an independent sender's use through this wrapper. End-to-end command confidentiality, authorization and AEAD security are separate claims.

### 17.3 Finite contract check

The deterministic script `audit_application_boundary_v43.py` checks a small executable model of the above public controller. For each fixed public event history and controller tape it varies counterfactual private AEAD and execution outcomes. It compares the complete added logical trace, including the next controller request. Its negative controls deliberately make receipt contents, receipt presence, receipt length, event order or the next request depend on a private outcome. A sixth control exposes execution status while holding AEAD acceptance fixed across successful, failed and queued execution labels. Each such violation is detected. The labels are test values for the wrapper model, not sampled ML-KEM executions.

The receipt records the script hash, finite cases and deterministic trace digests in `data/application-boundary-v43.json`. The replay adds no native observations. These checks test the example controller contract; they neither validate the native service's application integration nor certify the lemma, KDF, AEAD or an end-to-end command implementation. Implementing and auditing the protected application boundary remain deployment obligations.

# 18 Early-public generation and the recovery contribution

## 18.1 The observation created by public specialization

An ASA against unilateral KeyGen can disclose a useful private key before any later sharing step [7, 8]. Joint generation puts protected Bob's fresh contribution inside the computation that determines the public key. The split ML-KEM implementation then releases $\rho$ while the private state $S = \sigma \parallel z$ remains shared. This release enables public-matrix specialization before expansion, but also gives the substituted worker a key-dependent prefix on which to base an abort. Observed candidates, interrupted invocations and discarded reservations all remain part of its history. The prefix is already part of the eventual public key. Early release changes when it becomes observable and which candidates can be discarded before activation.

The security question is therefore whether a decoder can recover a useful private key after selecting among these staged observations. Uniform joint coins before selection are insufficient to settle this question. For example, choose a public set $Q$ of bounded-sampling-qualified prefixes with positive honest probability. A worker that continues exactly those prefixes changes their law to the honest law conditioned on $\rho \in Q$, including the mandatory public sampling predicate. A proof confined to completed keys would miss both this selection and the resources spent observing rejected candidates.

## 18.2 The two recovery reductions

The contribution is a recovery-preserving implementation placement for this concrete service. Section 20 separates its ML-KEM interface, placement-specific proof charges and measured circuit benefit from established ingredients. Direct D can also stop after observing a public key, so the reservation-selection argument is common to D and T. Section 13 collects its interface and deductions. The Alice-facing ideal emulator reconstructs the public prefix, public sampling refusals, original anchors and use responses from candidate public keys. The Bob-facing emulator preserves a different observation, including Bob's setup and application tapes, complementary shares, local Encaps coins and every component configuration. Computational anchor hiding and complete evaluator-view simulation make that second emulator possible without a candidate private key or opening.

In both directions the reduction embeds the KEM challenge in a uniformly chosen slot among all $\kappa$ generation reservations before selection. Unused, refused and interrupted slots remain in that budget. The hidden slot is independent of the unmarked execution and its final selected handle, giving the explicit $\kappa$ loss in useful-key recovery. The proof concerns the unconditional recovery event over the entire experiment. A key accepted after adversarial selection may have a biased distribution.

The Bob-direction proof also treats observations before delivery. A successful call's candidate anchor is fixed before any output-dependent view simulation and charged even if the call is later interrupted. Stopped-prefix closure retains exactly the visible messages, coins and configurations, while the owner burns the interrupted reservation. Uniform expected-time bounds cover valid and synthetic simulator outputs; separate strict-time cutoffs charge both PRG worlds and the final KEM reduction. These steps connect early-public specialization to both recovery experiments under the declared observation boundary.

Joint randomness, XOR sharing, committed-input consistency and malicious-secure 2PC supply the established ingredients. Our added work specifies this ML-KEM staged observation interface and proves its complete-view recovery consequences, including prefix-based selection and interrupted output-dependent views. The claim is instantiated for the stated service and its qualified component simulators. It carries the classical, sequential, protected-owner and private-delivery assumptions of the underlying proofs.

### 18.3 Architectures under explicit custody goals

The two experiments impose distinct constraints. A protected Bob-only dealer can satisfy the Alice-view recovery goal, but its full view contains the complete generated key. This remains true if it accepts Alice's seed contribution and later retains only $B$. Such a dealer fails Bob-view key privacy. Joint generation preserves the encapsulator's limited knowledge as well as protecting against substitution of Alice. Table 4 compares these goals, rather than only comparing prescribed saved-record formats.

Table 4: Architectures under Alice-side recovery and passive-Bob private-key privacy.

| Architecture | Generation and custody | Evidence and comparison scope |
|---|---|---|
| Unilateral Alice | Alice forms the complete key before sharing | The motivating ASA exposure remains. |
| Protected Bob only | Bob generates the complete key, even if later split | Satisfies the Alice-side goal under trust, but fails the Bob full-view goal. |
| Direct 2PC (D) | Joint generation with hiding anchors | Direct realization for both experiments; the concrete C38 function is evaluated under the qualified native test profile. |
| Split 2PC (T/L), SR | Joint generation; authenticated boundary or retained seed shares | Both privacy goals with new openings; the C38 circuitry includes opening inputs, PRG and linear-network work in every native request. |
| Self-Guarding | Guards a supplied protocol using trusted setup and verifiable computation | Generic protection under its own substitution model; no matched implementation benchmark here. |

Self-Guarding's Construction 4 wraps protocol computations using input encoding and verification. Definition 17 supports repeated executions, and Theorem 3 proves the compiled protocol under static malicious corruption and arbitrary substitutions, assuming correct, sound and input-private verifiable computation [11]. It can protect an appropriately specified historical service. Our contribution supplies the concrete ML-KEM service and recovery analysis with protected participating Bob and classical 2PC. The comparison concerns functionality and trust. Security and performance rankings require a common target and matched implementations.

### 18.4 State continuity and its implementation choices

A later call authenticates the actual share it consumes against the original generation. Section 15 establishes this correspondence for handoff, restoration and complete decapsulation. The domain includes rejecting ciphertexts and the original $z$, whose replacement can preserve successful decapsulation while changing implicit rejection [1, 12]. State authentication therefore maintains the original generated service after the recovery analysis establishes its privacy boundary.

The 64-byte state $\sigma \parallel z$ follows from deterministic ML-KEM generation with public $\rho$. T authenticates this sufficient representation before expansion. L authenticates expanded polynomials, and SR retains the compact state and pays reconstruction per use. These are implementation choices within the proved service relation. Minimal state length and a new standard seed encoding are outside the claim.

D provides the direct committed-state 2PC baseline with the same C38 anchor, output shares and complete-use target. U retains T's two-stage relation while expanding the public matrix through generic wires and checking a runtime copy of the original $\rho$. D/U mea-

sures net staging, handoff and public-input binding. U/T measures public-$\rho$ specialization, and T/L holds specialization fixed while moving the authenticated boundary. Section 19 gives the endpoint accounting for these secondary design choices. Corrected LP supplies the theoretical realization of the ordinary stage circuits. Native EMP supplies implementation costs under its separately stated component evidence and host conditions.

# 19 Committed-input tools and the staged recovery task

## 19.1 What the existing tools supply

Jarecki and Shmatikov give malicious-secure two-party computation on committed inputs in the common reference string model [13]. Their functionality permits commitments to be referenced by subsequent computation requests, with consistency across computations as an explicit application (Sec. 5). Chase, Ganesh and Mohassel discuss secure computation on committed or signed inputs [14]. Their Sec. 3.4 uses a shared committing OT to tie an evaluator's input in a garbled-circuit proof to its subsequent cut-and-choose computation. The extension to both parties is stated there as a separate question. Our check of an opening against the actual consumed share uses this established input-consistency principle.

These tools enforce a supplied relation. The application selects which history, descriptor and complete operation that relation preserves. Re-enrolling a fresh share on restoration would enforce a different policy correctly. The service here preserves the original generation, including implicit rejection under the original $z$. We use existing commitment and committed-input techniques to implement that policy.

## 19.2 The observation and recovery work beyond input consistency

Input consistency alone says which value a later call consumes. Our recovery target also specifies what an attacker observes before deciding whether to continue. The split construction publishes $\rho$ before activation, enabling public-matrix specialization. This changes the observation schedule for information contained in the final public key. A replaced Alice can condition abort and target selection on that prefix and all its component state. Passive Bob retains his entire execution view in the separate honest-Alice experiment, including setup coins and previously known Encaps keys. These observations must be reproduced in a reduction to the same honest-key recovery consequence.

Sections 10, 11 and 13 supply that analysis. The public-key emulators release each prefix and public refusal at the original time. A challenge public key is embedded among all reserved candidates before any selection, so aborted and unused attempts are charged by the factor $\kappa$. This avoids replacing a selected key by an unconditioned honest key. The Alice-facing emulation uses extracted openings for exact anchors. The Bob-facing emulation instead hides fresh anchors while preserving Bob's tapes, repeated values, known encapsulations and full component view. Its PRG hybrids charge latent successful anchors before output-dependent simulation. A stop before delivery therefore retains the correct prefix and its existing charge.

The resulting claim is the pair of recovery bounds for this staged ML-KEM service, under the observation, simulation and owner assumptions specified in the companion. It adds application-specific emulation and recovery arguments to the inherited input-consistency mechanism. XOR sharing supplies the joint entropy, and the 64-byte representation follows from ML-KEM. The contribution concerns how their staged exposure and historical use meet both recovery experiments. The endpoint and lifetime formulas below provide resource accounting for admitted representations.

**Relation to Self-Guarding.** Self-Guarding protects supplied next-message functions using trusted guarding and input-private verifiable computation [11]. Construction 4, Definition 17 and Theorem 3 provide reusable guarding under static malicious corruption and arbitrary substitution in the stand-alone model. In that model one party may be maliciously corrupted while the other party's core is substituted. It can protect an appropriate encoding of our historical service. Our protocol uses protected participating Bob and classical malicious-secure 2PC for a specified early-public service and its two recovery goals. Its theorem and trust model are those proved here. A cost or trust ranking requires instantiations of a common target.

### 19.3 Endpoint accounting for admitted implementation choices

The placement claim can be checked without selecting a particular commitment group or PRG. Fix the same gate basis, security targets, public-input treatment and explicitly composed endpoint circuits for each admitted profile. Count gates in the declared composition; cross-endpoint optimizations, if any, require a revised inventory. Let $C_0$ contain only common work, including the common final anchor construction. Let $K_j$ contain all remaining non-handoff gates of profile $j$, including its arithmetic, masks, routing, totalization and public specialization. Write $M_j$ and $V_j$ for the full costs of constructing and checking its handoff anchor. They include the message digest, opening processing, canonicality checks, commitment computation and comparison where present, with each gate charged exactly once.

**Proposition 2** (Endpoint cost of the selected placements)**.** *For the above composed circuits,*

$$\begin{aligned} G_D &= C_0 + K_D, \\ G_T &= C_0 + K_T + M_T + V_T, \\ G_L &= C_0 + K_L + M_L + V_L. \end{aligned} \tag{13}$$

*Consequently T has lower gate work than D precisely when*

$$M_T + V_T < K_D - K_T, \tag{14}$$

*and lower gate work than L precisely when*

$$K_T - K_L < (M_L + V_L) - (M_T + V_T). \tag{15}$$

*These comparisons preserve the original-state service only for placements that satisfy its full-domain completion and original-generation input conditions for every admitted request.*

*Proof.* D constructs a final anchor and has no generation handoff. T and L each add exactly one handoff construction and one actual-reinput verification. The final anchor work is common by hypothesis. Summing these disjoint inventories gives Eq. (13); subtraction gives Eqs. (14)–(15). Security admissibility is supplied by the sufficient-state argument, not by either cost inequality. □

**A directly testable consequence for the existing circuits.** The following specialization uses the selected padded ordinary-gate inventory, rather than assuming it for every implementation. Let $G_j^H$ denote its earlier digest-anchor circuit and $G_j^C$ the circuit obtained by replacing its anchor endpoints with a specified hiding, binding implementation. Let $e_F$ be the common final-anchor gate increment. Let $e_j^M, e_j^V$ be the signed increments for the handoff construction and verification, including replacement of the old comparison, new opening checks and any changed endpoint routing. Then

$$\begin{aligned} G_D^C &= G_D^H + e_F, \\ G_j^C &= G_j^H + e_F + e_j^M + e_j^V \quad (j = T, L). \end{aligned} \tag{16}$$

If other circuit pieces are changed, their increments must also be included. Using the selected H inventory of Sec. 24 gives

$$G_D^C - G_T^C = 31623249 - 2T_h - e_T^M - e_T^V, \tag{17}$$

$$G_L^C - G_T^C = 232959 + 2(L_h - T_h) + (e_L^M + e_L^V) - (e_T^M + e_T^V). \tag{18}$$

Here $T_h, L_h$ are the earlier single-endpoint handoff hash costs. If the underlying per-profile digest circuits are preserved in the H-to-C replacement and both cuts use the same fixed-length digest and identical post-digest commitment, opening-check and comparison circuits, with all width-specific routing already charged, then $e_T^M = e_L^M$ and $e_T^V = e_L^V$. Under these conditions their added commitment costs cancel in T/L. If domain, encoding or digest circuitry also changes, its signed increment remains in each endpoint cost and requires separate comparison. In this specified construction, $L_h \geq T_h$ therefore preserves T's lower total gate count. For D/T they do not cancel: T is cheaper exactly when $2T_h + e_T^M + e_T^V < 31623249$. The public-matrix specialization contributes to that selected base difference; the inequality does not attribute the entire benefit to the 64-byte boundary.

At a common LP cut-and-choose parameter $c$, multiplying by $4c$ preserves these table-ciphertext orderings. If profiles choose different parameters, compare $c_j G_j^C$ instead. Opening-input OT, rounds, peak memory and wall time remain separate coordinates. In particular, equal post-digest increments do not imply equal elapsed-time increments. The completed deterministic C38 inventory gives an L-minus-T generation AND gap of 2457600 for each of the six public fixtures. The C-minus-H AND increments for T and L are both 114048, preserving the earlier H AND difference. This cancellation concerns that gate basis only. The ordinary-gate inventory also includes the public XOR network and must be compared separately; no cancellation of all gate increments is inferred.

### 19.4 Separating staging from public-rho specialization

The generic-matrix control U retains T's two calls, 64-byte sufficient state, commitment domains, openings, output masks and public descriptor. Its second circuit expands the matrix from runtime $\rho$ wires, rather than folding the agreed public value into constants. Bob supplies this already-public string as a 256-bit holder copy. The circuit compares the copy with the immutable descriptor and combines that result with its acceptance flag. The same public bounded-sampling precheck applies to T and U; the generic circuit also retains its sampling-validity check. On accepted inputs their output bytes coincide. A changed holder copy has the specified failure output and cannot change the public parameters. This comparison changes the representation of public data, not the two private-key privacy goals or the authority over the stored descriptor.

Fix one emitted gate basis $\mathcal{B}$ and write $G_j^{\mathcal{B}}$ for the total generation gate count of $j \in \{D, U, T\}$ on the same public fixture. Let $E_\rho^{\mathcal{B}}$ be U's dedicated public-copy comparison and acceptance-combination cost in that basis. Define the signed net staging cost and the public-$\rho$ specialization saving by

$$E_{\text{stage}}^{\mathcal{B}} = G_U^{\mathcal{B}} - G_D^{\mathcal{B}} - E_\rho^{\mathcal{B}}, \tag{19}$$

$$S_{\text{pub}}^{\mathcal{B}} = G_U^{\mathcal{B}} - G_T^{\mathcal{B}} - E_\rho^{\mathcal{B}}. \tag{20}$$

The raw D/U difference is $E_{\text{stage}}^{\mathcal{B}} + E_\rho^{\mathcal{B}}$; the raw U/T difference is $S_{\text{pub}}^{\mathcal{B}} + E_\rho^{\mathcal{B}}$. Both corrected quantities remove the same dedicated check before applying the exact accounting identity

$$G_D^{\mathcal{B}} - G_T^{\mathcal{B}} = S_{\text{pub}}^{\mathcal{B}} - E_{\text{stage}}^{\mathcal{B}}. \tag{U39}$$

Consequently T saves gates over D exactly when $S^{\mathcal{B}}_{\text{pub}} > E^{\mathcal{B}}_{\text{stage}}$. The staging residual is signed; the identity imposes no nonnegativity assumption on it. $S^{\mathcal{B}}_{\text{pub}}$ includes bounded matrix expansion, products and all associated public-$\rho$ constant folding, including public-key serialization and hashing where the emitted circuit folds those fields. It is not an isolated matrix-product saving.

In the AND basis the dedicated equality uses 255 AND gates and combining its result adds one, so $E^{\text{AND}}_{\rho} = 256$. The holder copy also adds 256 input bits, whose OT cost is recorded separately from gates. The current six deterministic fixtures give $G^{\text{AND}}_{U} - G^{\text{AND}}_{D} = 230272$, hence $E^{\text{AND}}_{\text{stage}} = 230016$ in each fixture. The reproducible inventory records the complete per-fixture values in `data/matrix-cost-v39.json`. Apply the identities independently to the all-ordinary-gate basis using that basis's actual $E^{\mathcal{B}}_{\rho}$; the AND count cannot replace it. At a common LP parameter $c$, multiplying the ordinary-gate identity by $4c$ gives its table-ciphertext projection. Input OT and execution time require their own inventories and are excluded from that gate-based projection.

The native D/U and U/T contrasts time their complete implemented functions, including U's input binding and fresh preprocessing. A gate-count subtraction does not identify the seconds attributable to that check. Their measured differences therefore remain complete-function costs. T/L supplies the separate boundary-width comparison with public specialization and two calls held fixed. Together these controls separate the reasons for choosing a compact authenticated handoff from the larger public-specialization benefit, under the specified circuit implementation.

### 19.5 Lifetime retains a separate, measurable design choice

T and SR can use the same completed conservative T setup, so the retained representation changes without inventing a cheaper generation protocol. For a fixed digest length and an anchor construction with equal private opening size $u$ for both representations, Alice's required live records contain $1184 + u$ and $64 + u$ bytes before common metadata. SR saves 1120 bytes per key, a fraction $1120/(1184 + u)$ of T's record. The public anchor and spent history must be budgeted separately. An alternative commitment with another opening size changes these thresholds, even when the share payload still changes from 1184 to 64 bytes.

Let $N$ keys each complete $n$ uses over a declared horizon. With nonnegative prices $w_s, w_t, w_b$ for a stored private byte, a use second and a peer byte, let $\Delta t_C, \Delta b_C$ be SR-minus-T complete-use differences for the chosen C implementation and request mix. Common completed setup cancels, giving

$$\frac{J^{C}_{\text{SR}} - J^{C}_{T}}{N} = -1120w_s + n(w_t\Delta t_C + w_b\Delta b_C). \tag{21}$$

Thus positive per-use penalty gives a crossover at $1120w_s/(w_t\Delta t_C + w_b\Delta b_C)$; a nonpositive penalty gives no use-count crossover in this objective. Failed reservations add their actual execution costs. Old H timings do not determine the new $\Delta t_C, \Delta b_C$.

These formulas support the implementation choice within the proved service. They count authentication endpoints for a full-domain original-state representation and compare generation with retained-lifetime costs. The recovery analysis above supplies security admissibility, while the circuit and native inventories determine the resource tradeoff.

## 20 What the placement contributes

### 20.1 A public-key-emulatable specialization interface

The implementation exposes a useful ML-KEM-specific interface. The public seed $\rho$ both determines the matrix and is recoverable from the eventual public key. Its bounded-sampling

predicate is public. Consequently a public-key emulator can reproduce this early release and any public refusal without a generation seed or decapsulation key. The still-private $S = \sigma \parallel z$ suffices to regenerate the complete private core, including implicit rejection. T authenticates the actual share of this state consumed by its second call. These properties connect the location of public work, the state carried across calls and the two recovery experiments. An arbitrary secret-dependent early release need not have this interface.

Direct D is a valid realization of the same custody goals. It may expose its public key before protected activation and can then be interrupted. Thus public-key-dependent selection and the reservation factor $\kappa$ are common to D and T. T moves an observable point before matrix expansion, so a decision after that release may precede the second circuit's work. It also adds an externally retained generation handoff and its verification. The distinction concerns the observation and computation schedules; it does not require a new class of ASA available only against T.

Table 5: The two generation schedules under the same custody goals.

| Property | Direct D | Staged T |
|---|---|---|
| Public matrix work | The generic generation circuit computes it. | Public $\rho$ permits specialization before call two. |
| Observable stopping | Component prefixes and a public key may precede activation. | The first-call $\rho$ prefix additionally precedes matrix expansion. |
| Generation state | Final share and its anchor. | A 64-byte sufficient-state handoff, then the final share, each with its original anchor. |
| Continuation check | Later uses check the original final share. | Call two checks and consumes the original handoff share; later uses check the final share. |

### 20.2 An instance corollary for security and circuit cost

The following collects existing results for the selected placements. It is an instance corollary, not a new generic 2PC composition theorem. Let $R \in \{D, T\}$ denote the fixed placement, and retain the component, owner, output-delivery and stopping qualifications of Proposition 1. Component errors and reduction running times may depend on $R$ and on the fixed observer.

**Corollary 6** (Recovery and cost of the selected placement)**.** *For fixed budgets $\kappa \geq 1, \tau$, the two placements have fixed call and potential latent-anchor budgets*

| $R$ | $Q_R$ | $\overline{N}_R$ |
|---|---|---|
| $D$ | $\kappa + \tau$ | $\kappa$ |
| $T$ | $2\kappa + \tau$ | $2\kappa$. |

*For the C instance, $\sigma_{\text{bind}} = 128$. The number of freshly allocated successful anchors is at most $\overline{N}_R$. Uninvoked and functional-zero slots contribute zero hiding loss, while interrupted successful candidates retain their slot. For each separate recovery direction $j \in \{A, B\}$ and the cutoffs of the underlying theorems,*

$$p^R_{\text{rec},j} \leq \Delta^R_j + \kappa \left( \text{Adv}_{\text{KEM}}(D^R_{j,t_K}) + 2^{-\ell} + P^R_j / t_K \right), \tag{22}$$

*where*

$$\Delta_A^R = \sum_{i=1}^{Q_R} \epsilon_{\Pi,A,i}^R + \epsilon_H^R + \epsilon_p^R + 2^{-\sigma_{\mathrm{bind}}},$$
$$\Delta_B^R = \sum_{i=1}^{Q_R} \epsilon_{\Pi,B,i}^R + \Gamma_H^R(t_H),$$
$$\Gamma_H^R(t_H) = \sum_{a=1}^{\overline{N}_R} \left( \mathrm{Adv}_{\mathrm{PRG}}(D_{a,t_H}^{H,R}) + 2P_{H,a}^R/t_H \right).$$

*For an emitted circuit basis $\mathcal{B}$ on the same public fixture,*

$$G_D^{\mathcal{B}} - G_T^{\mathcal{B}} = S_{\mathrm{pub}}^{\mathcal{B}} - E_{\mathrm{stage}}^{\mathcal{B}}. \tag{23}$$

*Hence T reduces gate work precisely when its public-specialization saving exceeds its net staging cost. Both placements retain the two stated recovery guarantees under their respective error terms.*

*Proof.* Lemma 5 supplies the call budgets. Lemma 14 supplies the fixed potential-slot sum, including the two-world expected-time tails. Apply Proposition 1 once for each placement. Its public-key emulators account for each placement's actual release schedule; the reservation factor remains $\kappa$, and each view-transfer gap is paid once outside it. The component and expected-time errors keep their placement-specific values. Equation (U39), obtained from the D/U and U/T circuit contrasts, gives Eq. (23). The identity uses one gate basis, leaving time and security errors separate. □

The correct comparison therefore includes two kinds of overhead. T has an extra generation call and potential anchor per reservation. Its chosen circuits can nevertheless save work because public-matrix specialization removes more generic gates than staging introduces. The fixed public-fixture AND inventories give net staging 230016 ANDs and mean specialization 6384430.667 ANDs. The separate six-block native inventory gives the reported 48.3% net AND reduction. In the all-ordinary-gate basis used by the LP projection, means 59.715 and 32.597 million give a 45.4% net reduction. At common $c$, multiplication by $4c$ preserves that latter percentage for table ciphertexts. Input OT, rounds, memory and elapsed time require their own accounting.

### 20.3 Attribution of the technical result

The added capability is a concrete route to public-matrix specialization while retaining the two recovery targets for the resulting staged service. The cryptographic components supply their stated per-call properties. The service analysis establishes which ML-KEM state and public observation contract permit that placement, and the matched inventories measure its resource effect. This claim concerns the selected service and implementation; state-length minimality, a generic ASA compiler and a new fundamental reservation-guessing argument are outside its scope.

## 21 Concrete circuit and matched control evidence

The implementation evaluates the C38 stage functions, including public Toeplitz specialization, the SHAKE PRG, full 112-byte anchors, private openings and complete state checks. Alice is the garbler and Bob the evaluator. The adapter permutes external Bristol indices to preserve those roles; complementary shares and masks remain private inputs. Native shared output contains

Table 6: Established ingredients and the specified service contribution.

| Aspect | Established ingredient | What this work specifies, proves or measures |
|---|---|---|
| Joint custody | Joint entropy, XOR shares and malicious-secure 2PC. | Concrete ML-KEM shared-output and original-state service interfaces. |
| Across-call consistency | Committed-input computation and hiding/binding commitments. | Actual consumption of the authenticated sufficient state, retaining original implicit rejection. |
| Recovery | KEM security and component simulation. | The two complete-view public-key emulators, latent-anchor accounting and explicit placement budgets. |
| Resource choice | Public constant specialization and circuit inventories. | Matched D/U/T controls isolating handoff and public-specialization costs, with LP and native bases distinguished. |

public metadata and the success-gated masked Alice component. The ordinary functions also define the corrected LP realization, whose security and cost assumptions remain separate.

The frozen v20 EMP engine transmits 80-bit authentication tags, enables COT checking and uses unbiased shuffle, full-pool initialization and the fail-closed/padding repairs. Its fpre_threads = 1 creates a two-worker pool plus main thread; measurement adds a sampler. Triple batches have even size at least 320, with buckets 3 at size at least 280000, 4 at size at least 3100, and 5 below that threshold. The new control adapter preserves all 350 pinned backend sources. These source parameters and finite qualification checks describe the measured configuration. Corrected LP supplies the written simulation realization; EMP measurements establish functional and resource evidence under the stated component assumptions.

### 21.1 A generic-matrix control with the same staged relation

The new U circuit preserves T's first stage, 64-byte state, private openings, original domain tags, descriptors, final key, anchor relation and output adapter. Its second stage receives a Bob-held 32-byte runtime copy of the already public $\rho$ and checks every bit against the immutable public value. The matrix sampler and matrix products then operate on generic wires. Public bounded-sampling refusal occurs at the same stage boundary as T; U additionally gates its result on the generic sampler's validity wire. Its explicit public-copy check contributes $E_\rho$ and 256 input bits, which are reported separately.

Thus D/U compares staging, authenticated handoff and public-input binding under generic matrix treatment. U/T compares public-$\rho$ specialization under the same two-stage relation. This contrast includes matrix sampling, constant products and associated public-field folding, such as $\rho$ inside public-key hashing. Subtracting $E_\rho$ removes the dedicated runtime binding check; it does not turn the residual into a claim about only matrix multiplication. Section 19.4 defines the signed residuals in each declared gate basis.

### 21.2 Clear qualification and deterministic inventories

The new checker evaluates six public deterministic cases through D/U/T, compares their exact complete keys against OpenSSL 3.6.3, independently checks the commitment outputs and tests 36 changed-share, opening and public-$\rho$ cases. The retained C38 checker separately supplies six

full-key cases, 24 valid/invalid complete Decaps cases and 21 negative controls. Its independent SHAKE/Toeplitz known-answer vector checks byte packing. These finite checks establish functional agreement and targeted refusal behavior; the recovery reductions supply the security arguments.

A deterministic inventory records AND, XOR, inversion, ordinary gates, input bits, output bits and wires for six public fixtures. It includes all commitment endpoints and the U public-copy check. Two raw replays must be byte-identical. Public matrix operations are separated from associated public-field folding where the circuit instrumentation permits. LP projections use the full ordinary-gate inventory, while input OT and actual LP runtime require separate cost models and independent execution measurements.

### 21.3 Retained v39 six-block generation cohort

The retained v39 generation-only cohort has six independent seed blocks and covers all D/U/T execution orders. Within each block the role-only seed fixtures are replayed to obtain matched complete keys; masks, openings, preprocessing and workers are fresh for each profile. An out-of-protocol measurement controller prepares paired fixtures and performs oracle checks; each worker receives only its role's private frame. Temporary private fixtures are deleted after the block. U and T persist the same final and seed records, so their durable owner boundary also agrees. The earlier T/L and T/SR data remain separate cohorts and are never subtracted from these new timing observations.

Each request includes durable reservation, worker launch, construction, conversion, fresh preprocessing, termination and durable activation. Compilation, fixture preparation, independent oracle checks and recorded post-run trace analysis are untimed. Native input/MAC/pool-tail probes and U success, changed-share, changed-opening and changed-public-copy guards must pass before formal sampling. Qualification-only attempts are excluded and identified in their own raw directories. The complete six-block cohort is required before analysis; a failed reserved block cannot be silently replaced or resumed into the cohort.

Memory uses a requested 100-ms common host clock for both workers. Footprint includes compressed charges, shared RSS is not deduplicated, and actual-PID `wait4` records each worker's CPU and lifetime peak RSS. Host-wide paging describes the declared host and cannot identify a causal paging fraction. Intervals use 10000 paired bootstrap resamples of six independent key blocks, preserving their within-block dependence. The source package contains raw receipts, source manifests and the commands needed for independent offline reconciliation of the complete native cohort.

### 21.4 Exact controlled circuit decomposition

Six deterministic public fixtures give exactly 12,761,465 ANDs for D and 12,991,737 for U. The raw increase is 230,272 ANDs (1.8044%). The public-copy equality and its transfer-valid combination use 256 ANDs; the corrected staging/handoff residual is 230,016 ANDs. Its 256 additional public runtime input bits remain separate input-OT work.

| Gate basis | D | U | T |
|---|---|---|---|
| AND | 12,761,465.00 | 12,991,737.00 | 6,607,050.33 |
| XOR | 38,697,697.00 | 39,845,766.00 | 21,390,154.17 |
| Inversion | 8,256,016.00 | 8,491,144.67 | 4,600,254.00 |
| All ordinary | 59,715,178.00 | 61,328,647.67 | 32,597,458.50 |

Values in the table are descriptive means over six fixed public fixtures, rather than confidence intervals. Every generated ordinary gate is counted; public setup coefficients fold into XOR networks, and full private openings and commitment checks remain in each stage.

| Corrected contrast/component | ANDs | Ordinary gates |
|---|---|---|
| Public-copy binding | 256.00 | 638.33 |
| Net staging/handoff | 230,016.00 | 1,612,831.33 |
| Matrix sampler removed | 3,977,000.00 | 19,882,205.00 |
| Public product specialization | 2,407,137.67 | 8,844,832.83 |
| Associated fields/validity | 293.00 | 3,513.00 |
| Total public-rho specialization | 6,384,430.67 | 28,730,550.83 |

The matrix sampler and product sections have separate inventory-only scopes. The remaining signed residual includes public-field constant folding and generic sampler-valid output gating. Their exact sum gives the total public-rho specialization contrast after removing the dedicated binding check. These identities hold independently in each gate basis and do not assign native seconds to individual source sections or individual circuit gates.

At common $c = 532$, the theoretical LP generation table-ciphertext projections $4cG$ for D/U/T are 127.074/130.507/69.367 billion. This is a circuit-size projection, leaving input OT, other checks, rounds, memory and actual LP execution to their separate cost models.

### 21.5 Retained v39 generation measurements

| Metric | D | U (generic) | T (public) |
|---|---|---|---|
| ANDs (M) | 12.76 | 12.99 | 6.60 |
| Traffic (GiB) | 6.21 | 6.34 | 3.23 |
| KeyGen (s) | 140.65 | 138.57 | 14.46 |
| 95% interval | [133.16,148.13] | [121.93,153.92] | [13.82,15.18] |
| Worker CPU (s) | 60.78 | 62.33 | 18.75 |
| Footprint (GiB) | 17.54 | 17.92 | 10.99 |
| Largest worker RSS (GiB) | 4.94 | 5.02 | 4.56 |

The paired U-minus-D latency is -2.08 s with 95% block-bootstrap interval [-17.92,10.98]; U-minus-T is 124.11 s with interval [107.97,138.81]. The contrast times the complete implemented functions, including runtime input binding and durable record handling. It therefore measures the operational choice, while the deterministic component inventory supplies its gate decomposition.

The D/T ratio is 9.73 for wall time and 3.24 for summed actual-worker CPU. These ratios describe this six-block cohort on the declared 16-GiB host. Paging and compression observations are environmental evidence; they do not determine a causal fraction of the time difference.

All six formal blocks pass the independent raw/source audit. Their exact profile sequence covers all six permutations, every block has three successful requests, and no failed formal attempt is substituted. Source pins, qualification hashes, worker-role circuit counts, traffic and timing exclusions are reconciled offline. Online oracle/full-core receipts remain identified as receipts because the private fixtures were deliberately deleted after checking.

### 21.6 What the latency gap measures

The following diagnostic reanalyses the complete v39 16-GiB cohort. It adds no native observations and is separate from a replication on a memory-sufficient host. The replay script verifies the original 117 source/raw hashes, request timing deductions, actual worker PIDs, phase totals and aligned memory samples before computing the results.

| Measure | D | T |
|---|---|---|
| Request wall time (s) | 140.653 | 14.463 |
| Summed worker CPU (s) | 60.783 | 18.754 |
| Dependent-phase wall (s) | 75.974 | 9.216 |
| Dependent-phase CPU (s) | 21.034 | 8.057 |
| Aligned footprint peak (GiB) | 17.539 | 10.988 |
| Aligned compressed-charge peak (GiB) | 15.253 | 8.293 |

Ratios of D/T means are 1.933 for ANDs, 1.921 for peer traffic, 3.241 for summed worker CPU and 9.725 for wall time. Paired block bootstrap intervals for the last two ratios are [3.138,3.326] and [9.147,10.370]. Resampling preserves all observations within the same six independent key blocks. The CPU total sums two concurrent workers and their threads, whereas wall time follows the request boundary. Their difference is not a measured waiting interval or a paging-time estimate. The phase observations locate much of the elapsed gap in the backend's independent and dependent phases, which include communication, memory activity and scheduling as well as gate processing.

Both profiles show substantial compressed-memory charges. T therefore does not provide an unpressured control. Across the six whole D/U/T blocks, host VM counters record 481.092 GiB of swap-ins and 603.090 GiB of swap-outs. These intervals also include U, controller/oracle work and unrelated host activity, so they cannot assign swap traffic or delay to D or T. Worker major-fault counts are similar and do not measure compression or time blocked on swap I/O. The diagnostics establish workload reduction and the observed host behaviour. They leave the causal paging fraction and the memory-sufficient D/T latency ratio undetermined.

### 21.7 A separate memory-sufficient replication

The supplied replication harness keeps the frozen C38 stage functions, 350 backend source pins, 80-bit tags, COT checks, shuffle, full ML-KEM-768 workload and original request timing. Its preregistered design uses six new independent seed blocks, with three D/T and three T/D orders. The two profiles in a block generate exactly matching complete keys while using fresh masks, openings, preprocessing and worker processes. It records host VM counters around every request and continuously during the cohort, in addition to the original worker CPU, memory and phase records. Capacity and pressure criteria are fixed before sampling. All attempts and failures remain archived; a contaminated cohort cannot be silently filtered into a memory-sufficient result.

The capacity target is a single macOS arm64 host with at least 32 GiB physical memory and sufficient idle headroom. These are provisioning conditions, followed by checks of actual swap and compression activity. The current 16-GiB host fails the capacity gate before worker launch. No memory-sufficient timing result is included in this version. A replication on a different CPU or operating system must retain that additional host difference when interpreting any contrast with the original cohort.

### 21.8 Existing-data order and deletion sensitivity

This analysis uses exactly the six independent seed/key blocks and 18 D/U/T KeyGens already measured in v39 on the 16-GiB Apple M5 host. It adds zero native observations and zero complete uses. Each complete block retains its matched D/U/T keys and fresh preprocessing, masks, openings and workers. All six execution permutations occur once; thus each profile is observed twice at positions 1, 2 and 3. These are descriptive comparisons: with one block per permutation, seed, temporal host state and order cannot be isolated as causal effects.

The analyzer verifies all 117 raw-file SHA-256 pins in the v40 latency diagnostic and its source hash, independently reconciles timing deductions, actual worker PID/CPU records, two-way peer traffic, stage gate counts and aligned sampled memory peaks, then cross-checks all 18 diagnostic rows. Full hashes and exact unrounded numbers are in `data/sensitivity-v41.json`; the chronological and position ledgers are also supplied as CSV. Floating-point reductions use `math.fsum` in explicit block/role order. The replay accepts an alternate output directory and emits deterministic bytes, without executing a benchmark or accessing private fixtures.

The statistical unit is an independent block, not a profile request, worker, phase, memory sample, resample or repeated analysis. The 10000 deterministic paired bootstrap draws preserve each full D/U/T triple and add no observations. Their percentile intervals are exploratory and conditional on this small host/cohort; they do not validate a population model, a causal order effect, or equivalence of D and U. The six leave-one-block estimates use five blocks each and are not six independent replications. Early/late summaries describe chronology without identifying a drift mechanism.

**Complete block ledger.** Blocks follow measured time order, with wall and summed-worker CPU values in D/U/T order and all request durations expressed in seconds.

| Block | Order | Wall D/U/T | CPU D/U/T | U−D | D/T |
|---|---|---|---|---|---|
| 1 | DUT | 146.045 / 147.045 / 15.022 | 63.986 / 63.194 / 19.262 | +1.000 | 9.722 |
| 2 | UTD | 146.295 / 158.311 / 16.051 | 62.823 / 64.553 / 19.568 | +12.016 | 9.115 |
| 3 | TDU | 149.110 / 163.711 / 14.256 | 62.574 / 74.293 / 18.579 | +14.601 | 10.460 |
| 4 | TUD | 149.151 / 113.454 / 13.464 | 62.564 / 54.026 / 18.817 | -35.697 | 11.078 |
| 5 | UDT | 126.187 / 137.833 / 14.336 | 57.812 / 61.576 / 17.892 | +11.646 | 8.802 |
| 6 | DTU | 127.129 / 111.076 / 13.649 | 54.939 / 56.321 / 18.409 | -16.053 | 9.314 |

The UTC block starts are 16:30:12, 16:35:20, 16:40:41, 16:46:08, 16:50:44, 16:55:23 on 21 October 2026; the last block ends at 16:59:35 UTC.

**Execution-position descriptions.** Each cell contains only two original observations. Ranges are minima and maxima, not intervals for an order effect.

| Profile | Pos. | Blocks | Wall mean | Wall range | CPU mean | CPU range |
|---|---|---|---|---|---|---|
| D | 1 | 1,6 | 136.587 | [127.129,146.045] | 59.462 | [54.939,63.986] |
| D | 2 | 3,5 | 137.648 | [126.187,149.110] | 60.193 | [57.812,62.574] |
| D | 3 | 2,4 | 147.723 | [146.295,149.151] | 62.693 | [62.564,62.823] |
| U | 1 | 2,5 | 148.072 | [137.833,158.311] | 63.064 | [61.576,64.553] |
| U | 2 | 1,4 | 130.250 | [113.454,147.045] | 58.610 | [54.026,63.194] |
| U | 3 | 3,6 | 137.394 | [111.076,163.711] | 65.307 | [56.321,74.293] |
| T | 1 | 3,4 | 13.860 | [13.464,14.256] | 18.698 | [18.579,18.817] |
| T | 2 | 2,6 | 14.850 | [13.649,16.051] | 18.989 | [18.409,19.568] |
| T | 3 | 1,5 | 14.679 | [14.336,15.022] | 18.577 | [17.892,19.262] |

**All leave-one-block analyses.** Every deletion retains the five complete D/U/T triples. Ratios are ratios of arithmetic means, not means of per-block ratios.

| Omitted | $n$ | U−D wall | D/T wall | D/T CPU | D/T bytes | D/T ANDs |
|---|---|---|---|---|---|---|
| None | 6 | -2.081 | 9.725 | 3.241 | 1.920952 | 1.932529 |
| 1 | 5 | -2.697 | 9.726 | 3.224 | 1.920743 | 1.932318 |
| 2 | 5 | -4.901 | 9.864 | 3.247 | 1.921135 | 1.932715 |
| 3 | 5 | -5.418 | 9.581 | 3.216 | 1.920883 | 1.932459 |
| 4 | 5 | +4.642 | 9.477 | 3.224 | 1.920748 | 1.932323 |
| 5 | 5 | -4.826 | 9.908 | 3.243 | 1.921266 | 1.932847 |
| 6 | 5 | +0.713 | 9.802 | 3.291 | 1.920936 | 1.932512 |

The full U−D wall difference is -2.081092 s; its exploratory paired-block interval is [-17.923616,10.980100] s. The six leave-one-block means range from -5.417588 to +4.642020 s, crossing zero. This cohort does not establish the sign of a U-versus-D wall advantage. The D/T wall ratio of means is 9.725111, with leave-one-block range [9.476639,9.907782] and exploratory interval [9.146863,10.369999]. The observed D/T separation persists under every single deletion; this is robustness within the recorded host/cohort.

**Early/late observations, without drift attribution.** Blocks 1–3 versus 4–6 are a descriptive chronological split. The early and late wall means (seconds) are D: 147.150 / 134.156; U: 156.356 / 120.788; T: 15.110 / 13.816. The corresponding U−D means are +9.206 / -13.368 s and D/T ratios are 9.739 / 9.710. Order, seeds and changing host state are not separately identified. No causal time or order coefficient is estimated.

**Unchanged timing and exclusion boundaries.** The request boundary is durable reservation through worker termination and durable activation. It includes launch, construction, conversion and fresh preprocessing. Compilation and fixture preparation precede the timed request; the archived OpenSSL oracle and post-run profile analysis intervals are explicitly deducted and reconciled for every request. Qualification-only records, the four U pilot/guard attempts, component probes, clear-function checks, earlier generation cohorts and earlier complete-use cohorts are excluded from these timing summaries. No measured block or request in the six-block cohort is filtered out; each deletion is an explicitly labelled sensitivity analysis. Temporary private fixtures were removed by the original cohort and are neither restored nor archived by this analysis.

Summed CPU adds two concurrent workers and their threads; subtracting it from wall time does not measure waiting or paging time. Aligned footprint and compressed charges remain sampled process-accounting observations, with controller work excluded and shared RSS not deduplicated. Host swap counters surround whole D/U/T blocks, include oracle/controller and unrelated activity, and cannot allocate traffic or delay to a profile. Both D and T showed compression; no memory-sufficient control, high-memory run, cross-host timing result, causal paging fraction, LP runtime measurement, or native security certificate is added or inferred here.

### 21.9 Retained C38 lifetime and boundary cohort

The following tables retain the earlier complete C38 cohort: six independent blocks, 18 D/T/L KeyGens and 24 T/SR complete uses. Their sampling, source and receipt hashes are unchanged. D/T in this cohort combines staging and specialization; the new D/U/T control above supplies the decomposition. T/L fixes specialization and compares the authenticated boundary. Each valid/invalid use pair shares its independently generated key and retains fresh masks, preprocessing and workers. These use measurements support the expanded/seed lifetime comparison, rather than a new generation-cohort timing estimate.

**Ordinary-gate cost bridge.** The independently generated inventory counts ordinary Boolean gates $G = \#\mathrm{AND} + \#\mathrm{XOR} + \#\mathrm{INV}$ after the specified public-constant folding. Component counters are inclusive and overlapping; only complete stage totals are summed. Six deterministic public fixtures give descriptive means and ranges, rather than independent cryptographic or timing samples. The C generation means are 59.705, 32.571 and 44.973 million ordinary gates for D, T and L. Complete expanded and retained use average 46.769 and 50.992 million ordinary gates under an explicitly declared equal valid/invalid mix. For the illustrative reservation budget $(\kappa, \tau) = (1, 12)$ and $\lambda = 128$, $Q_R = d_R\kappa + \tau$ is 13 for D and 14 for the two-call generation profiles, including SR.

Thus $c = 4(128 + 4 + 1) = 532$ suffices for the stated cut-and-choose subtotal. Table 7 applies $4cG_{\text{life}}$ to $G_{\text{life}} = \kappa G_{\text{KG}} + \tau G_{\text{use}}$, with six repetitions of each counted valid/invalid public-fixture circuit. It counts projected garbled-table ciphertexts, not native traffic bytes, measured wall time or a total LP implementation cost. Input OT, consistency checks, subprotocol setup and owner costs are excluded, and no native EMP timing is scaled into an LP runtime. These are conditional projections for the six public circuit fixtures, not universal per-ciphertext resource bounds.

Table 7: Illustrative corrected-LP table-ciphertext projection from all ordinary gates. The complete reservation budget and the six-public-fixture projection scope are stated explicitly.

| Profile | $G_{\text{KG}}$ (M) | $G_{\text{use}}$ (M) | $Q_R$ | $4cG_{\text{life}}$ ($10^{12}$) |
|---|---|---|---|---|
| D | 59.705 | 46.769 | 13 | 1.321 |
| T | 32.571 | 46.769 | 14 | 1.264 |
| L | 44.973 | 46.769 | 14 | 1.290 |
| SR | 32.571 | 50.992 | 14 | 1.371 |

**Native timing and memory scope.** Each native group has six independent seed/key blocks. Valid/invalid requests share their block's generated key and remain paired in all resampling. Actual worker peaks come from parent-side Darwin `wait4` on the recorded worker PIDs, with no wrapper process. Self-sampling requests 100 ms; the largest observed gap and cross-worker alignment age are 0.251 and 0.251 s. Aggregate footprint includes compressed charges, and summed RSS double-counts shared mappings; neither is exclusive physical RAM. Host-wide paging deltas bracket whole blocks, including small outside-timer oracle work, and are not attributed solely to these workers. The adjusted reservation-to-durable-completion latency excludes the individually recorded external oracle and post-run profile-analysis intervals.

**Retained records.** Alice's secret share-plus-opening payload is 1200/80 bytes for T/SR; Bob's corresponding secret shares use 1184/64 bytes. The versioned test-controller files also contain public descriptors and encoding metadata: measured Alice files are 4421/3302 bytes and Bob files 4419/3300 bytes. These byte counts distinguish the 1120-byte secret-payload saving from the full serialized-file saving under the versioned controller metadata encoding.

## Guide to the retained hash-anchor cost appendices

The following appendices retain the H construction's inventories and measurements for reproducibility. Their numerical gate differences, byte counts and timings exclude the new PRG commitments and opening inputs. The C38 ordinary-gate inventory above supplies its own concrete LP table-count projection. The older gate figures below are retained as historical H references. The semantic sufficient-state conditions apply to C only after adding the opening to each private state and the new binding premise. The present-tense protocol and anchor assumptions in these retained appendices refer to H. Their historical SR regeneration map $G(S) = X$, denoted Regen in the C38 discussion and distinct from the commitment PRG, also supplies the functional completion for C; C integrity and Bob full-view privacy use the new commitment and evaluator-view proofs above.

## 22 LP cost model and its relation to the EMP measurements

**Two levels of cost evidence.** The security construction uses the corrected full Lindell–Pinkas (LP) protocol [3], Protocol 4.1 with its specified subprotocols. For a circuit of the *mathematical* stage function, let $G_i$ denote the number of ordinary Boolean gates after choosing an explicit binary truth-table representation, $\ell_i$ the equal padded private-input length of each party, and $o_i$ its fixed evaluator-output length. The formulae below apply to these $G_i$. The EMP experiments instead provide frozen *reference-stage inventories* $G_i^{\text{shape}}$. Substitution of those inventories into the formulae is a conditional workload projection; it is not a proof that the native compiler or its generated functions refine every mathematical stage on every input. In particular, the theoretical totalization of malformed L states and parameterized hash families must be included in the chosen mathematical circuit. We do not silently set their circuit-cost difference to zero. A realization can use the formulae with its actual $G_i$; the numerical rows below explicitly use $G_i^{\text{shape}}$.

**Counting all gates.** The retained worker records `c.gates` before conversion to the EMP format. This counter includes every emitted AND, XOR and INV, including output-routing XOR gates; known public constants are folded before gate emission. Thus it is not the AND-only count used to discuss EMP work. The dedicated routing gate for each output also satisfies Appendix A's terminal-output-wire convention; it is already counted and is not added again. Each emitted AND/XOR can be represented by its ordinary two-input truth table; an INV is a two-input function that ignores its second argument. This elementary topology translation retains one gate per emitted operation and permits the conservative four-row LP representation. It does not invoke free XOR, half gates, EMP authentication, or an optimized garbling theorem. The small variations across the six paired public instances are retained rather than replaced by an AND-based estimate.

**Component formulae and finite parameters.** Use a common $c = 532$ for every call in the illustrative $(\lambda, \kappa, \tau) = (128, 1, 12)$ comparison. This covers the cut-and-choose subtotal for every profile's call budget $Q_R \leq 14$; it is not a claim that the *total* error is at most $2^{-128}$. Set the Appendix B aggregation parameter $L = \lambda$, choose a group of order greater than $2^L$ satisfying the stated assumptions, and scale encryption, knowledge and extractor parameters. For each call, LP Section 4.3 gives the following ordinary gate-table component:

$$N_{\text{table},i} = 4cG_i, \qquad B_{\text{table},i} = 4cG_i\,t, \qquad r_{\text{core},i} = 12, \tag{24}$$

where $B$ is in *bits* and $t$ is the bit length of one complete *nested* table ciphertext. The symbol $t$ is not a wire-label length. The published $6.5cG_i$ symmetric-operation accounting units are likewise not measured time or a count of single PRF calls. For example, the literal Appendix A construction uses two encryptions per row. Constructing $c$ circuits, decrypting all four rows in the $c/2$ check circuits, and trying all four rows in the $c/2$ evaluation circuits takes at most $16cG_i$ primitive encryption/decryption calls. PRF expansion, output decoding and all public-key operations must also be included when accounting for the full protocol.

For the Appendix A illustrative encryption $E_k(m) = (r, F_k(r) \oplus (m\|0^z))$, with an $r$-bit nonce, $z$ range-check bits and a $k$-bit label, a nested table row has $t = k + 2(r + z)$ bits. Setting all three lengths to 128 gives $t = 640$ bits, or 80 bytes, only as a serialization example. It neither fixes a DDH group nor establishes a concrete 128-bit bound for the whole service. Under this example, the table-only means are 9.181/4.848/6.769 TiB for D/T/L, respectively; these are analytical projections, exclude other communication, and are not measured LP traffic.

**Input and public-key costs are not free.** Let $w$ bound the encoding length of the group elements, field scalars and keys included in LP's group-element accounting convention. The reported Section 4.3 input/proof component has $(7c\ell_i + 22\ell_i + 7c + 5)$ such accounting units. A faithful bit model must also include aggregation challenges, output tables, the extractor seed, the set $J$, application descriptors, the authenticated relay, and channel framing. In particular, Protocol B.4 is called $\ell_i$ times with $\eta = c$ in batch CCOT and $\ell_i$ times with $\eta = c/2$ in input consistency. These calls alone contribute exactly

$$N_{\text{short},i} = 6c\ell_i \quad L\text{-bit exponentiations}, \qquad B_{\gamma,i} = \tfrac{3}{2}c\ell_i L \quad \text{bits}. \tag{25}$$

There are also $2\ell_i$ one-out-of-two tuple proofs and the batch setup proof. This protocol-step vector, together with their specified group, replaces the published conversion of 40-bit short exponentiations into 160-bit exponentiation equivalents. We do not reuse that fixed-parameter conversion after changing $L$. Consequently this group-unit expression alone is neither a complete accounting of the transmitted bits nor a model of the protocol's running time.

**The output adapter and equal input lengths.** The reference inventories already contain the accepted-output masking $E = o_A \oplus P$: Alice's private pad $P$ is explicitly part of her input and its XOR gates have already been counted. Bob's retained share mask is likewise already an input and need not be duplicated in the evaluator output. Thus this mapping needs neither a second pad nor a second set of pad-XOR gates. A circuit of the exact mathematical refusal semantics may still change other gates; its cost remains in $G_i$ rather than being identified with $G_i^{\text{shape}}$. The equal private-input lengths per party are D: 9984; T: $(1024, 9984)$; L: $(19200, 28160)$; and CheckedUse: 9728 bits. No further dummy padding is needed for these interfaces. Bob relays only $E$ and intended public metadata, never his retained private mask. Relay lengths including these public fields are D: 2433; T: $(160, 2433)$; L: $(2432, 2433)$; and Use: 33 bytes. Relays are additional to the 12 core LP message rounds of each call; descriptor agreement and owner synchronization are additional as well. For T versus D, the split adds 160 relay bytes. The per-party extra input is $11008 - 9984 = 1024$ bits (128 bytes), so the total over both parties is 256 bytes. For L versus D, the total extra input over both parties is 9344 bytes and the relay adds 2432 bytes.

**Projection results and retained paired data.** Table 8 averages the six retained paired KeyGen inventories and applies Eq. (24). It does not scale measured EMP bytes by $c$. The gate-table component is smaller for T than D in every block; the ratio of mean totals gives a 47.19% reduction. L adds 39.61% relative to T. The largest stage falls from 30.738 to 22.676 million ordinary gates between T and L, but this predicts neither LP resident memory nor paging. LP can choose different storage/streaming strategies and its labels have different sizes.

For repeated use the unchanged CheckedUse computation matters. The three retained valid-use pilot shapes contain 46.394 million ordinary gates and 9.844 million ANDs, with only a few output/context-dependent gate differences. The analytical workload of one mean KeyGen plus 12 copies of its retained valid-use shape has 1310.890/1251.349/1277.741 billion gate-table entries for D/T/L. This is a quota calculation, not 12 newly measured uses. Thus the T/D gate-table benefit is about 4.54% for that workload; the KeyGen-only percentage must not be extrapolated to the whole service budget.

The portable part of the design is the mathematical split point, stage count, party-local input/share/pad widths, and the ability to specialize the public matrix computation. Under a *common chosen reference topology*, $4c\sum_i G_i$ directly preserves the corresponding gate-table ratios. The emitted topology and concrete constant folding, EMP's AND-only accounting, role as-

| Reference-inventory projection | D | T | L |
|---|---|---|---|
| All gates (M) | 59.294 | 31.313 | 43.716 |
| Largest stage (M) | 59.294 | 30.738 | 22.676 |
| Gate-table entries ($10^9$) | 126.177 | 66.635 | 93.028 |
| $\sum_i \ell_i$ (bits/party) | 9984 | 11008 | 47360 |
| LP core rounds | 12 | 24 | 24 |
| Relay incl. public fields (B) | 2433 | 2593 | 4865 |

Table 8: LP workload projection from six frozen reference-stage inventories, with $c = 532$. Entries are complete table ciphertexts, not bytes. The numerical inventories are not certified circuits of the parameterized mathematical functionality.

signment, authentication and OT implementation, measured peer bytes, elapsed time and memory remain backend-specific. In particular, the 10.05-fold EMP KeyGen wall-time ratio is not an LP speedup. This cost bridge supplies a reproducible analytical comparator without equating the two implementations or turning the EMP runs into a security certificate.

**Reproduction.** Run `python3 lp_cost_bridge_v25.py` from the source package. The standard-library-only script verifies 48 frozen source/trace hashes, checks Alice/Bob stage-count agreement, retains all six paired rows and their exact rational differences, and writes `data/lp-cost-v25.json` and the displayed table. The JSON lists all units, parameter choices, full symbolic terms, and the original source hashes. It runs no benchmark and changes no historical raw data.

# 23 Authenticated seed retention: an alternative with the same goal

The threat model alone does not force 2PC. Protected Bob can retain the complete generation seed or expanded key, authenticate the original handle in his protected database, and execute native decapsulation. Against malicious Alice this can preserve both confidentiality and the original-key relation. It reconstructs the key and decapsulated secret in Bob's prescribed worker. Consequently it does not meet the additional operational split-custody policy used in this paper. That policy, rather than a malicious-Bob theorem, motivates two-party evaluation.

There is also a smaller-state alternative that preserves the policy. This section defines it explicitly and compares its complete use circuit with the expanded-share interface. The 64-byte state is not claimed to be novel or minimal. The comparison identifies a storage/online-computation choice, rather than an unconditional advantage of materializing expanded shares.

## 23.1 Retained shared-seed protocol SR

Use the unmodified two-call T KeyGen, including its original transfer and final anchors, public refusal policy and protected final-completion rule. Retain the *original* $(A_S, B_S, D_S, \rho)$ after successful completion, with $S = \sigma \parallel z = A_S \oplus B_S$. Both prescribed workers may discard their expanded core shares after activation; Bob no longer needs the final full-share anchor $C$. The immutable original $ek$, generation context and original $D_S$ remain authoritative. A malicious Alice may retain every previous output; security never assumes that she erases the original expanded share.

For each reserved use, authenticate Alice's actual 64-byte reinput $x_S$ under the original $D_S$, in the same 2PC invocation that consumes it. Let

$$G(\sigma \parallel z) = \mathsf{Encode}_{12}(\mathsf{NTT}(s_0), \mathsf{NTT}(s_1), \mathsf{NTT}(s_2)) \parallel z, \quad s_j = \mathsf{CBD}_2(\mathsf{PRF}_\sigma(j)).$$

Here the PRF and sampling are precisely the ML-KEM-768 secret-noise computation. The function returns

$$U = \mathsf{TotalDecaps}(G(x_S \oplus B_S), ek, c) \oplus R,$$

with fresh uniform Bob mask $R$, an independent Alice output pad, and the same public authentication flag and zeroed failure payload as CheckedUse. Ciphertext rejection remains internal. Restore retains $D_S$ and never freshly enrolls $x_S$. The descriptor binds the original context, $\rho$, $ek$, operation and fresh SID. All reservation, worker-reaping and nonrollback owner rules remain unchanged. A use never updates the long-lived share or its anchor.

Regeneration does *not* repeat error-vector sampling or public matrix products: the original $ek$ is retained, while decapsulation needs only $\hat{s}$ and $z$. This distinction materially lowers the cost of an honest seed-retention baseline. The original T KeyGen is a conservative common setup cost; an optimized SR setup could omit the subsequently discarded expanded anchor/share output, but no time or saving for such an optimization is claimed.

**Lemma 17** (Original-key integrity of retained seed)**.** *Under the same stage-security, hash, owner and static-Alice assumptions as the framework theorem, SR realizes the corresponding exact-origin service with the same output correctness goal. Its fixed reservation budget is* $Q_{\mathrm{SR}} = 2\kappa + \tau$. *The ideal interface retains the original seed shares instead of requiring an expanded share at use.*

*Proof.* The first two calls are the existing T KeyGen. In a no-collision execution, acceptance of the actual $x_S$ against the retained original $D_S$ implies $x_S = A_S$. The same fixed $B_S$ therefore yields the originally generated $S$. By substitution in ML-KEM KeyGen, $G(S) = X$ for the original private core: its three secret-noise polynomials and $z$ are unchanged. The retained $ek$ is the one authenticated by T's protected final completion. Thus the delivered shares satisfy $U \oplus R = \mathsf{TotalDecaps}(X, ek, c)$, including implicit rejection. Any changed accepted seed share supplies the transfer-domain collision already covered by the global collision finder. Fresh masks/pads and independent intermediate and expanded share masks permit the same joint-view coupling, even if Alice retains both old shares. The forward local-simulation lemma applies to the same $2\kappa + \tau$ sequential bounded slots. Owner reservation induction and descriptor-collision accounting are unchanged. The conclusion equates the authorized application behavior; it neither identifies the persistent-share representations nor certifies the native EMP implementation of either protocol. □

### 23.2 Exact native-reference circuit comparison

The standalone probe `seed_alternative_v26.py` reuses the frozen reference circuit builder and complete Decaps body. Its generated adapter only replaces the expanded-input decoding/XOR wrapper by supplied core wires; it does not change re-encryption, comparison or implicit rejection. The expanded function follows the measured worker's CheckedUse construction. Both functions retain its private canonical-core guard and failure gating, use the same public key/ciphertext, and receive matched fresh use masks and output pads. The retained-seed guard is conservative because $G$ already produces canonical coefficients; no minimality claim is made.

Three fresh keys, each with a valid and a one-bit-mutated ciphertext, give six complete clear-circuit comparisons with OpenSSL. Both prescribed output share pairs match complete decapsulation. Three seed-share mutations (in $\sigma$, $z$ and the last byte) and one expanded-share $z$

mutation produce zeroed refusal outputs. This establishes these test instances and gate inventories, not backend malicious security or an exhaustive test of the seed interface. The subsequent native experiment below separately measures complete requests and memory. No seed, key, private share or ciphertext fixture is serialized. Only aggregate gate inventories, test outcomes and source hashes are retained in `data/seed-alternative-v26.json`.

The per-party required private share changes from 1184 to 64 bytes, saving 1120 bytes (94.59%). Both retain the public $ek$, context and one 64-byte anchor, so these common items and owner history are excluded. This is required prescribed storage, not an erasure guarantee for malicious Alice. Padded CheckedUse private input falls from 9728 to 768 bits per party, including the fresh 256-bit mask/pad in both cases; the masked relay stays 33 bytes.

The invariant AND decomposition is

$$\begin{aligned} \mathsf{SecretNoiseNTT}_3 &= 1{,}520{,}316, \\ \mathsf{Auth}_{1184} - \mathsf{Auth}_{64} &= 652{,}800 - 76{,}800 = 576{,}000, \\ \mathsf{AND}_{\mathrm{SR}} - \mathsf{AND}_{\mathrm{expanded}} &= 944{,}316. \end{aligned}$$

The use-time full-share hash costs 652800 ANDs because its public-key digest is already public and two initial constant blocks fold away. The distinct KeyGen final-anchor count is 691200: its public-key digest is still computed on circuit wires. Substituting the latter into the use comparison would underestimate the SR premium by 38400 ANDs.

Across the six tested public instances the exact all-gate inventory and its relative increase are reported in the retained JSON. Public constant folding can change total XOR/INV counts between instances, whereas the extra 944316 ANDs are invariant in these tests. These are native-reference Boolean counts, not the parameterized mathematical-family sizes in the theoretical LP bridge. They must not be transferred into an LP implementation speedup. Given the same conservative KeyGen, $n$ uses add $944316n$ reference ANDs to SR while saving 1120 required private bytes per party. For infrequent use or many stored keys SR may be preferable; the expanded interface pays more storage to avoid repeated secret-noise/NTT materialization.

### 23.3 Complete same-backend retained-handle experiment

The v30 native adapter makes the two lifetime choices executable under the same service boundary. It uses the frozen v20 repaired EMP engine without changing its COT, tag, bucket, challenge or shuffle parameters. The build checks all 350 frozen backend source pins. Both variants compile the same worker translation unit with the same arm64 Clang options; one additional public macro selects the SR state/output adapter and circuit builder. Full malicious security of this native engine remains a separate component assumption; the LP realization and its proof do not certify these binaries or their platform interfaces.

Both setups execute the original two T KeyGen circuits, including the final expanded-state computation and authentication. Only protected successful final completion releases the persistent record. T stores the expanded private share and its original full-share anchor. SR instead returns the original 64-byte first-stage share and its original transfer anchor, with the same generated $ek$ and context; it does not freshly enroll a restored share. SR's Bob record actually stores 272 bytes, comprising the same 208-byte header and 64 private bytes, versus 1392 bytes for T. Its local owner-to-worker IPC envelope is padded with 1120 verified zeros to reuse the existing launcher. Those zeros are not persisted and are not circuit private input. The workers receive 768 private bits per party in SR, including the 256-bit fresh pad/mask, versus 9728 bits for T.

For use, the SR builder is the previously independently clear-evaluated retained-state function, with the same complete Decaps core. It authenticates the actual seed-share input against

the original transfer-domain anchor before using those wires to regenerate the three secret polynomials/NTT and $z$. Public $ek$, the context, ciphertext and original anchor retain the same descriptor/owner protections. It conservatively retains the private canonical guard, masks and output pad. The owner adapter uses the same SQLite transaction, SID/quota consumption, protected completion, lease/reaping and durable output paths. A profile cannot select the other profile's existing handle. These are reviewed source correspondences and tested paths, not a native refinement theorem for the new adapter or the underlying backend.

Qualification precedes formal sampling. Six small backend probes exercise constant-output input-marker diagnostics, MAC tamper refusal and complete bucket-pool handling across both compiled profiles. Finite marker tests do not establish input privacy. A full excluded pilot checks matched valid/invalid uses, SR mutations in $\sigma$, $z$ and its final byte, a T $z$ mutation, both reopened-owner uses and SID replay refusal. All four altered shares are refused through native authentication, retire the reservation without an output and leave the original record unchanged. Reopened owners produce the original implicit-rejection value; duplicate SID refusal launches no worker. Oracle checks also verify actual stored sizes and that the retained seed and original transfer anchor derive from the original generation, rather than later input.

The predeclared formal cohort has six independent key blocks. Each block uses one role-private seed pair for the paired T/SR setups; role-only inherited pipes keep the other party's contribution outside each worker. The test controller reconstructs the reference key only for independent OpenSSL checks, outside workers and timers. Both profiles receive the same valid ciphertext and the same one-bit-mutated ciphertext, whose original-key rejection result is independently checked. Profile order alternates, and blocks divisible by three run invalid before valid. Each request has fresh masks, preprocessing and workers; each profile/block has a fresh owner. All 24 complete use outputs match independent Decaps. The 12 setup KeyGens, qualification, probes, pilot and oracle work are excluded from the use statistics. Private temporary fixtures are removed and never serialized into the public evidence archive.

| Case | Profile | Time (s) [95% CI] | CPU (s) | Traffic (GiB) | Footprint (GiB) | RSS (GiB) |
|---|---|---|---|---|---|---|
| Valid | T | 48.44 [46.32,50.69] | 37.11 | 4.787 | 13.86 | 4.77 |
| Valid | SR | 66.33 [61.91,70.84] | 42.83 | 5.245 | 15.05 | 4.89 |
| Invalid | T | 48.83 [46.30,51.46] | 37.43 | 4.787 | 13.84 | 4.85 |
| Invalid | SR | 62.67 [59.33,66.10] | 43.57 | 5.245 | 15.01 | 4.89 |

Each row has six independent key blocks. Time spans reservation through durable output and complete joins. CPU sums both actual workers; RSS is the mean of the largest single-worker actual-PID lifetime peak per request, not a sum of independent maxima. Footprint sums synchronized samples and includes compression. Intervals use 10000 percentile-bootstrap resamples of key blocks. They are conditional on this host/run and are not population guarantees.

For a 50/50 valid/invalid mix, T/SR means are 48.63/64.50 s and 4.787/5.245 GiB traffic. The paired time difference is 15.87 s (95% interval [14.07,17.89]); SR is slower in this cohort. SR reduces actual Bob files from 1392 to 272 bytes (208 common header bytes), while increasing temporary footprint. Intervals resample the six independent key blocks.

The paired equal-mixture CPU difference is 5.93 s, with 95% interval [4.89,6.85]. The paired footprint difference is 1.19 GiB, with interval [1.00,1.44]. Observed maximum sampling gap and alignment age are 0.299 and 0.229 s, respectively; requested sampling was 100 ms. System-wide swap deltas are retained per request, without attributing them solely to these workers. No VM reset, paging-free environment or WAN result is claimed.

| Phase | T wall (s) | SR wall (s) | T CPU (s) | SR CPU (s) |
|---|---|---|---|---|
| Preparation | 1.16 | 1.02 | 1.92 | 1.60 |
| Independent preprocessing | 12.56 | 16.31 | 18.81 | 22.57 |
| Dependent preprocessing | 32.13 | 40.94 | 14.84 | 17.01 |
| Online | 2.39 | 5.84 | 1.06 | 1.42 |
| Other / owner / join | 0.39 | 0.39 | 0.64 | 0.60 |

These equal-mixture phase means follow Bob's successive wall boundaries plus the owner/join residual; CPU sums both actual workers, with startup/exit residual in Other. Preparation includes circuit construction, conversion and backend initialization. Differences include waiting and the observed host VM state; their additive attribution is not a causal separation of computation, compression and paging within this measured service boundary.

The mean paired SR increment in ANDs is 944,316. The mean paired SR increment in all gates is 4,213,236. The mean paired SR increment in peer traffic is 491,876,279 bytes. The extra ANDs reconcile to the independently clear-evaluated 944316 difference in every matched ciphertext pair. Smaller input lowers input-related work but does not remove secret-noise/NTT or all-gate garbling/preprocessing work. Their combined complete-request effect is measured here, without inferring causation from Boolean gate inventories alone.

The results are one matched lifetime comparison, not an independent-backend ranking or a general performance ordering. Native seed retention is now measured at the complete-request boundary; an optimized seed-only setup, LP implementation, concurrent service and a different storage/use workload remain unmeasured. The selection procedure uses these observed costs only after both mathematical profiles meet the same original-handle relation and the same operational custody policy under the stated assumptions.

## 24 Constructive cost of the mathematical KeyGen functions

The frozen EMP inventories are no longer the only cost bridge. We give a separate, deliberately unoptimized Boolean construction of the mathematical KeyGen stages. It uses the bounded algorithms in Sec. 2, not an assertion that the native generator refines them. Its ordinary two-input truth-table gates include XOR, INV, constants and terminal identity gates; constant and unary truth tables ignore inputs. Constants are materialized twice per stage and no constant folding, expression elimination, free XOR or half gates is used. Elementary networks are emitted to their stated budgets; unused carry, multiplication and padding gates are retained. Thus the counts below specify *selected padded circuits*, rather than subtracting independent upper bounds to infer a performance improvement. No claim of optimality, measured LP runtime or resident memory follows.

**A uniform public specialization.** For T/L completion, a single public polynomial-time algorithm recomputes the four-block matrix from the agreed $\rho$, checks refusal, and generates the known-factor multiplication networks. Both parties bind the same function, indices and public descriptor before invocation. The specialization and padding depend only on public data. The generator has fixed polynomial size bounds and does not choose unrelated nonuniform functions. Each public-matrix known-factor multiply is padded to 22 modular-addition networks, even for a zero factor. Alternatively the same public matrix can remain ordinary circuit inputs; the numerical extra cost of that option is given below. Neither circuit choice imports any correctness assertion about the native compiler into the mathematical security theorem.

**Elementary networks and deterministic expansion.** A width-$w$ ripple adder emits six gates per bit, using $s = a \oplus b \oplus r$ and $r' = a \oplus ((a \oplus b) \wedge (a \oplus r))$. Sharing $a \oplus b$ needs five functional gates; a sixth, inert padding gate per bit is retained explicitly in this construction. Subtracting uses another $w$ inversions; a bit selector uses three gates. Canonical 12-bit modular addition/subtraction uses a 13-bit add, subtract and selector, costing $13(6 + 7 + 3) = 208$ gates. A generic multiply uses 12 selected additions and 11 doublings, costing $12(12 + 208) + 11 \cdot 208 = 4928$ gates. Known-factor double-and-add uses $\mathrm{popcount}(a) + \lfloor \log_2 a \rfloor \leq 22$ additions for $0 < a < 3329$, and zero for $a = 0$, before public padding.

The explicit Keccak schedule costs 201216 gates per permutation: each of 24 rounds uses 1600 column-accumulation XORs, 320 parity XORs, 1600 state XORs, 4800 $\chi$ gates and 64 round-constant XORs. Absorption emits one XOR per rate bit; rotations and byte encoding are wire permutations. Six centered-binomial noise samplers cost 1539456 gates. The six NTTs cost 19670976 gates: there are 896 butterflies per transform, with the fixed standard twiddle powers counted individually before padding. The joint seed XOR and its expansion cost 202304 gates.

For D, each of nine private matrix polynomials uses four XOF blocks, 448 candidate comparisons and a 512-slot bitonic network. Its 11520 comparators use 77 gates for a 10-bit key comparison and $4 \cdot 22$ gates to exchange the key/value bits. Stream positions are included in sorting keys, so accepted coefficients retain stream order. Candidate-validity inversions and final validity gates are included. The total is 24734034 gates, including XOF expansion. D's matrix products cost 28938240 gates. The public known-factor T/L products cost 22044672 gates. These are constructive counts of the mathematical bounded algorithms, with the same canonical field representation; they are not the reference EMP section counters.

**Full input domains and failure payloads.** L completion checks all 1536 input coefficients using 13-bit subtraction and a conjunction tree. It computes its fixed Boolean arithmetic on every input and internally selects a fixed zero pair $(ek, X)$ when any coefficient is noncanonical; the private predicate is not a public authentication flag. The totalization costs $1536 \cdot 91 + 1535 + 18944 = 160255$ gates. Thus undefined native behavior is not needed. The analogous canonicalization/dummy selection for the common CheckedUse core costs $768 \cdot 91 + 767 + 9472 = 80127$ gates; its remaining cost is common to all architectures. Final generation failure gates every payload bit before release. The final masks, Alice output pad, actual handoff equality check and distinct terminal output wires are all counted.

**Explicit hash costs.** Let $F, P, T_h, L_h$ be the chosen ordinary circuit sizes of final anchoring, public-key hashing, 64-byte handoff hashing and 2336-byte handoff hashing, respectively, including their complete encodings and public indices. With 512-bit framework hash outputs, the selected circuits have exactly

$$G_D = 75142876 + F + P, \tag{26}$$

$$G_T = 43519627 + F + P + 2T_h, \tag{27}$$

$$G_L = 43752586 + F + P + 2L_h. \tag{28}$$

These are gates, not ANDs. The nonframework-hash stage subtotals are

$$\begin{aligned} b_D &= 75142876, \\ (b_{T1}, b_{T2}) &= (204610, 43315017), \\ (b_{L1}, b_{L2}) &= (21469570, 22283016). \end{aligned}$$

For an arbitrary framework digest length $h_c$, the respective bases are $75141852+2h_c$, $43516555+6h_c$ and $43749514 + 6h_c$; changes in other public encodings and growing hash parameters are carried by $F, P, T_h, L_h$. No numerical gate bound follows from collision resistance alone. A selected family must supply all four costs at the relevant parameter values; fixed SHA3/BLAKE2b costs do not account for the parameter growth required by the asymptotic theorem.

Equation (28) supplies an exact crossover for these circuits: T uses fewer table gates than D if and only if $2T_h < 31623249$. Also $G_L - G_T = 232959 + 2(L_h - T_h)$. Hence an early cut has no universal backend-independent percentage benefit; its handoff hash cost matters. Keeping the public matrix as ordinary circuit inputs adds 6893568 gates to each T/L completion, changing the T/D condition to $2T_h < 24729681$. Thus the public-input treatment is quantified rather than hidden behind a reference-topology identity.

**A growing-family cost sensitivity.** For a concrete serialization example, use separate indexed collision-resistant compression families for $H_c, H_p$ with 512-bit state/block and circuit budgets $C_c, C_p$. For messages shorter than $2^{64}$ bits, length strengthening appends `0x80`, zero padding and a 64-bit bit length. A 32-byte fixed domain and 64-byte context give final/T/L message lengths 1344/192/2464 bytes; public-key hashing has 1184 bytes. Consequently $F = 22C_c$, $T_h = 4C_c$, $L_h = 39C_c$ and $P = 19C_p$. The compression index is a public parameter of its circuit, not an unaccounted secret input. When actual indices/context encodings have other lengths, Eq. (28) applies with their actual costs. For an asymptotic family the state, encoding and circuit budgets grow, including a length field wide enough for its polynomial message bound; fixed 512-bit numbers here describe only this finite cost slice.

| $C_c = C_p$ | $G_D$ (M) | $G_T$ (M) | $G_L$ (M) | T/D saving |
|---|---|---|---|---|
| 100000 | 79.243 | 48.420 | 55.653 | 38.90% |
| 500000 | 95.643 | 68.020 | 103.253 | 28.88% |
| 1000000 | 116.143 | 92.520 | 162.753 | 20.34% |
| 4000000 | 239.143 | 239.520 | 519.753 | −0.16% |

The compression-cost crossover is $C_c < 3952906.125$ gates, independent of the common $C_p$ term. These are explicit sensitivity points, not a chosen compression implementation, empirical estimates or concrete 128-bit security assertions. Length-strengthened fixed-domain chaining reduces any output collision to a compression collision by tracing equal final states backward to the first unequal compression input, with the length field handling unequal-length encodings. The proof therefore continues to charge the stated collision-resistance assumption.

**The numerical difference from reference shapes.** For this serialization, subtracting the frozen mean reference inventories gives

$$\begin{aligned} G_D - G_D^{\text{shape}} &= 15849019\tfrac{1}{3} + 22C_c + 19C_p, \\ G_T - G_T^{\text{shape}} &= 12206239\tfrac{2}{3} + 30C_c + 19C_p, \\ G_L - G_L^{\text{shape}} &= 36540\tfrac{1}{2} + 100C_c + 19C_p. \end{aligned}$$

This is a quantified comparison against a new selected construction, not a compiler refinement claim or an upper bound for every conceivable realization. Multiply each full $G_R$, including hash costs, by $4c$ for LP's table component. The old 47.19% reference-shape saving and 10.05-fold measured EMP wall-time ratio cannot replace these expressions. Common-use cost is added equally to D/T/L for a whole service workload; KeyGen savings alone do not determine its percentage.

**Reproduction and remaining boundary.** Run `lp_math_cost_v26.py` with Python 3. It verifies the 48 source pins, recomputes the construction schedule, hash-call coefficients, crossover and reference deltas, and writes `data/lp-math-cost-v26.json`. It also exhaustively checks all reachable canonical sums/differences, all 12-bit canonicality inputs, all public factors and the six-bit ripple recurrence. These arithmetic controls support the explicit lowering identities; they do not mechanically verify the entire cryptographic compiler. The remaining configurable quantity is the selected growing hash family's circuit budget, explicitly isolated above, and LP is not executed in the present analytical cost evaluation.

# 25 Authenticated-cut and retained-lifetime design conditions

The sufficient-state corollary first filters representations by the complete original-handle relation. This section derives cost conditions for the remaining placements. Its scope is the declared circuits and resource metrics. The distinction matters because a later cut can reduce the largest stage while increasing the duplicated authentication work, and smaller persistent state can increase complete-request cost. The declared interfaces and cost inventory determine these conditions.

## 25.1 Why an expanded handoff can lose total work

Fix one gate basis and two admissible generation cuts with the same final anchor/public-key hashing. Write $H_j$ for the cost of one handoff hash and $K_j$ for all remaining profile-specific gates, including encoding, canonicality, masking, output routing and public-input treatment. If the handoff is hashed when produced and again on its actual reinput, its total is

$$G_j = C + K_j + 2H_j,$$

where $C$ contains only common work. Thus an early cut $e$ beats a later cut $l$ in this metric precisely when

$$K_e - K_l < 2(H_l - H_e). \tag{29}$$

The inequality compares duplicated authentication with the remaining work saved by moving the cut. Both representations must satisfy the original-handle and full-domain obligations.

**Proposition 3** (Gate ordering of the selected ML-KEM cuts)**.** *For the independent padded ordinary-gate construction of Sec. 24, using the same hashes and public specialization,*

$$G_L - G_T = 232959 + 2(L_h - T_h), \tag{30}$$

$$G_D - G_T = 31623249 - 2T_h. \tag{31}$$

*Consequently $T$ has lower total gate work than $L$ whenever $L_h \geq T_h$. It has lower gate work than $D$ exactly when $2T_h < 31623249$. At common LP parameter $c$, this ordering also holds for $4cG_j$ table ciphertexts, including all declared framework hash costs.*

*Proof.* The constructive inventory gives $G_D = 75142876 + F + P$, $G_T = 43519627 + F + P + 2T_h$ and $G_L = 43752586 + F + P + 2L_h$. Subtracting cancels the common final-anchor and public-key hashes $F, P$. The differences of the remaining bases are $43752586 - 43519627 = 232959$ and $75142876 - 43519627 = 31623249$. These yield Eqs. (30)–(31). Positive $4c$ preserves the ordering. □

In the explicit finite 512-bit serialization example, $T_h = 4C_c$ and $L_h = 39C_c$, so

$$G_L - G_T = 232959 + 70C_c > 0 \quad (C_c \geq 0).$$

The early cut therefore has strictly fewer gates in this selected construction for every nonnegative compression-circuit cost in that slice. The T/D crossover is $C_c < 3952906.125$. Other encodings and growing hash families use their actual $T_h, L_h$ in the general conditions. These are conditional circuit-cost conclusions, with the hash-family assumption and parameter-growth cost intact.

T and L have two generation calls, so equal reservation bounds select the same LP parameter. For D versus T a common conservative $c$ can be chosen from the maximum call budget. If each profile instead minimizes its own parameter, the table comparison becomes $c_T G_T < c_D G_D$, equivalently

$$2T_h < \frac{c_D}{c_T}(75142876 + F + P) - (43519627 + F + P).$$

This keeps a lower call count's security-parameter benefit in the comparison. Core rounds, relays and other cryptographic costs remain additional coordinates.

**Independent native correspondence.** The native AND metric differs from the LP ordinary-gate metric. In the frozen EMP circuits, T/L have the same core arithmetic. Their two hand-off hashes *together* cost 153600 and 2611200 ANDs, giving $2611200 - 153600 = 2457600$ extra L ANDs. This is the complete measured paired T/L AND gap. The native result verifies the duplicated-hash explanation within that backend. The positive 232959 ordinary-gate term comes from the separate padded mathematical lowering. Neither count is substituted for the other. L's smaller largest stage consequently supplies a different optimization coordinate from total work. Its footprint interval spans zero relative to T in the matched cohort, so a memory advantage does not follow from the smaller largest stage.

### 25.2 When keeping seeds is worth reconstruction

T and SR meet the same original-operation and split-custody goals after the same completed conservative T setup. Let $N$ keys remain live over the declared horizon and each complete $n$ uses with fixed valid fraction $v$. Let $w_s, w_t, w_b$ be nonnegative prices per private-share byte over that horizon, per use second and per peer byte, in one common cost unit. Write $\Delta t(v), \Delta b(v)$ for SR-minus-T complete-use costs under the matched request boundary. Define

$$d(v) = w_t \Delta t(v) + w_b \Delta b(v).$$

The storage-and-use objective has exact algebraic difference

$$\frac{J_{\mathrm{SR}} - J_T}{N} = -1120 w_s + n d(v). \tag{32}$$

Here setup cost is common and conditioned on completion. Failed attempts add their actual executed work, accounted for over the declared horizon as separate reserved attempts.

**Proposition 4** (Storage filter and use-count crossover)**.** *With per-party private-payload budget $\mathcal{B}$, the interval $64N \le \mathcal{B} < 1184N$ admits SR alone within the T/SR pair. Below $64N$ neither fits, and at or above $1184N$ both pass this storage constraint. When both fit and $d(v) > 0$, the crossover is*

$$n^*(v) = \frac{1120 w_s}{d(v)}. \tag{33}$$

*SR has lower $J$ for $n < n^*$ and T for $n > n^*$, with equality at the boundary. If $d = 0$, only the stored-byte term separates them. If $d < 0$, SR is weakly cheaper for all $n \ge 0$ in this objective.*

*Proof.* Multiply the required widths 64 and 1184 by $N$ and compare with $\mathcal{B}$. Both protocols share setup, so subtracting their objectives cancels it and leaves Eq. (32). Its affine dependence on $n$ has slope $d(v)$ and intercept $-1120w_s$. Solving for its zero gives Eq. (33) and the stated sign cases. Feasibility is applied before cost ordering, so an infeasible cheaper point remains inadmissible. □

These budgets count prescribed private-share payloads. The measured Bob records include 208 common header bytes, giving 272/1392 bytes for SR/T. A budget for those serialized records instead uses thresholds $272N$ and $1392N$. Both also retain public-key files and owner history. Such common costs cancel in the objective difference but consume their actual storage budget. A hard temporary-memory constraint is an independent feasibility check. Sampled footprint maxima are observations of this host rather than hard capacity bounds.

In the six-block equal-mix cohort, $\Delta t = 15.8672$ seconds and $\Delta b = 491876278.67$ peer bytes. All six paired block differences are positive in both coordinates. SR saves persistent state while costing more complete-use time and traffic in this cohort, despite 8960 fewer private-input bits per party. The smaller input's OT benefit is therefore already included in the measured complete request. Choosing $w_s = 0$ and any positive time or traffic price selects T whenever both pass storage. A positive storage price allows SR only below the workload-dependent crossover or when T fails the storage budget. The resulting frontier is conditional on the declared costs, workload and capacity constraints for the matched experimental backend.

**Uncertainty and reproducibility.** For declared $w_t, w_b$, form six paired block values $d_i = w_t\Delta t_i + w_b\Delta b_i$ before resampling. This preserves the correlation of time and traffic, and the valid/invalid cases within a key. When $w_t > 0, w_b = 0$ at equal mix, the point crossover per unit ratio $w_s/w_t$ is $1120/15.8672 = 70.586$. Transforming the positive six-block 95% time interval gives endpoint range $[1120/17.8857, 1120/14.0732] = [62.620, 79.584]$ for that same ratio. This is a transformed uncertainty range for the local projection, conditional on these prices and run conditions. It is not a measured multi-use workload.

The runnable `derive_design_v32.py` recomputes the base differences, finite-serialization crossover and per-block lifetime inputs from the frozen inventories and raw use rows. `select_state_v30.py` remains the general storage/workload selector. The new calculator checks the comparison conditions and their sign reversals, with scope metadata that separates the analytical LP model from native measurements. Security admissibility is supplied by the written proof and is not inferred from either calculator.

# References


[1] National Institute of Standards and Technology, "Module-lattice-based key-encapsulation mechanism standard," National Institute of Standards and Technology, FIPS 203, Aug. 2024. [Online]. Available: https://doi.org/10.6028/NIST.FIPS.203

[2] J. Katz and Y. Lindell, "Handling expected polynomial-time strategies in simulation-based security proofs," *Journal of Cryptology*, vol. 21, no. 3, pp. 303–349, 2008. [Online]. Available: https://www.cs.umd.edu/~jkatz/papers/expected-full.pdf

[3] Y. Lindell and B. Pinkas, "Secure two-party computation via cut-and-choose oblivious transfer," *Journal of Cryptology*, vol. 25, no. 4, pp. 680–722, 2012, corrected full version, Oct. 29, 2014. [Online]. Available: https://www.cs.columbia.edu/~tal/6261/SP16/LP_cac.pdf

[4] Y. Lindell, "Parallel coin-tossing and constant-round secure two-party computation," *Journal of Cryptology*, vol. 16, no. 3, pp. 143–184, 2003. [Online]. Available: https://eprint.iacr.org/2001/107

[5] M. Naor, "Bit commitment using pseudorandomness," *Journal of Cryptology*, vol. 4, no. 2, pp. 151–158, 1991. [Online]. Available: https://www.wisdom.weizmann.ac.il/~naor/PAPERS/bit.pdf

[6] P. Mohassel, M. Rosulek, and Y. Zhang, "Fast and secure three-party computation: The garbled circuit approach," in *Proceedings of the 22nd ACM SIGSAC Conference on Computer and Communications Security*, 2015, pp. 591–602. [Online]. Available: https://eprint.iacr.org/2015/931

[7] M. Bellare, K. G. Paterson, and P. Rogaway, "Security of symmetric encryption against mass surveillance," in *CRYPTO*, 2014, pp. 1–19. [Online]. Available: https://doi.org/10.1007/978-3-662-44371-2_1

[8] M. Bellare, J. Jaeger, and D. Kane, "Mass-surveillance without the state: Strongly undetectable algorithm-substitution attacks," in *ACM CCS*, 2015, pp. 1431–1440. [Online]. Available: https://doi.org/10.1145/2810103.2813681

[9] J. B. Almeida *et al.*, "Formally verifying Kyber episode V: Machine-checked IND-CCA security and correctness of ML-KEM in EasyCrypt," IACR ePrint 2024/843, 2024. [Online]. Available: https://eprint.iacr.org/2024/843

[10] Amazon Web Services, "ImportKeyMaterial," AWS Key Management Service API Reference, accessed: Oct. 20, 2026. [Online]. Available: https://docs.aws.amazon.com/kms/latest/APIReference/API_ImportKeyMaterial.html

[11] D. Friolo, A. Reale, and D. Venturi, "Self-guarding arbitrary cryptographic primitives and 2PC protocols," in *IEEE CSF*, 2026, pp. 1–15. [Online]. Available: https://doi.org/10.1109/CSF68417.2026.00040

[12] S. Turner, P. Kampanakis, J. Massimo, and B. Westerbaan, "Internet X.509 public key infrastructure—algorithm identifiers for the module-lattice-based key-encapsulation mechanism (ML-KEM)," Internet Engineering Task Force, RFC 9935, Mar. 2026. [Online]. Available: https://www.rfc-editor.org/rfc/rfc9935.html

[13] S. Jarecki and V. Shmatikov, "Efficient two-party secure computation on committed inputs," in *Advances in Cryptology—EUROCRYPT 2007*, ser. Lecture Notes in Computer Science, vol. 4515. Springer, 2007, pp. 97–114. [Online]. Available: https://www.cs.cornell.edu/~shmat/shmat_eurocrypt07.pdf

[14] M. Chase, C. Ganesh, and P. Mohassel, "Efficient zero-knowledge proof of algebraic and non-algebraic statements with applications to privacy preserving credentials," in *Advances in Cryptology—CRYPTO 2016, Part III*, ser. Lecture Notes in Computer Science, vol. 9816. Springer, 2016, pp. 499–530. [Online]. Available: https://eprint.iacr.org/2016/583